

**ARBAMINCH UNIVERSITY**  
**COLLEGE OF BUSINESS AND ECONOMICS**  
**DEPARTMENT OF ECONOMICS**



**Econometrics I (Econ 3061)**  
**MODULE**

**AMU 2023**

## *Introduction to the*

The principal objective of the course, “*Econometrics I*”, is to provide an elementary but comprehensive introduction to the art and science of econometrics. It enables students to see how economic theory, statistical and mathematical methods are combined in the analysis of economic data, with a purpose of giving empirical content to economic theories and verify or refute them.

This course includes four chapters. The first chapter introduces students with the definition and some fundamental conceptualization of econometrics. In chapter two a fairly detailed treatment of the simple classical linear regression model will be made. In this chapter students will be introduced with the basic logic, concepts, assumptions, estimation methods, and interpretations of the simple classical linear regression models and their applications in economic science. Chapter three, which deals with Multiple Regression Models, is basically the extension of the simple regression models. But in chapter three attempts will be made to expand the linear regression model by incorporating more than one explanatory variables or regressors to the model. Chapter four pays due attention to violations of CLRM assumptions, their consequences, and the remedial measures. Specifically, Autocorrelation, Heteroscedasticity, and Multicollinearity problems will be given much focus.

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## **Chapter One**

### **Introduction**

#### **1.1 Definition and scope of econometrics**

The economic theories we learn in various economics courses suggest many relationships among economic variables. For instance, in microeconomics we learn demand and supply models in which the quantities demanded and supplied of a good depend on its price. In macroeconomics, we study ‘investment function’ to explain the amount of aggregate investment in the economy as the rate of interest changes; and ‘consumption function’ that relates aggregate consumption to the level of aggregate disposable income.

Each of such specifications involves a relationship among economic variables. As economists, we may be interested in questions such as: If one variable changes in a certain magnitude, by how much will another variable change? Also, given that we know the value of one variable; can we forecast or predict the corresponding value of another? The purpose of studying the relationships among economic variables and attempting to answer questions of the type raised here, is to help us understand the real economic world we live in.

However, economic theories that postulate the relationships between economic variables have to be checked against data obtained from the real world. If empirical data verify the relationship proposed by economic theory, we accept the theory as valid. If the theory is incompatible with the observed behavior, we either reject the theory or in the light of the empirical evidence of the data, modify the theory. To provide a better understanding of economic relationships and a better

guidance for economic policy making we also need to know the quantitative relationships between the different economic variables. We obtain these quantitative measurements taken from the real world. *The field of knowledge which helps us to carryout such an evaluation of economic theories in empirical terms is econometrics.*

Distance students! Having said the background statement in our attempt for defining ‘*ECONOMETRICS*’, we may now formally define what econometrics is.

### *WHAT IS ECONOMETRICS?*

Literally interpreted, econometrics means “economic measurement”, but the scope of econometrics is much broader as described by leading econometricians. Various econometricians used different ways of wordings to define econometrics. But if we distill the fundamental features/concepts of all the definitions, we may obtain the following definition.

*“Econometrics is the science which integrates economic theory, economic statistics, and mathematical economics to investigate the empirical support of the general schematic law established by economic theory. It is a special type of economic analysis and research in which the general economic theories, formulated in mathematical terms, is combined with empirical measurements of economic phenomena. Starting from the relationships of economic theory, we express them in mathematical terms so that they can be measured. We then use specific methods, called econometric methods in order to obtain numerical estimates of the coefficients of the economic relationships.”*

Measurement is an important aspect of econometrics. However, the scope of econometrics is much broader than measurement. As D.Intriligator rightly stated

the “*metric*” part of the word econometrics signifies ‘*measurement*’, and hence econometrics is basically concerned with measuring of economic relationships.

In short, econometrics may be considered as the integration of economics, mathematics, and statistics for the purpose of providing numerical values for the parameters of economic relationships and verifying economic theories.

### 1.2 Econometrics vs. mathematical economics

Mathematical economics states economic theory in terms of mathematical symbols. There is no essential difference between mathematical economics and economic theory. Both state the same relationships, but while economic theory use verbal exposition, mathematical symbols. Both express economic relationships in an exact or deterministic form. Neither mathematical economics nor economic theory allows for random elements which might affect the relationship and make it stochastic. Furthermore, they do not provide numerical values for the coefficients of economic relationships.

Econometrics differs from mathematical economics in that, although econometrics presupposes, the economic relationships to be expressed in mathematical forms, it does not assume exact or deterministic relationship. Econometrics assumes random relationships among economic variables. Econometric methods are designed to take into account random disturbances which relate deviations from exact behavioral patterns suggested by economic theory and mathematical economics. Furthermore, econometric methods provide numerical values of the coefficients of economic relationships.

### 1.3 Econometrics vs. statistics

Econometrics differs from both mathematical statistics and economic statistics. An economic statistician gathers empirical data, records them, tabulates them or

charts them, and attempts to describe the pattern in their development over time and perhaps detect some relationship between various economic magnitudes. Economic statistics is mainly a descriptive aspect of economics. It does not provide explanations of the development of the various variables and it does not provide measurements the coefficients of economic relationships.

Mathematical (or inferential) statistics deals with the method of measurement which are developed on the basis of controlled experiments. But statistical methods of measurement are not appropriate for a number of economic relationships because for most economic relationships controlled or carefully planned experiments cannot be designed due to the fact that the nature of relationships among economic variables are stochastic or random. Yet the fundamental ideas of inferential statistics are applicable in econometrics, but they must be adapted to the problem economic life. Econometric methods are adjusted so that they may become appropriate for the measurement of economic relationships which are stochastic. The adjustment consists primarily in specifying the stochastic (random) elements that are supposed to operate in the real world and enter into the determination of the observed data.

#### 1.4 Economic models vs. econometric models

##### *i) Economic models:*

Any economic theory is an observation from the real world. For one reason, the immense complexity of the real world economy makes it impossible for us to understand all interrelationships at once. Another reason is that all the interrelationships are not equally important as such for the understanding of the economic phenomenon under study. The sensible procedure is therefore, to pick up the important factors and relationships relevant to our problem and to focus our attention on these alone. Such a deliberately simplified analytical framework is called on economic model. It is an organized set of relationships that describes the

functioning of an economic entity under a set of simplifying assumptions. All economic reasoning is ultimately based on models. Economic models consist of the following three basic structural elements.

1. A set of variables
2. A list of fundamental relationships and
3. A number of strategic coefficients

*ii) Econometric models:*

The most important characteristic of economic relationships is that they contain a random element which is ignored by mathematical economic models which postulate exact relationships between economic variables.

Example: Economic theory postulates that the demand for a commodity depends on its price, on the prices of other related commodities, on consumers' income and on tastes. This is an exact relationship which can be written mathematically as:

$$Q = b_0 + b_1P + b_2P_0 + b_3Y + b_4t$$

The above demand equation is exact. However, many more factors may affect demand. In econometrics the influence of these 'other' factors is taken into account by the introduction into the economic relationships of random variable. In our example, the demand function studied with the tools of econometrics would be of the stochastic form:

$$Q = b_0 + b_1P + b_2P_0 + b_3Y + b_4t + u$$

where  $u$  stands for the random factors which affect the quantity demanded.

### 1.5 Methodology of econometrics

Econometric research is concerned with the measurement of the parameters of economic relationships and with the predication of the values of economic variables. The relationships of economic theory which can be measured with

econometric techniques are relationships in which some variables are postulated as causes of the variation of other variables. Starting with the postulated theoretical relationships among economic variables, econometric research or inquiry generally proceeds along the following lines/stages.

1. Specification the model
2. Estimation of the model
3. Evaluation of the estimates
4. Evaluation of he forecasting power of the estimated model

### *1. Specification of the model*

In this step the econometrician has to express the relationships between economic variables in mathematical form. This step involves the determination of three important tasks:

- i) the dependent and independent (explanatory) variables which will be included in the model.
- ii) the *a priori* theoretical expectations about the size and sign of the parameters of the function.
- iii) the mathematical form of the model (number of equations, specific form of the equations, etc.)

Note: The specification of the econometric model will be based on economic theory and on any available information related to the phenomena under investigation. Thus, specification of the econometric model presupposes knowledge of economic theory and familiarity with the particular phenomenon being studied.

Specification of the model is the most important and the most difficult stage of any econometric research. It is often the weakest point of most econometric applications. In this stage there exists enormous degree of likelihood of

committing errors or incorrectly specifying the model. Some of the common reasons for incorrect specification of the econometric models are:

1. the imperfections, looseness of statements in economic theories.
2. the limitation of our knowledge of the factors which are operative in any particular case.
3. the formidable obstacles presented by data requirements in the estimation of large models.

The most common errors of specification are:

- a. Omissions of some important variables from the function.
- b. The omissions of some equations (for example, in simultaneous equations model).
- c. The mistaken mathematical form of the functions.

## *2. Estimation of the model*

This is purely a technical stage which requires knowledge of the various econometric methods, their assumptions and the economic implications for the estimates of the parameters. This stage includes the following activities.

- a. Gathering of the data on the variables included in the model.
- b. Examination of the identification conditions of the function (especially for simultaneous equations models).
- c. Examination of the aggregations problems involved in the variables of the function.
- d. Examination of the degree of correlation between the explanatory variables (i.e. examination of the problem of multicollinearity).
- e. Choice of appropriate economic techniques for estimation, i.e. to decide a specific econometric method to be applied in estimation; such as, OLS, MLM, Logit, and Probit.

### 3. *Evaluation of the estimates*

This stage consists of deciding whether the estimates of the parameters are *theoretically meaningful and statistically satisfactory*. This stage enables the econometrician to evaluate the results of calculations and determine the reliability of the results. For this purpose we use various criteria which may be classified into three groups:

- i. *Economic a priori criteria*: These criteria are determined by economic theory and refer to the size and sign of the parameters of economic relationships.
- ii. *Statistical criteria (first-order tests)*: These are determined by statistical theory and aim at the evaluation of the statistical reliability of the estimates of the parameters of the model. Correlation coefficient test, standard error test, t-test, F-test, and  $R^2$ -test are some of the most commonly used statistical tests.
- iii. *Econometric criteria (second-order tests)*: These are set by the theory of econometrics and aim at the investigation of whether the assumptions of the econometric method employed are satisfied or not in any particular case. The econometric criteria serve as a second order test (as test of the statistical tests) i.e. they determine the reliability of the statistical criteria; they help us establish whether the estimates have the desirable properties of unbiasedness, consistency etc. Econometric criteria aim at the detection of the violation or validity of the assumptions of the various econometric techniques.

### 4) *Evaluation of the forecasting power of the model:*

Forecasting is one of the aims of econometric research. However, before using an estimated model for forecasting by some way or another the predictive power of the model. It is possible that the model may be economically

meaningful and statistically and econometrically correct for the sample period for which the model has been estimated; yet it may not be suitable for forecasting due to various factors (reasons). Therefore, this stage involves the investigation of the stability of the estimates and their sensitivity to changes in the size of the sample. Consequently, we must establish whether the estimated function performs adequately outside the sample of data. i.e. we must test an extra sample performance the model.

### 1.6 Desirable properties of an econometric model

An econometric model is a model whose parameters have been estimated with some appropriate econometric technique. The ‘goodness’ of an econometric model is judged customarily according to the following desirable properties.

1. *Theoretical plausibility.* The model should be compatible with the postulates of economic theory. It must describe adequately the economic phenomena to which it relates.
2. *Explanatory ability.* The model should be able to explain the observations of the actual world. It must be consistent with the observed behaviour of the economic variables whose relationship it determines.
3. *Accuracy of the estimates of the parameters.* The estimates of the coefficients should be accurate in the sense that they should approximate as best as possible the true parameters of the structural model. The estimates should if possible possess the desirable properties of unbiasedness, consistency and efficiency.

4. *Forecasting ability*. The model should produce satisfactory predictions of future values of the dependent (endogenous) variables.
5. *Simplicity*. The model should represent the economic relationships with maximum simplicity. The fewer the equations and the simpler their mathematical form, the better the model is considered, *ceteris paribus* (that is to say provided that the other desirable properties are not affected by the simplifications of the model).

### 1.7 Goals of Econometrics

Three main goals of Econometrics are identified:

- i) *Analysis* i.e. testing economic theory
- ii) *Policy making* i.e. Obtaining numerical estimates of the coefficients of economic relationships for policy simulations.
- iii) *Forecasting* i.e. using the numerical estimates of the coefficients in order to forecast the future values of economic magnitudes.

#### *Review questions*

- How would you define econometrics?
- How does it differ from mathematical economics and statistics?
- Describe the main steps involved in any econometrics research.
- Differentiate between economic and econometric model.
- What are the goals of econometrics?

## **Chapter Two**

### **THE CLASSICAL REGRESSION ANALYSIS**

#### **[The Simple Linear Regression Model]**

Economic theories are mainly concerned with the relationships among various economic variables. These relationships, when phrased in mathematical terms, can predict the effect of one variable on another. The functional relationships of these variables define the dependence of one variable upon the other variable (s) in the specific form. The specific functional forms may be linear, quadratic, logarithmic, exponential, hyperbolic, or any other form.

In this chapter we shall consider a simple linear regression model, i.e. a relationship between two variables related in a linear form. We shall first discuss two important forms of relation: *stochastic* and *non-stochastic*, among which we shall be using the former in econometric analysis.

#### *2.1. Stochastic and Non-stochastic Relationships*

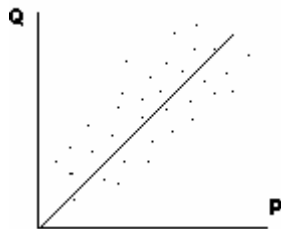
A relationship between  $X$  and  $Y$ , characterized as  $Y = f(X)$  is said to be deterministic or non-stochastic if for each value of the independent variable ( $X$ ) there is one and only one corresponding value of dependent variable ( $Y$ ). On the other hand, a relationship between  $X$  and  $Y$  is said to be stochastic if for a particular value of  $X$  there is a whole probabilistic distribution of values of  $Y$ . In such a case, for any given value of  $X$ , the dependent variable  $Y$  assumes some specific value only with some probability. Let's illustrate the distinction between stochastic and non stochastic relationships with the help of a supply function.

Assuming that the supply for a certain commodity depends on its price (other determinants taken to be constant) and the function being linear, the relationship can be put as:

$$Q = f(P) = \alpha + \beta P \text{ ----- (2.1)}$$

The above relationship between P and Q is such that for a particular value of P, there is only one corresponding value of Q. This is, therefore, a deterministic (non-stochastic) relationship since for each price there is always only one corresponding quantity supplied. This implies that all the variation in Y is due solely to changes in X, and that there are no other factors affecting the dependent variable.

If this were true all the points of price-quantity pairs, if plotted on a two-dimensional plane, would fall on a straight line. However, if we gather observations on the quantity actually supplied in the market at various prices and we plot them on a diagram we see that they do not fall on a straight line.



**Fig 2.a The scatter diagram**

The derivation of the observation from the line may be attributed to several factors.

- a. Omission of variables from the function
- b. Random behavior of human beings
- c. Imperfect specification of the mathematical form of the model
- d. Error of aggregation
- e. Error of measurement

In order to take into account the above sources of errors we introduce in econometric functions a random variable which is usually denoted by the letter ‘u’ or ‘ $\epsilon$ ’ and is called error term or random disturbance or stochastic term of the function, so called because u is supposed to ‘disturb’ the exact linear relationship which is assumed to exist between X and Y. By introducing this random variable in the function the model is rendered stochastic of the form:

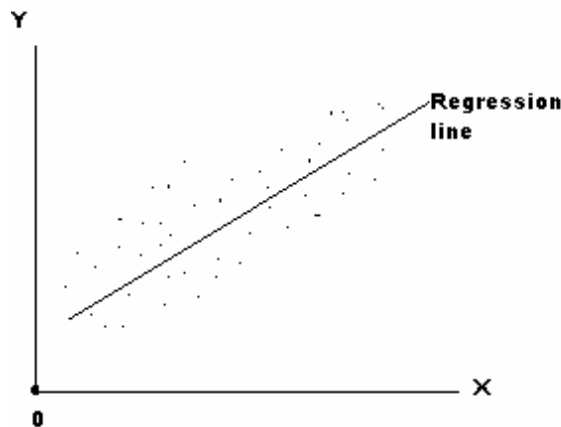
$$Y_i = \alpha + \beta X + u_i \dots\dots\dots (2.2)$$

Thus a stochastic model is a model in which the dependent variable is not only determined by the explanatory variable(s) included in the model but also by others which are not included in the model.

## 2.2. Simple Linear Regression model.

The above stochastic relationship (2.2) with one explanatory variable is called simple linear regression model.

The true relationship which connects the variables involved is split into two parts: a part represented by a line and a part represented by the random term ‘u’.



**Fig 2.b the scatter diagram and the regression line**

The scatter of observations represents the true relationship between Y and X. The line represents the exact part of the relationship and the deviation of the observation from the line represents the random component of the relationship.

- Were it not for the errors in the model, we would observe all the points on the line  $Y'_1, Y'_2, \dots, Y'_n$  corresponding to  $X_1, X_2, \dots, X_n$ . However because of the random disturbance, we observe  $Y_1, Y_2, \dots, Y_n$  corresponding to  $X_1, X_2, \dots, X_n$ . These points diverge from the regression line by  $u_1, u_2, \dots, u_n$ .

$$\underset{\text{the dependent var iable}}{Y_i} = \underset{\text{the regression line}}{\alpha + \beta x_i} + \underset{\text{random Var iable}}{u_i}$$

- The first component in the bracket is the part of Y explained by the changes in X and the second is the part of Y not explained by X, that is to say the change in Y is due to the random influence of  $u_i$ .

### 2.2.1 Assumptions of the Classical Linear Stochastic Regression Model.

The classicals made important assumption in their analysis of regression. The most important of these assumptions are discussed below.

#### 1. *The model is linear in parameters.*

The classicals assumed that the model should be linear in the parameters regardless of whether the explanatory and the dependent variables are linear or not. This is because if the parameters are non-linear it is difficult to estimate them since their value is not known but you are given with the data of the dependent and independent variable.

Example 1.  $Y = \alpha + \beta x + u$  is linear in both parameters and the variables, so it

Satisfies the assumption

2.  $\ln Y = \alpha + \beta \ln x + u$  is linear only in the parameters. Since the  
the classicals worry on the parameters, the model satisfies  
the assumption.

Dear distance students! Check yourself whether the following models satisfy the  
above assumption and give your answer to your tutor.

$$a. \ln Y^2 = \alpha + \beta \ln X^2 + U_i$$

$$b. Y_i = \sqrt{\alpha + \beta X_i} + U_i$$

2.  $U_i$  is a random real variable

This means that the value which  $u$  may assume in any one period depends on  
chance; it may be positive, negative or zero. Every value has a certain probability  
of being assumed by  $u$  in any particular instance.

2. *The mean value of the random variable( $U$ ) in any particular period is  
zero*

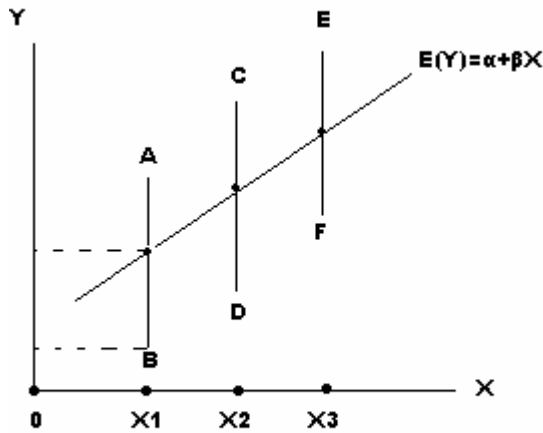
This means that for each value of  $x$ , the random variable( $u$ ) may assume  
various values, some greater than zero and some smaller than zero, but if we  
considered all the possible and negative values of  $u$ , for any given value of  $X$ ,  
they would have on average value equal to zero. In other words the positive  
and negative values of  $u$  cancel each other.

$$\text{Mathematically, } E(U_i) = 0 \dots\dots\dots (2.3)$$

3. The variance of the random variable( $U$ ) is constant in each period (The assumption of homoscedasticity)

For all values of  $X$ , the  $u$ 's will show the same dispersion around their mean. In Fig.2.c this assumption is denoted by the fact that the values that  $u$  can assume lie within the same limits, irrespective of the value of  $X$ . For  $X_1$ ,  $u$  can assume any value within the range  $AB$ ; for  $X_2$ ,  $u$  can assume any value within the range  $CD$  which is equal to  $AB$  and so on.

Graphically;



**Fig. 2.c Homoscedastic Variance**

Mathematically;

$Var(U_i) = E[U_i - E(U_i)]^2 = E(U_i)^2 = \sigma^2$  (Since  $E(U_i) = 0$ ). This constant variance is called *homoscedasticity* assumption and the constant variance itself is called homoscedastic variance.

4. *The random variable (U) has a normal distribution*

This means the values of u (for each x) have a bell shaped symmetrical distribution about their zero mean and constant variance  $\sigma^2$ , i.e.

$$U_i \sim N(0, \sigma^2) \dots\dots\dots 2.4$$

5. *The random terms of different observations (U, U) are independent.*

*(The assumption of no autocorrelation)*

This means the value which the random term assumed in one period does not depend on the value which it assumed in any other period.

Algebraically,

$$\begin{aligned} \text{Cov}(u_i, u_j) &= E[(u_i - E(u_i))(u_j - E(u_j))] \\ &= E(u_i u_j) = 0 \dots\dots\dots (2.5) \end{aligned}$$

6. *The  $X_i$  are a set of fixed values in the hypothetical process of repeated sampling which underlies the linear regression model.*

- This means that, in taking large number of samples on Y and X, the  $X_i$  values are the same in all samples, but the  $u_i$  values do differ from sample to sample, and so of course do the values of  $y_i$ .

7. *The random variable (U) is independent of the explanatory variables.*

This means there is no correlation between the random variable and the explanatory variable. If two variables are unrelated their covariance is zero.

$$\text{Hence } \text{Cov}(X_i, U_i) = 0 \dots\dots\dots (2.6)$$

Proof:-

$$\text{cov}(XU) = E[(X_i - E(X_i))(U_i - E(U_i))]$$

$$\begin{aligned}
&= E[(X_i - E(X_i))(U_i)] \text{ given } E(U_i) = 0 \\
&= E(X_i U_i) - E(X_i)E(U_i) \\
&= E(X_i U_i) \\
&= X_i E(U_i), \text{ given that the } x_i \text{ are fixed} \\
&= 0
\end{aligned}$$

8. *The explanatory variables are measured without error*

- U absorbs the influence of omitted variables and possibly errors of measurement in the y's. i.e., we will assume that the regressors are error free, while y values may or may not include errors of measurement.

Dear students! We can now use the above assumptions to derive the following basic concepts.

A. *The dependent variable  $Y_i$  is normally distributed.*

$$\text{i.e. } \therefore Y_i \sim N[(\alpha + \beta x_i), \sigma^2] \dots \dots \dots (2.7)$$

Proof:

$$\text{Mean: } E(Y) = E(\alpha + \beta x_i + u_i)$$

$$= \alpha + \beta x_i \text{ Since } E(u_i) = 0$$

$$\text{Variance: } \text{Var}(Y_i) = E(Y_i - E(Y_i))^2$$

$$= E(\alpha + \beta x_i + u_i - (\alpha + \beta x_i))^2$$

$$= E(u_i)^2$$

$$= \sigma^2 \text{ (since } E(u_i)^2 = \sigma^2)$$

$$\therefore \underline{\underline{\text{var}(Y_i) = \sigma^2}} \dots \dots \dots (2.8)$$

The shape of the distribution of  $Y_i$  is determined by the shape of the distribution of  $u_i$  which is normal by assumption 4. Since  $\alpha$  and  $\beta$ , being constant, they don't

affect the distribution of  $y_i$ . Furthermore, the values of the explanatory variable,  $x_i$ , are a set of fixed values by assumption 5 and therefore don't affect the shape of the distribution of  $y_i$ .

$$\therefore Y_i \sim N(\alpha + \beta x_i, \sigma^2)$$

*B. successive values of the dependent variable are independent, i.e*

$$\text{Cov}(Y_i, Y_j) = 0$$

Proof:

$$\begin{aligned} \text{Cov}(Y_i, Y_j) &= E\{[Y_i - E(Y_i)][Y_j - E(Y_j)]\} \\ &= E\{[\alpha + \beta X_i + U_i - E(\alpha + \beta X_i + U_i)][\alpha + \beta X_j + U_j - E(\alpha + \beta X_j + U_j)]\} \\ &\quad (\text{Since } Y_i = \alpha + \beta X_i + U_i \text{ and } Y_j = \alpha + \beta X_j + U_j) \\ &= E[(\alpha + \beta X_i + U_i - \alpha - \beta X_i)(\alpha + \beta X_j + U_j - \alpha - \beta X_j)], \text{Since } E(u_i) = 0 \\ &= E(U_i U_j) = 0 \quad (\text{from equation (2.5)}) \end{aligned}$$

Therefore,  $\text{Cov}(Y_i, Y_j) = 0$ .

## 2.2.2 Methods of estimation

Specifying the model and stating its underlying assumptions are the first stage of any econometric application. The next step is the estimation of the numerical values of the parameters of economic relationships. The parameters of the simple linear regression model can be estimated by various methods. Three of the most commonly used methods are:

1. Ordinary least square method (OLS)
2. Maximum likelihood method (MLM)
3. Method of moments (MM)

But, here we will deal with the OLS and the MLM methods of estimation.

### 2.2.2.1 The ordinary least square (OLS) method

The model  $Y_i = \alpha + \beta X_i + U_i$  is called the true relationship between Y and X because Y and X represent their respective population value, and  $\alpha$  and  $\beta$  are called the true parameters since they are estimated from the population value of Y and X. But it is difficult to obtain the population value of Y and X because of technical or economic reasons. So we are forced to take the sample value of Y and X. The parameters estimated from the sample value of Y and X are called the estimators of the true parameters  $\alpha$  and  $\beta$  and are symbolized as  $\hat{\alpha}$  and  $\hat{\beta}$ .

The model  $Y_i = \hat{\alpha} + \hat{\beta} X_i + e_i$ , is called estimated relationship between Y and X since  $\hat{\alpha}$  and  $\hat{\beta}$  are estimated from the sample of Y and X and  $e_i$  represents the sample counterpart of the population random disturbance  $U_i$ .

Estimation of  $\alpha$  and  $\beta$  by least square method (OLS) or classical least square (CLS) involves finding values for the estimates  $\hat{\alpha}$  and  $\hat{\beta}$  which will minimize the sum of square of the squared residuals ( $\sum e_i^2$ ).

From the estimated relationship  $Y_i = \hat{\alpha} + \hat{\beta} X_i + e_i$ , we obtain:

$$e_i = Y_i - (\hat{\alpha} + \hat{\beta} X_i) \dots \dots \dots (2.6)$$

$$\sum e_i^2 = \sum (Y_i - \hat{\alpha} - \hat{\beta} X_i)^2 \dots \dots \dots (2.7)$$

To find the values of  $\hat{\alpha}$  and  $\hat{\beta}$  that minimize this sum, we have to partially differentiate  $\sum e_i^2$  with respect to  $\hat{\alpha}$  and  $\hat{\beta}$  and set the partial derivatives equal to zero.

$$1. \quad \frac{\partial \sum e_i^2}{\partial \hat{\alpha}} = -2 \sum (Y_i - \hat{\alpha} - \hat{\beta} X_i) = 0 \dots \dots \dots (2.8)$$

Rearranging this expression we will get:  $\boxed{\sum Y_i = n\hat{\alpha} + \hat{\beta} \sum X_i} \dots \dots \dots (2.9)$

If you divide (2.9) by 'n' and rearrange, we get

$$\hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X} \dots \dots \dots (2.10)$$

$$2. \frac{\partial \sum e_i^2}{\partial \hat{\beta}} = -2 \sum X_i (Y_i - \hat{\alpha} - \hat{\beta} X_i) = 0 \dots\dots\dots (2.11)$$

Note: at this point that the term in the parenthesis in equation 2.8 and 2.11 is the residual,  $e = Y_i - \hat{\alpha} - \hat{\beta} X_i$ . Hence it is possible to rewrite (2.8) and (2.11) as  $-\sum e_i = 0$  and  $-\sum X_i e_i = 0$ . It follows that;

$$\sum e_i = 0 \text{ and } \sum X_i e_i = 0 \dots\dots\dots (2.12)$$

If we rearrange equation (2.11) we obtain;

$$\boxed{\sum Y_i X_i = \hat{\alpha} \sum X_i + \hat{\beta} \sum X_i^2} \dots\dots\dots (2.13)$$

Equation (2.9) and (2.13) are called the Normal Equations. Substituting the values of  $\hat{\alpha}$  from (2.10) to (2.13), we get:

$$\begin{aligned} \sum Y_i X_i &= \sum X_i (\bar{Y} - \hat{\beta} \bar{X}) + \hat{\beta} \sum X_i^2 \\ &= \bar{Y} \sum X_i - \hat{\beta} \bar{X} \sum X_i + \hat{\beta} \sum X_i^2 \\ \sum Y_i X_i - \bar{Y} \sum X_i &= \hat{\beta} (\sum X_i^2 - \bar{X} \sum X_i) \\ \sum XY - n \bar{X} \bar{Y} &= \hat{\beta} (\sum X_i^2 - n \bar{X}^2) \\ \boxed{\hat{\beta} = \frac{\sum XY - n \bar{X} \bar{Y}}{\sum X_i^2 - n \bar{X}^2}} &\dots\dots\dots (2.14) \end{aligned}$$

Equation (2.14) can be rewritten in somewhat different way as follows;

$$\begin{aligned} \sum (X - \bar{X})(Y - \bar{Y}) &= \sum (XY - X\bar{Y} - \bar{X}Y + \bar{X}\bar{Y}) \\ &= \sum XY - \bar{Y} \sum X - \bar{X} \sum Y + n \bar{X} \bar{Y} \\ &= \sum XY - n \bar{Y} \bar{X} - n \bar{X} \bar{Y} + n \bar{X} \bar{Y} \\ \sum (X - \bar{X})(Y - \bar{Y}) &= \sum XY - n \bar{X} \bar{Y} \dots\dots\dots (2.15) \end{aligned}$$

$$\sum (X - \bar{X})^2 = \sum X^2 - n \bar{X}^2 \dots\dots\dots (2.16)$$

Substituting (2.15) and (2.16) in (2.14), we get

$$\hat{\beta} = \frac{\Sigma(X - \bar{X})(Y - \bar{Y})}{\Sigma(X - \bar{X})^2}$$

Now, denoting  $(X_i - \bar{X})$  as  $x_i$ , and  $(Y_i - \bar{Y})$  as  $y_i$  we get;

$$\boxed{\hat{\beta} = \frac{\Sigma x_i y_i}{\Sigma x_i^2}} \dots\dots\dots (2.17)$$

The expression in (2.17) to estimate the parameter coefficient is termed is the formula in *deviation form*.

#### 2.2.2.2 Estimation of a function with zero intercept

Suppose it is desired to fit the line  $Y_i = \alpha + \beta X_i + U_i$ , subject to the restriction  $\alpha = 0$ . To estimate  $\hat{\beta}$ , the problem is put in a form of restricted minimization problem and then Lagrange method is applied.

$$\text{We minimize: } \Sigma e_i^2 = \sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta} X_i)^2$$

$$\text{Subject to: } \hat{\alpha} = 0$$

The composite function then becomes

$$Z = \Sigma (Y_i - \hat{\alpha} - \hat{\beta} X_i)^2 - \lambda \hat{\alpha}, \quad \text{where } \lambda \text{ is a Lagrange multiplier.}$$

We minimize the function with respect to  $\hat{\alpha}, \hat{\beta}$ , and  $\lambda$

$$\frac{\partial Z}{\partial \hat{\alpha}} = -2 \Sigma (Y_i - \hat{\alpha} - \hat{\beta} X_i) - \lambda = 0 \text{-----(i)}$$

$$\frac{\partial Z}{\partial \hat{\beta}} = -2 \Sigma (Y_i - \hat{\alpha} - \hat{\beta} X_i) (X_i) = 0 \text{-----(ii)}$$

$$\frac{\partial Z}{\partial \lambda} = -2 \hat{\alpha} = 0 \text{-----(iii)}$$

Substituting (iii) in (ii) and rearranging we obtain:

$$\begin{aligned} \Sigma X_i (Y_i - \hat{\beta} X_i) &= 0 \\ \Sigma Y_i X_i - \hat{\beta} \Sigma X_i^2 &= 0 \end{aligned}$$

$$\hat{\beta} = \frac{\sum X_i Y_i}{\sum X_i^2} \dots \dots \dots (2.18)$$

This formula involves the *actual values* (observations) of the variables and not their deviation forms, as in the case of unrestricted value of  $\hat{\beta}$ .

### 2.2.2.3. Statistical Properties of Least Square Estimators

There are various econometric methods with which we may obtain the estimates of the parameters of economic relationships. We would like to an estimate to be as close as the value of the true population parameters i.e. to vary within only a small range around the true parameter. *How are we to choose among the different econometric methods, the one that gives ‘good’ estimates?* We need some criteria for judging the ‘goodness’ of an estimate.

‘Closeness’ of the estimate to the population parameter is measured by the mean and variance or standard deviation of the *sampling distribution* of the estimates of the different econometric methods. We assume the usual process of repeated sampling i.e. we assume that we get a very large number of samples each of size ‘n’; we compute the estimates  $\hat{\beta}$ ’s from each sample, and for each econometric method and we form their distribution. We next compare the *mean (expected value)* and the *variances* of these distributions and we choose among the alternative estimates the one whose distribution is concentrated as close as possible around the population parameter.

### PROPERTIES OF OLS ESTIMATORS

The ideal or optimum properties that the OLS estimates possess may be summarized by well known theorem known as the *Gauss-Markov Theorem*.

*Statement of the theorem:* “Given the assumptions of the classical linear regression model, the OLS estimators, in the class of linear and unbiased estimators, have the minimum variance, i.e. the OLS estimators are BLUE.

According to the this theorem, under the basic assumptions of the classical linear regression model, the least squares estimators are linear, unbiased and have minimum variance (i.e. are best of all linear unbiased estimators). Some times the theorem referred as the BLUE theorem i.e. Best, Linear, Unbiased Estimator. An estimator is called BLUE if:

- Linear*: a linear function of the a random variable, such as, the dependent variable Y.
- Unbiased*: its average or expected value is equal to the true population parameter.
- Minimum variance*: It has a minimum variance in the class of linear and unbiased estimators. An unbiased estimator with the least variance is known as an efficient estimator.

According to the Gauss-Markov theorem, the OLS estimators possess all the BLUE properties. The detailed proof of these properties are presented below  
Dear colleague lets proof these properties one by one.

*a. Linearity: (for  $\hat{\beta}$ )*

**Proposition:**  $\hat{\alpha}$  &  $\hat{\beta}$  are linear in Y.

Proof: From (2.17) of the OLS estimator of  $\hat{\beta}$  is given by:

$$\hat{\beta} = \frac{\sum x_i y_i}{\sum x_i^2} = \frac{\sum x_i (Y - \bar{Y})}{\sum x_i^2} = \frac{\sum x_i Y - \bar{Y} \sum x_i}{\sum x_i^2},$$

$$(\text{but } \sum x_i = \sum (X - \bar{X}) = \sum X - n\bar{X} = n\bar{X} - n\bar{X} = 0)$$

$$\Rightarrow \hat{\beta} = \frac{\sum x_i Y}{\sum x_i^2}; \quad \text{Now, let } \frac{x_i}{\sum x_i^2} = K_i \quad (i = 1, 2, \dots, n)$$

$$\therefore \hat{\beta} = \sum K_i Y \text{ ----- (2.19)}$$

$$\Rightarrow \hat{\beta} = K_1 Y_1 + K_2 Y_2 + K_3 Y_3 + \text{-----} + K_n Y_n$$

$$\therefore \hat{\beta} \text{ is linear in Y}$$

*Check yourself question:*

Show that  $\hat{\alpha}$  is linear in  $Y$ ? Hint:  $\hat{\alpha} = \sum (\frac{1}{n} - \bar{X}k_i) Y_i$ . Derive this relationship between  $\hat{\alpha}$  and  $Y$ .

*b. Unbiasedness:*

**Proposition:**  $\hat{\alpha}$  &  $\hat{\beta}$  are the unbiased estimators of the true parameters  $\alpha$  &  $\beta$

From your statistics course, you may recall that if  $\hat{\theta}$  is an estimator of  $\theta$  then  $E(\hat{\theta}) - \theta = \text{the amount of bias}$  and if  $\hat{\theta}$  is the unbiased estimator of  $\theta$  then bias = 0 i.e.  $E(\hat{\theta}) - \theta = 0 \Rightarrow E(\hat{\theta}) = \theta$

In our case,  $\hat{\alpha}$  &  $\hat{\beta}$  are estimators of the true parameters  $\alpha$  &  $\beta$ . To show that they are the unbiased estimators of their respective parameters means to prove that:

$$E(\hat{\beta}) = \beta \quad \text{and} \quad E(\hat{\alpha}) = \alpha$$

- **Proof (1):** Prove that  $\hat{\beta}$  is unbiased i.e.  $E(\hat{\beta}) = \beta$ .

We know that  $\hat{\beta} = \sum k_i Y_i = \sum k_i (\alpha + \beta X_i + U_i)$

$$= \alpha \sum k_i + \beta \sum k_i X_i + \sum k_i u_i,$$

$$\text{but } \sum k_i = 0 \quad \text{and} \quad \sum k_i X_i = 1$$

$$\sum k_i = \frac{\sum x_i}{\sum x_i^2} = \frac{\sum (X - \bar{X})}{\sum x_i^2} = \frac{\sum X - n\bar{X}}{\sum x_i^2} = \frac{n\bar{X} - n\bar{X}}{\sum x_i^2} = 0$$

$$\Rightarrow \sum k_i = 0 \dots\dots\dots (2.20)$$

$$\begin{aligned} \sum k_i X_i &= \frac{\sum x_i X_i}{\sum x_i^2} = \frac{\sum (X - \bar{X}) X_i}{\sum x_i^2} \\ &= \frac{\sum X^2 - \bar{X} \sum X}{\sum X^2 - n\bar{X}^2} = \frac{\sum X^2 - n\bar{X}^2}{\sum X^2 - n\bar{X}^2} = 1 \end{aligned}$$

$$\Rightarrow \sum k_i X_i = 1 \dots\dots\dots (2.21)$$

$$\hat{\beta} = \beta + \sum k_i u_i \Rightarrow \hat{\beta} - \beta = \sum k_i u_i \dots\dots\dots (2.22)$$

$$E(\hat{\beta}) = E(\beta) + \sum k_i E(u_i), \text{ Since } k_i \text{ are fixed}$$

$$E(\hat{\beta}) = \beta, \quad \text{since } E(u_i) = 0$$

Therefore,  $\hat{\beta}$  is unbiased estimator of  $\beta$ .

- **Proof(2):** prove that  $\hat{\alpha}$  is unbiased i.e.:  $E(\hat{\alpha}) = \alpha$

From the proof of linearity property under 2.2.2.3 (a), we know that:

$$\begin{aligned}\hat{\alpha} &= \Sigma \left( \frac{1}{n} - \bar{X}k_i \right) Y_i \\ &= \Sigma \left[ \left( \frac{1}{n} - \bar{X}k_i \right) (\alpha + \beta X_i + U_i) \right], \quad \text{Since } Y_i = \alpha + \beta X_i + U_i \\ &= \alpha + \beta \frac{1}{n} \Sigma X_i + \frac{1}{n} \Sigma u_i - \alpha \bar{X} \Sigma k_i - \beta \bar{X} \Sigma k_i X_i - \bar{X} \Sigma k_i u_i \\ &= \alpha + \frac{1}{n} \Sigma u_i - \bar{X} \Sigma k_i u_i, \quad \Rightarrow \hat{\alpha} - \alpha = \frac{1}{n} \Sigma u_i - \bar{X} \Sigma k_i u_i \\ &= \Sigma \left( \frac{1}{n} - \bar{X}k_i \right) u_i \dots\dots\dots (2.23)\end{aligned}$$

$$E(\hat{\alpha}) = \alpha + \frac{1}{n} \Sigma E(u_i) - \bar{X} \Sigma k_i E(u_i)$$

$$E(\hat{\alpha}) = \alpha \dots\dots\dots (2.24)$$

$\therefore \hat{\alpha}$  is an unbiased estimator of  $\alpha$ .

### c. Minimum variance of $\hat{\alpha}$ and $\hat{\beta}$

Now, we have to establish that out of the class of linear and unbiased estimators of  $\alpha$  and  $\beta$ ,  $\hat{\alpha}$  and  $\hat{\beta}$  possess the smallest sampling variances. For this, we shall first obtain variance of  $\hat{\alpha}$  and  $\hat{\beta}$  and then establish that each has the minimum variance in comparison of the variances of other linear and unbiased estimators obtained by any other econometric methods than OLS.

#### a. Variance of $\hat{\beta}$

$$\text{var}(\hat{\beta}) = E(\hat{\beta} - E(\hat{\beta}))^2 = E(\hat{\beta} - \beta)^2 \dots\dots\dots (2.25)$$

Substitute (2.22) in (2.25) and we get

$$\begin{aligned}\text{var}(\hat{\beta}) &= E \left( \sum k_i u_i \right)^2 \\ &= E \left[ k_1^2 u_1^2 + k_2^2 u_2^2 + \dots\dots\dots + k_n^2 u_n^2 + 2k_1 k_2 u_1 u_2 + \dots\dots\dots + 2k_{n-1} k_n u_{n-1} u_n \right]\end{aligned}$$

$$\begin{aligned}
&= E[k_1^2 u_1^2 + k_2^2 u_2^2 + \dots + k_n^2 u_n^2] + E[2k_1 k_2 u_1 u_2 + \dots + 2k_{n-1} k_n u_{n-1} u_n] \\
&= E(\sum_i k_i^2 u_i^2) + E(\sum_{i \neq j} 2k_i k_j u_i u_j) \\
&= \sum_i k_i^2 E(u_i^2) + 2 \sum_{i \neq j} k_i k_j E(u_i u_j) = \sigma^2 \sum_i k_i^2 \quad (\text{Since } E(u_i u_j) = 0) \\
&\quad \Sigma k_i = \frac{\Sigma x_i}{\Sigma x_i^2}, \text{ and therefore, } \Sigma k_i^2 = \frac{\Sigma x_i^2}{(\Sigma x_i^2)^2} = \frac{1}{\Sigma x_i^2} \\
&\boxed{\therefore \text{var}(\hat{\beta}) = \sigma^2 \Sigma k_i^2 = \frac{\sigma^2}{\Sigma x_i^2}} \dots \dots \dots (2.26)
\end{aligned}$$

**b. Variance of  $\hat{\alpha}$**

$$\begin{aligned}
\text{var}(\hat{\alpha}) &= E((\hat{\alpha} - E(\alpha))^2) \\
&= E(\hat{\alpha} - \alpha)^2 \dots \dots \dots (2.27)
\end{aligned}$$

Substituting equation (2.23) in (2.27), we get

$$\begin{aligned}
\text{var}(\hat{\alpha}) &= E\left[\sum \left(\frac{1}{n} - \bar{X}k_i\right)^2 u_i^2\right] \\
&= \sum \left(\frac{1}{n} - \bar{X}k_i\right)^2 E(u_i^2) \\
&= \sigma^2 \sum \left(\frac{1}{n} - \bar{X}k_i\right)^2 \\
&= \sigma^2 \sum \left(\frac{1}{n^2} - \frac{2}{n} \bar{X}k_i + \bar{X}^2 k_i^2\right) \\
&= \sigma^2 \sum \left(\frac{1}{n} - \frac{2\bar{X}}{n} \Sigma k_i + \bar{X}^2 \Sigma k_i^2\right), \text{ Since } \Sigma k_i = 0 \\
&= \sigma^2 \left(\frac{1}{n} + \bar{X}^2 \Sigma k_i^2\right) \\
&= \sigma^2 \left(\frac{1}{n} + \frac{\Sigma x_i^2}{\Sigma x_i^2}\right), \text{ Since } \Sigma k_i^2 = \frac{\Sigma x_i^2}{(\Sigma x_i^2)^2} = \frac{1}{\Sigma x_i^2}
\end{aligned}$$

Again:

$$\begin{aligned}
&\frac{1}{n} + \frac{\Sigma x_i^2}{\Sigma x_i^2} = \frac{\Sigma x_i^2 + n\bar{X}^2}{n\Sigma x_i^2} = \left| \frac{\Sigma x_i^2}{n\Sigma x_i^2} \right| \\
\therefore \text{var}(\hat{\alpha}) &= \sigma^2 \left( \frac{1}{n} + \frac{\Sigma x_i^2}{\Sigma x_i^2} \right) = \sigma^2 \left( \frac{\Sigma x_i^2}{n\Sigma x_i^2} \right) \dots \dots \dots (2.28)
\end{aligned}$$

Dear student! We have computed the variances OLS estimators. Now, it is time to check whether these variances of OLS estimators do possess minimum variance property compared to the variances other estimators of the true  $\alpha$  and  $\beta$ , other than  $\hat{\alpha}$  and  $\hat{\beta}$ .

To establish that  $\hat{\alpha}$  and  $\hat{\beta}$  possess minimum variance property, we compare their variances with that of the variances of some other alternative linear and unbiased estimators of  $\alpha$  and  $\beta$ , say  $\alpha^*$  and  $\beta^*$ . Now, we want to prove that any other linear and unbiased estimator of the true population parameter obtained from any other econometric method has larger variance than that OLS estimators.

Lets first show minimum variance of  $\hat{\beta}$  and then that of  $\hat{\alpha}$ .

### 1. Minimum variance of $\hat{\beta}$

Suppose:  $\beta^*$  an alternative linear and unbiased estimator of  $\beta$  and;

$$\text{Let } \beta^* = \sum w_i Y_i \dots\dots\dots (2.29)$$

$$\text{where, } w_i \neq k_i ; \text{ but: } w_i = k_i + c_i$$

$$\beta^* = \sum w_i (\alpha + \beta X_i + u_i) \text{ Since } Y_i = \alpha + \beta X_i + U_i$$

$$= \alpha \sum w_i + \beta \sum w_i X_i + \sum w_i u_i$$

$$\therefore E(\beta^*) = \alpha \sum w_i + \beta \sum w_i X_i, \text{ since } E(u_i) = 0$$

Since  $\beta^*$  is assumed to be an unbiased estimator, then for  $\beta^*$  to be an unbiased estimator of  $\beta$ , there must be true that  $\sum w_i = 1$  and  $\sum w_i X_i = 1$  in the above equation.

$$\text{But, } w_i = k_i + c_i$$

$$\sum w_i = \sum (k_i + c_i) = \sum k_i + \sum c_i$$

$$\text{Therefore, } \sum c_i = 0 \text{ since } \sum k_i = \sum w_i = 1$$

$$\text{Again } \sum w_i X_i = \sum (k_i + c_i) X_i = \sum k_i X_i + \sum c_i X_i$$

$$\text{Since } \sum w_i X_i = 1 \text{ and } \sum k_i X_i = 1 \Rightarrow \sum c_i X_i = 0.$$

From these values we can drive  $\sum c_i x_i = 0$ , where  $x_i = X_i - \bar{X}$

$$\sum c_i x_i = \sum c_i (X_i - \bar{X}) = \sum c_i X_i - \bar{X} \sum c_i$$

$$\text{Since } \sum c_i x_i = 1 \quad \sum c_i = 0 \Rightarrow \sum c_i X_i = 0$$

Thus, from the above calculations we can summarize the following results.

$$\sum w_i = 0, \quad \sum w_i x_i = 1, \quad \sum c_i = 0, \quad \sum c_i X_i = 0$$

To prove whether  $\hat{\beta}^*$  has minimum variance or not let's compute  $\text{var}(\beta^*)$  to compare with  $\text{var}(\hat{\beta})$ .

$$\text{var}(\beta^*) = \text{var}(\sum w_i Y_i)$$

$$= \sum w_i^2 \text{var}(Y_i)$$

$$\therefore \text{var}(\beta^*) = \sigma^2 \sum w_i^2 \text{ since } \text{Var}(Y_i) = \sigma^2$$

$$\text{But, } \sum w_i^2 = \sum (k_i + c_i)^2 = \sum k_i^2 + 2 \sum k_i c_i + \sum c_i^2$$

$$\Rightarrow \sum w_i^2 = \sum k_i^2 + \sum c_i^2 \text{ Since } \sum k_i c_i = \frac{\sum c_i x_i}{\sum x_i^2} = 0$$

$$\text{Therefore, } \text{var}(\beta^*) = \sigma^2 (\sum k_i^2 + \sum c_i^2) \Rightarrow \sigma^2 \sum k_i^2 + \sigma^2 \sum c_i^2$$

$$\boxed{\text{var}(\beta^*) = \text{var}(\hat{\beta}) + \sigma^2 \sum c_i^2}$$

Given that  $c_i$  is an arbitrary constant,  $\sigma^2 \sum c_i^2$  is a positive i.e it is greater than zero.

Thus  $\text{var}(\beta^*) > \text{var}(\hat{\beta})$ . This proves that  $\hat{\beta}^*$  possesses minimum variance property.

In the similar way we can prove that the least square estimate of the constant intercept ( $\hat{\alpha}$ ) possesses minimum variance.

## 2. Minimum Variance of $\hat{\alpha}$

We take a new estimator  $\alpha^*$ , which we assume to be a linear and unbiased estimator of function of  $X$ . The least square estimator  $\hat{\alpha}$  is given by:

$$\hat{\alpha} = \sum (\mathcal{Y}_n - \bar{X} k_i) Y_i$$

By analogy with that the proof of the minimum variance property of  $\hat{\beta}^*$ , let's use the weights  $w_i = c_i + k_i$  Consequently;

$$\alpha^* = \sum (\mathcal{Y}_n - \bar{X} w_i) Y_i$$

Since we want  $\alpha^*$  to be an unbiased estimator of the true  $\alpha$ , that is,  $E(\alpha^*) = \alpha$ , we substitute for  $Y = \alpha + \beta x_i + u_i$  in  $\alpha^*$  and find the expected value of  $\alpha^*$ .

$$\begin{aligned}\alpha^* &= \Sigma(1/n - \bar{X}w_i)(\alpha + \beta X_i + u_i) \\ &= \Sigma\left(\frac{\alpha}{n} + \frac{\beta X_i}{n} + \frac{u_i}{n} - \bar{X}w_i \alpha - \beta \bar{X} \Sigma w_i X_i - \bar{X} \Sigma w_i u_i\right) \\ \alpha^* &= \alpha + \beta \bar{X} + \Sigma u_i / n - \alpha \bar{X} \Sigma w_i - \beta \bar{X} \Sigma w_i X_i - \bar{X} \Sigma w_i u_i\end{aligned}$$

For  $\alpha^*$  to be an unbiased estimator of the true  $\alpha$ , the following must hold.

$$\Sigma(w_i) = 0, \quad \Sigma(w_i X_i) = 1 \quad \text{and} \quad \Sigma(w_i u_i) = 0$$

i.e., if  $\Sigma w_i = 0$ , and  $\Sigma w_i X_i = 1$ . These conditions imply that  $\Sigma c_i = 0$  and  $\Sigma c_i X_i = 0$ .

As in the case of  $\hat{\beta}$ , we need to compute  $\text{Var}(\alpha^*)$  to compare with  $\text{var}(\hat{\alpha})$

$$\begin{aligned}\text{var}(\alpha^*) &= \text{var}\left(\Sigma(1/n - \bar{X}w_i)Y_i\right) \\ &= \Sigma(1/n - \bar{X}w_i)^2 \text{var}(Y_i) \\ &= \sigma^2 \Sigma(1/n - \bar{X}w_i)^2 \\ &= \sigma^2 \Sigma(1/n^2 + \bar{X}^2 w_i^2 - 2\bar{X}w_i/n) \\ &= \sigma^2 (1/n^2 + \Sigma \bar{X}^2 w_i^2 - 2\bar{X} \Sigma w_i/n) \\ \text{var}(\alpha^*) &= \sigma^2 \left(1/n + \bar{X}^2 \Sigma w_i^2\right), \text{ Since } \Sigma w_i = 0 \\ &\quad \text{but } \Sigma w_i^2 = \Sigma k_i^2 + \Sigma c_i^2 \\ \Rightarrow \text{var}(\alpha^*) &= \sigma^2 \left(1/n + \bar{X}^2 (\Sigma k_i^2 + \Sigma c_i^2)\right) \\ \text{var}(\alpha^*) &= \sigma^2 \left(1/n + \frac{\bar{X}^2}{\Sigma X_i^2}\right) + \sigma^2 \bar{X}^2 \Sigma c_i^2 \\ &= \sigma^2 \left(\frac{\Sigma X_i^2}{n \Sigma X_i^2}\right) + \sigma^2 \bar{X}^2 \Sigma c_i^2\end{aligned}$$

The first term in the bracket is  $\text{var}(\hat{\alpha})$ , hence

$$\boxed{\text{var}(\alpha^*) = \text{var}(\hat{\alpha}) + \sigma^2 \bar{X}^2 \Sigma c_i^2}$$

$$\Rightarrow \text{var}(\alpha^*) > \text{var}(\hat{\alpha}), \text{ Since } \sigma^2 \bar{X}^2 \Sigma c_i^2 > 0$$

Therefore, we have proved that the least square estimators of linear regression model are best, linear and unbiased (BLU) estimators.

*The variance of the random variable ( $U_i$ )*

Dear student! You may observe that the variances of the OLS estimates involve  $\sigma^2$ , which is the population variance of the random disturbance term. But it is difficult to obtain the population data of the disturbance term because of technical and economic reasons. Hence it is difficult to compute  $\sigma^2$ ; this implies that variances of OLS estimates are also difficult to compute. But we can compute these variances if we take the unbiased estimate of  $\sigma^2$  which is  $\hat{\sigma}^2$  computed from the sample value of the disturbance term  $e_i$  from the expression:

$$\hat{\sigma}^2_u = \frac{\sum e_i^2}{n-2} \dots\dots\dots 2.30$$

To use  $\hat{\sigma}^2$  in the expressions for the variances of  $\hat{\alpha}$  and  $\hat{\beta}$ , we have to prove

whether  $\hat{\sigma}^2$  is the unbiased estimator of  $\sigma^2$ , i.e.,  $E(\hat{\sigma}^2) = E\left(\frac{\sum e_i^2}{n-2}\right) = \sigma^2$

To prove this we have to compute  $\sum e_i^2$  from the expressions of  $\hat{Y}$ ,  $\hat{y}$ ,  $\hat{y}$  and  $e_i$ .

*Proof:*

$$\begin{aligned} Y_i &= \hat{\alpha} + \hat{\beta}X_i + e_i \\ \hat{Y} &= \hat{\alpha} + \hat{\beta}x \\ \Rightarrow Y &= \hat{Y} + e_i \dots\dots\dots (2.31) \end{aligned}$$

$$\Rightarrow e_i = Y_i - \hat{Y} \dots\dots\dots (2.32)$$

Summing (2.31) will result the following expression

$$\begin{aligned} \sum Y_i &= \sum y_i + \sum e_i \\ \sum Y_i &= \sum \hat{Y}_i \quad \text{since } (\sum e_i) = 0 \end{aligned}$$

Dividing both sides the above by 'n' will give us

$$\frac{\sum Y}{n} = \frac{\sum \hat{Y}}{n} \rightarrow \bar{Y} = \bar{\hat{Y}} \text{------(2.33)}$$

Putting (2.31) and (2.33) together and subtract

$$\begin{aligned} Y &= \hat{Y} + e \\ Y &= \bar{\hat{Y}} \\ \Rightarrow (Y - \bar{Y}) &= (\hat{Y} - \bar{\hat{Y}}) + e \\ \Rightarrow y_i &= \hat{y}_i + e \text{..... (2.34)} \end{aligned}$$

From (2.34):

$$e_i = y_i - \hat{y}_i \text{.....(2.35)}$$

Where the y's are in deviation form.

Now, we have to express  $y_i$  and  $\hat{y}_i$  in other expression as derived below.

From:  $Y_i = \alpha + \beta X_i + U_i$

$$\bar{Y} = \alpha + \beta \bar{X} + \bar{U}$$

We get, by subtraction

$$\begin{aligned} y_i &= (Y_i - \bar{Y}) = \beta_i (X_i - \bar{X}) + (U_i - \bar{U}) = \beta x_i + (U - \bar{U}) \\ \Rightarrow y_i &= \beta x + (U - \bar{U}) \text{.....(2.36)} \end{aligned}$$

Note that we assumed earlier that ,  $E(u) = 0$  , i.e in taking a very large number samples we expect U to have a mean value of zero, but in any particular single sample  $\bar{U}$  is not necessarily zero.

Similarly: From;

$$\left. \begin{aligned} \hat{Y} &= \hat{\alpha} + \beta \hat{x} \\ \bar{\hat{Y}} &= \hat{\alpha} + \beta \bar{\hat{x}} \end{aligned} \right\}$$

We get, by subtraction

$$\begin{aligned} \hat{Y} - \bar{\hat{Y}} &= \beta (\hat{x} - \bar{\hat{x}}) \\ \Rightarrow \hat{y} &= \beta \hat{x} \text{..... (2.37)} \end{aligned}$$

Substituting (2.36) and (2.37) in (2.35) we get

$$\begin{aligned}
e_i &= \beta x_i + (u_i - \bar{u}) - \hat{\beta} x_i \\
&= (u_i - \bar{u}) - (\hat{\beta} - \beta) x_i
\end{aligned}$$

The summation over the n sample values of the squares of the residuals over the ‘n’ samples yields:

$$\begin{aligned}
\Sigma e_i^2 &= \Sigma [(u_i - \bar{u}) - (\hat{\beta} - \beta) x_i]^2 \\
&= \Sigma [(u_i - \bar{u})^2 + (\hat{\beta} - \beta)^2 x_i^2 - 2(u_i - \bar{u})(\hat{\beta} - \beta) x_i] \\
&= \Sigma (u_i - \bar{u})^2 + (\hat{\beta} - \beta)^2 \Sigma x_i^2 - 2[(\hat{\beta} - \beta) \Sigma x_i (u_i - \bar{u})]
\end{aligned}$$

Taking expected values we have:

$$E(\Sigma e_i^2) = E[\Sigma (u_i - \bar{u})^2] + E[(\hat{\beta} - \beta)^2 \Sigma x_i^2] - 2E[(\hat{\beta} - \beta) \Sigma x_i (u_i - \bar{u})] \dots\dots\dots (2.38)$$

The right hand side terms of (2.38) may be rearranged as follows

$$\begin{aligned}
\text{a. } E[\Sigma (u_i - \bar{u})^2] &= E(\Sigma u_i^2 - 2\bar{u} \Sigma u_i) \\
&= E\left(\Sigma u_i^2 - \frac{(\Sigma u_i)^2}{n}\right) \\
&= \Sigma E(u_i^2) - \frac{1}{n} E(\Sigma u_i)^2 \\
&= n\sigma_u^2 - \frac{1}{n} E(u_1 + u_2 + \dots + u_i)^2 \text{ since } E(u_i^2) = \sigma_u^2 \\
&= n\sigma_u^2 - \frac{1}{n} E(\Sigma u_i^2 + 2\Sigma u_i u_j) \\
&= n\sigma_u^2 - \frac{1}{n} (\Sigma E(u_i^2) + 2\Sigma u_i u_j) \quad i \neq j \\
&= n\sigma_u^2 - \frac{1}{n} n\sigma_u^2 - \frac{2}{n} \Sigma E(u_i u_j) \\
&= n\sigma_u^2 - \sigma_u^2 \text{ (given } E(u_i u_j) = 0) \\
&= \sigma_u^2 (n-1) \dots\dots\dots (2.39)
\end{aligned}$$

$$\text{b. } E[(\hat{\beta} - \beta)^2 \Sigma x_i^2] = \Sigma x_i^2 E(\hat{\beta} - \beta)^2$$

Given that the X's are fixed in all samples and we know that

$$E(\hat{\beta} - \beta)^2 = \text{var}(\hat{\beta}) = \sigma_u^2 \frac{1}{\Sigma x_i^2}$$

$$\text{Hence } \sum_i x_i^2 \cdot E(\hat{\beta} - \beta)^2 = \sum_i x_i^2 \cdot \sigma_u^2 \frac{1}{\sum x_i^2}$$

$$\boxed{\sum x_i^2 \cdot E(\hat{\beta} - \beta)^2 = \sigma_u^2} \dots \dots \dots (2.40)$$

$$\begin{aligned} \text{c. } -2 E[(\hat{\beta} - \beta) \sum x_i (u_i - \bar{u})] &= -2 E[(\hat{\beta} - \beta) (\sum x_i u_i - \bar{u} \sum x_i)] \\ &= -2 E[(\hat{\beta} - \beta) (\sum x_i u_i)] , \text{ since } \sum x_i = 0 \end{aligned}$$

But from (2.22) ,  $(\hat{\beta} - \beta) = \sum k_i u_i$  and substitute it in the above expression, we will get:

$$\begin{aligned} -2 E[(\hat{\beta} - \beta) \sum x_i (u_i - \bar{u})] &= -2 E(\sum k_i u_i) (\sum x_i u_i) \\ &= -2 E \left[ \left( \frac{\sum x_i u_i}{\sum x_i^2} \right) (\sum x_i u_i) \right] , \text{ since } k_i = \frac{x_i}{\sum x_i^2} \\ &= -2 E \left[ \frac{(\sum x_i u_i)^2}{\sum x_i^2} \right] \\ &= -2 E \left[ \frac{\sum x_i^2 u_i^2 + 2 \sum x_i x_j u_i u_j}{\sum x_i^2} \right] \\ &= -2 \left[ \frac{\sum x_i^2 E(u_i^2) + 2 \sum (x_i x_j) E(u_i u_j)}{\sum x_i^2} \right] \quad i \neq j \\ &= -2 \frac{\sum x_i^2 E(u_i^2)}{\sum x_i^2} \quad (\text{given } E(u_i u_j) = 0) \end{aligned}$$

$$\boxed{= -2 E(u_i^2) = -2 \sigma_u^2} \dots \dots \dots (2.41)$$

Consequently, Equation (2.38) can be written in terms of (2.39), (2.40) and (2.41) as follows:  $E(\sum e_i^2) = (n-1)\sigma_u^2 + \sigma_u^2 - 2\sigma_u^2 = (n-2)\sigma_u^2 \dots \dots \dots (2.42)$

From which we get

$$E \left( \frac{\sum e_i^2}{(n-2)} \right) = E(\sigma_u^2) = \sigma_u^2 \dots \dots \dots (2.43)$$

$$\text{Since } \sigma_u^2 = \frac{\sum e_i^2}{n-2}$$

Thus,  $\hat{\sigma}^2 = \frac{\sum e_i^2}{n-2}$  is unbiased estimate of the true variance of the error term ( $\sigma^2$ ).

Dear student! The conclusion that we can drive from the above proof is that we can substitute  $\hat{\sigma}^2 = \frac{\sum e_i^2}{n-2}$  for ( $\sigma^2$ ) in the variance expression of  $\hat{\alpha}$  and  $\hat{\beta}$ , since  $E(\hat{\sigma}^2) = \sigma^2$ . Hence the formula of variance of  $\hat{\alpha}$  and  $\hat{\beta}$  becomes;

$$Var(\hat{\beta}) = \frac{\hat{\sigma}^2}{\sum x_i^2} = \frac{\sum e_i^2}{(n-2) \sum x_i^2} \quad (2.44)$$

$$Var(\hat{\alpha}) = \hat{\sigma}^2 \left( \frac{\sum X_i^2}{n \sum x_i^2} \right) = \frac{\sum e_i^2 \sum X_i^2}{n(n-2) \sum x_i^2} \quad (2.45)$$

Note:  $\sum e_i^2$  can be computed as  $\sum e_i^2 = \sum y_i^2 - \hat{\beta}^2 \sum x_i y_i$

Dear Student! Do not worry about the derivation of this expression! we will perform the derivation of it in our subsequent subtopic.

#### 2.2.2.4. Statistical test of Significance of the OLS Estimators (First Order tests)

After the estimation of the parameters and the determination of the least square regression line, we need to know how ‘good’ is the fit of this line to the sample observation of Y and X, that is to say we need to measure the dispersion of observations around the regression line. This knowledge is essential because the closer the observation to the line, the better the goodness of fit, i.e. the better is the explanation of the variations of Y by the changes in the explanatory variables.

We divide the available criteria into three groups: *the theoretical a priori criteria*, *the statistical criteria*, and *the econometric criteria*. Under this section, our focus is on statistical criteria (first order tests). The two most commonly used first order tests in econometric analysis are:

- i. The coefficient of determination (the square of the correlation coefficient i.e.  $R^2$ ). This test is used for judging the explanatory power of the independent variable(s).
- ii. The standard error tests of the estimators. This test is used for judging the statistical reliability of the estimates of the regression coefficients.

### 1. TESTS OF THE 'GOODNESS OF FIT' WITH $R^2$

$r^2$  shows the percentage of total variation of the dependent variable that can be explained by the changes in the explanatory variable(s) included in the model. To elaborate this let's draw a horizontal line corresponding to the mean value of the dependent variable  $\bar{Y}$ . (see figure 'd' below). By fitting the line  $\hat{Y} = \hat{\alpha}_0 + \hat{\beta}_1 X$  we try to obtain the explanation of the variation of the dependent variable  $Y$  produced by the changes of the explanatory variable  $X$ .

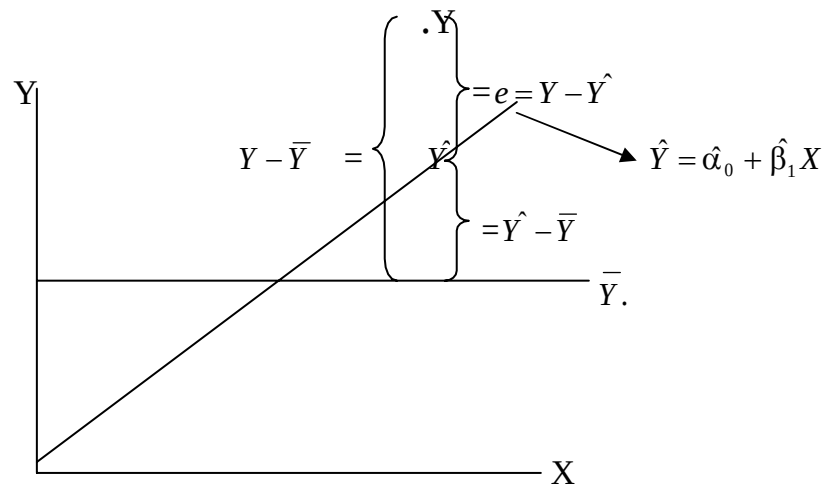


Figure 'd'. Actual and estimated values of the dependent variable  $Y$ .

As can be seen from fig.(d) above,  $Y - \bar{Y}$  represents measures the variation of the sample observation value of the dependent variable around the mean. However the variation in  $Y$  that can be attributed the influence of  $X$ , (i.e. the regression line) is given by the vertical distance  $\hat{Y} - \bar{Y}$ . The part of the total variation in  $Y$  about

$\bar{Y}$  that can't be attributed to X is equal to  $e = Y - \hat{Y}$  which is referred to as the residual variation.

In summary:

$e_i = Y_i - \hat{Y}$  = deviation of the observation  $Y_i$  from the regression line.

$y_i = Y - \bar{Y}$  = deviation of Y from its mean.

$\hat{y} = \hat{Y} - \bar{Y}$  = deviation of the regressed (predicted) value ( $\hat{Y}$ ) from the mean.

Now, we may write the observed Y as the sum of the predicted value ( $\hat{Y}$ ) and the residual term ( $e_i$ ).

$$\begin{array}{ccccc} Y_i & = & \hat{Y} & + & e_i \\ \text{Observed } Y_i & & \text{predicted } Y_i & & \text{Residual} \end{array}$$

From equation (2.34) we can have the above equation but in deviation form

$y = \hat{y} + e$ . By squaring and summing both sides, we obtain the following expression:

$$\sum y^2 = \sum (\hat{y}^2 + e^2)$$

$$\sum y^2 = \sum (\hat{y}^2 + e^2 + 2\hat{y}e)$$

$$= \sum \hat{y}^2 + \sum e^2 + 2\sum \hat{y}e$$

$$\text{But } \sum \hat{y}e = \sum e(\hat{Y} - \bar{Y}) = \sum e(\hat{\alpha} + \hat{\beta}x_i - \bar{Y})$$

$$= \hat{\alpha}\sum e_i + \hat{\beta}\sum ex_i - \hat{Y}\sum e_i$$

$$(\text{but } \sum e_i = 0, \sum ex_i = 0)$$

$$\Rightarrow \sum \hat{y}e = 0 \dots\dots\dots (2.46)$$

Therefore;

$$\boxed{\begin{array}{ccccc} \sum y^2 & = & \sum \hat{y}^2 & + & \sum e^2 \\ \text{Total} & & \text{Explained} & & \text{Unexplained} \\ \text{var iation} & & \text{var iation} & & \text{var iation} \end{array}} \dots\dots\dots (2.47)$$

OR,

|                                                   |   |                                                       |   |                                                       |
|---------------------------------------------------|---|-------------------------------------------------------|---|-------------------------------------------------------|
| Total sum of<br>square<br><del>1-2-3</del><br>TSS | = | Explained sum<br>of square<br><del>1-2-3</del><br>ESS | + | Re sidual sum<br>of square<br><del>1-2-3</del><br>RSS |
|---------------------------------------------------|---|-------------------------------------------------------|---|-------------------------------------------------------|

i.e

|                   |             |
|-------------------|-------------|
| $TSS = ESS + RSS$ | .....(2.48) |
|-------------------|-------------|

Mathematically; the explained variation as a percentage of the total variation is explained as:

|                                                         |             |
|---------------------------------------------------------|-------------|
| $\frac{ESS}{TSS} = \frac{\Sigma \hat{y}^2}{\Sigma y^2}$ | .....(2.49) |
|---------------------------------------------------------|-------------|

From equation (2.37) we have  $\hat{y} = \beta \hat{x}$ . Squaring and summing both sides give us

$$\Sigma \hat{y}^2 = \beta^2 \Sigma x^2 \text{ ----- (2.50)}$$

We can substitute (2.50) in (2.49) and obtain:

$$ESS / TSS = \frac{\beta^2 \Sigma x^2}{\Sigma y^2} \text{ .....(2.51)}$$

$$= \left( \frac{\Sigma xy}{\Sigma x^2} \right)^2 \frac{\Sigma x_i^2}{\Sigma y^2}, \text{ Since } \hat{\beta} = \frac{\Sigma x_i y_i}{\Sigma x_i^2}$$

$$= \frac{\Sigma xy}{\Sigma x^2} \frac{\Sigma xy}{\Sigma y^2} \text{ .....(2.52)}$$

Comparing (2.52) with the formula of the correlation coefficient:

$$r = \text{Cov}(X, Y) / \sigma_x \sigma_y = \Sigma xy / n \sigma_x \sigma_y = \Sigma xy / (\Sigma x^2 \Sigma y^2)^{1/2} \text{ .....(2.53)}$$

Squaring (2.53) will result in: 

|                                                 |
|-------------------------------------------------|
| $r^2 = (\Sigma xy)^2 / (\Sigma x^2 \Sigma y^2)$ |
|-------------------------------------------------|

 .....(2.54)

Comparing (2.52) and (2.54), we see exactly the expressions. Therefore:

|                                                                             |
|-----------------------------------------------------------------------------|
| $ESS/TSS = \frac{\Sigma xy}{\Sigma x^2} \frac{\Sigma xy}{\Sigma y^2} = r^2$ |
|-----------------------------------------------------------------------------|

From (2.48),  $RSS = TSS - ESS$ . Hence  $R^2$  becomes;

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS} = 1 - \frac{\Sigma e_i^2}{\Sigma y^2} \text{ .....(2.55)}$$

From equation (2.55) we can drive;

$$RSS = \sum_i e_i^2 = \sum_i y_i^2 (1 - R^2) \text{------(2.56)}$$

*The limit of  $R^2$ :* The value of  $R^2$  falls between zero and one. i.e.  $0 \leq R^2 \leq 1$ .

- *Interpretation of  $R^2$*

Suppose  $R^2 = 0.9$ , this means that the regression line gives a good fit to the observed data since this line explains 90% of the total variation of the Y value around their mean. The remaining 10% of the total variation in Y is unaccounted for by the regression line and is attributed to the factors included in the disturbance variable  $u_i$ .

*Check yourself question:*

a. Show that  $0 \leq R^2 \leq 1$ .

b. Show that the square of the coefficient of correlation is equal to ESS/TSS.

*Exercise:*

Suppose  $r_{xy}$  = is the correlation coefficient between Y and X and is give by:

$$= \frac{\sum x_i y_i}{\sqrt{\sum x_i^2} \sqrt{\sum y_i^2}}$$

And let  $r_{yy}^2$  = the square of the correlation coefficient between Y and  $\hat{Y}$ , and is

given by: 
$$r_{yy}^2 = \frac{(\sum y \hat{y})^2}{\sum y^2 \sum \hat{y}^2}$$

Show that:     i)  $r_{yy}^2 = R^2$                       ii)  $r_{yy} = r_{yx}$

## 2. TESTING THE SIGNIFICANCE OF OLS PARAMETERS

To test the significance of the OLS parameter estimators we need the following:

- Variance of the parameter estimators
- Unbiased estimator of  $\sigma^2$

- The assumption of normality of the distribution of error term.

We have already derived that:

- $\text{var}(\hat{\beta}) = \frac{\hat{\sigma}^2}{\sum x^2}$
- $\text{var}(\hat{\alpha}) = \frac{\hat{\sigma}^2 \sum X^2}{n \sum x^2}$
- $\hat{\sigma}^2 = \frac{\sum e^2}{n-2} = \frac{RSS}{n-2}$

For the purpose of estimation of the parameters the assumption of normality is not used, but we use this assumption to test the significance of the parameter estimators; because the testing methods or procedures are based on the assumption of the normality assumption of the disturbance term. Hence before we discuss on the various testing methods it is important to see whether the parameters are normally distributed or not.

We have already assumed that the error term is normally distributed with mean zero and variance  $\sigma^2$ , i.e.  $U_i \sim N(0, \sigma^2)$ . Similarly, we also proved that  $Y_i \sim N[(\alpha + \beta x), \sigma^2]$ . Now, we want to show the following:

1.  $\hat{\beta} \sim N\left(\beta, \frac{\sigma^2}{\sum x^2}\right)$
2.  $\hat{\alpha} \sim N\left(\alpha, \frac{\sigma^2 \sum X^2}{n \sum x^2}\right)$

To show whether  $\hat{\alpha}$  and  $\hat{\beta}$  are normally distributed or not, we need to make use of one property of normal distribution. “..... any linear function of a normally distributed variable is itself normally distributed.”

$$\hat{\beta} = \sum k_i Y_i = k_1 Y_1 + k_2 Y_{2i} + \dots + k_n Y_n$$

$$\hat{\alpha} = \sum w_i Y_i = w_1 Y_1 + w_2 Y_{2i} + \dots + w_n Y_n$$

Since  $\hat{\alpha}$  and  $\hat{\beta}$  are linear in  $Y$ , it follows that

$$\hat{\beta} \sim N\left(\beta, \frac{\sigma^2}{\sum X^2}\right) \quad ; \quad \hat{\alpha} \sim N\left(\alpha, \frac{\sigma^2 \sum X^2}{n \sum X^2}\right)$$

The OLS estimates  $\hat{\alpha}$  and  $\hat{\beta}$  are obtained from a sample of observations on Y and X. Since sampling errors are inevitable in all estimates, it is necessary to apply test of significance in order to measure the size of the error and determine the degree of confidence in order to measure the validity of these estimates. This can be done by using various tests. The most common ones are:

- i) *Standard error test*   ii) *Student's t-test*   iii) *Confidence interval*

All of these testing procedures reach on the same conclusion. Let us now see these testing methods one by one.

- i) *Standard error test*

This test helps us decide whether the estimates  $\hat{\alpha}$  and  $\hat{\beta}$  are significantly different from zero, i.e. whether the sample from which they have been estimated might have come from a population whose true parameters are zero.  $\alpha = 0$  and / or  $\beta = 0$ .

Formally we test the null hypothesis

$$H_0 : \beta_i = 0 \text{ against the alternative hypothesis } H_1 : \beta_i \neq 0$$

The standard error test may be outlined as follows.

*First:* Compute standard error of the parameters.

$$SE(\hat{\beta}) = \sqrt{\text{var}(\hat{\beta})}$$

$$SE(\hat{\alpha}) = \sqrt{\text{var}(\hat{\alpha})}$$

*Second:* compare the standard errors with the numerical values of  $\hat{\alpha}$  and  $\hat{\beta}$ .

*Decision rule:*

- If  $SE(\hat{\beta}_i) > \frac{1}{2}\hat{\beta}_i$ , accept the null hypothesis and reject the alternative hypothesis. We conclude that  $\hat{\beta}_i$  is statistically insignificant.

- If  $SE(\hat{\beta}_i) < \frac{1}{2}\hat{\beta}_i$ , reject the null hypothesis and accept the alternative hypothesis. We conclude that  $\hat{\beta}_i$  is statistically significant.

The acceptance or rejection of the null hypothesis has definite economic meaning. Namely, the acceptance of the null hypothesis  $\beta = 0$  (the slope parameter is zero) implies that the explanatory variable to which this estimate relates does not in fact influence the dependent variable Y and should not be included in the function, since the conducted test provided evidence that changes in X leave Y unaffected. In other words acceptance of  $H_0$  implies that the relationship between Y and X is in fact  $Y = \alpha + (0)x = \alpha$ , i.e. there is no relationship between X and Y.

*Numerical example:* Suppose that from a sample of size  $n=30$ , we estimate the following supply function.

$$Q = 120 + 0.6p + e_i$$

$$SE : (1.7) \quad (0.025)$$

Test the significance of the slope parameter at 5% level of significance using the standard error test.

$$SE(\hat{\beta}) = 0.025$$

$$(\hat{\beta}) = 0.6$$

$$\frac{1}{2}\hat{\beta} = 0.3$$

This implies that  $SE(\hat{\beta}_i) < \frac{1}{2}\hat{\beta}_i$ . The implication is  $\hat{\beta}$  is statistically significant at 5% level of significance.

*Note: The standard error test is an approximated test (which is approximated from the z-test and t-test) and implies a two tail test conducted at 5% level of significance.*

#### ii) Student's t-test

Like the standard error test, this test is also important to test the significance of the parameters. From your statistics, any variable X can be transformed into t using the general formula:

$$t = \frac{\bar{X} - \mu}{s_x}, \text{ with } n-1 \text{ degree of freedom.}$$

Where  $\mu_i$  = value of the population mean

$s_x$  = sample estimate of the population standard deviation

$$s_x = \sqrt{\frac{\sum (X - \bar{X})^2}{n-1}}$$

$n$  = sample size

We can derive the t-value of the OLS estimates

$$\left. \begin{aligned} t_{\hat{\beta}} &= \frac{\hat{\beta}_i - \beta}{SE(\hat{\beta})} \\ t_{\hat{\alpha}} &= \frac{\hat{\alpha} - \alpha}{SE(\hat{\alpha})} \end{aligned} \right\} \text{ with } n-k \text{ degree of freedom.}$$

Where:

SE = is standard error

$k$  = number of parameters in the model.

Since we have two parameters in simple linear regression with intercept different from zero, our degree of freedom is  $n-2$ . Like the standard error test we formally test the hypothesis:  $H_0 : \beta_i = 0$  against the alternative  $H_1 : \beta_i \neq 0$  for the slope parameter; and  $H_0 : \alpha = 0$  against the alternative  $H_1 : \alpha \neq 0$  for the intercept.

To undertake the above test we follow the following steps.

*Step 1:* Compute  $t^*$ , which is called the computed value of  $t$ , by taking the value of  $\beta$  in the null hypothesis. In our case  $\beta = 0$ , then  $t^*$  becomes:

$$t^* = \frac{\hat{\beta} - 0}{SE(\hat{\beta})} = \frac{\hat{\beta}}{SE(\hat{\beta})}$$

*Step 2:* Choose level of significance. Level of significance is the probability of making ‘wrong’ decision, i.e. the probability of rejecting the hypothesis when it is actually true or the probability of committing a type I error. It is customary in

econometric research to choose the 5% or the 1% level of significance. This means that in making our decision we allow (tolerate) five times out of a hundred to be 'wrong' i.e. reject the hypothesis when it is actually true.

*Step 3:* Check whether there is one tail test or two tail test. If the inequality sign in the alternative hypothesis is  $\neq$ , then it implies a two tail test and divide the chosen level of significance by two; decide the critical region or critical value of  $t$  called  $t_c$ . But if the inequality sign is either  $>$  or  $<$  then it indicates one tail test and there is no need to divide the chosen level of significance by two to obtain the critical value of  $t$  from the  $t$ -table.

Example:

If we have  $H_0 : \beta_i = 0$

against:  $H_1 : \beta_i \neq 0$

Then this is a two tail test. If the level of significance is 5%, divide it by two to obtain critical value of  $t$  from the  $t$ -table.

*Step 4:* Obtain critical value of  $t$ , called  $t_c$  at  $\alpha/2$  and  $n-2$  degree of freedom for two tail test.

*Step 5:* Compare  $t^*$  (the computed value of  $t$ ) and  $t_c$  (critical value of  $t$ )

- If  $t^* > t_c$ , reject  $H_0$  and accept  $H_1$ . The conclusion is  $\hat{\beta}$  is statistically significant.
- If  $t^* < t_c$ , accept  $H_0$  and reject  $H_1$ . The conclusion is  $\hat{\beta}$  is statistically insignificant.

Numerical Example:

Suppose that from a sample size  $n=20$  we estimate the following consumption function:

$$C = 100 + 0.70Y + e$$

(75.5) (0.21)

The values in the brackets are standard errors. We want to test the null hypothesis:  $H_0 : \beta_i = 0$  against the alternative  $H_1 : \beta_i \neq 0$  using the t-test at 5% level of significance.

a. the t-value for the test statistic is:

$$t^* = \frac{\hat{\beta} - 0}{SE(\hat{\beta})} = \frac{\hat{\beta}}{SE(\hat{\beta})} = \frac{0.70}{0.21} \cong 3.3$$

b. Since the alternative hypothesis ( $H_1$ ) is stated by inequality sign ( $\neq$ ), it is a two tail test, hence we divide  $\alpha/2 = 0.05/2 = 0.025$  to obtain the critical value of 't' at  $\alpha/2=0.025$  and 18 degree of freedom (df) i.e. ( $n-2=20-2$ ). From the table 'tc' at 0.025 level of significance and 18 df is 2.10.

c. Since  $t^*=3.3$  and  $tc=2.1$ ,  $t^*>tc$ . It implies that  $\hat{\beta}$  is statistically significant.

### iii) *Confidence interval*

Rejection of the null hypothesis doesn't mean that our estimate  $\hat{\alpha}$  and  $\hat{\beta}$  is the correct estimate of the true population parameter  $\alpha$  and  $\beta$ . It simply means that our estimate comes from a sample drawn from a population whose parameter  $\beta$  is different from zero.

In order to define how close the estimate to the true parameter, we must construct confidence interval for the true parameter, in other words we must establish limiting values around the estimate within which the true parameter is expected to lie within a certain "degree of confidence". In this respect we say that with a given probability the population parameter will be within the defined confidence interval (confidence limits).

We choose a probability in advance and refer to it as confidence level (interval coefficient). It is customary in econometrics to choose the 95% confidence level. This means that in repeated sampling the confidence limits, computed from the

sample, would include the true population parameter in 95% of the cases. In the other 5% of the cases the population parameter will fall outside the confidence interval.

In a two-tail test at  $\alpha$  level of significance, the probability of obtaining the specific t-value either  $-t_c$  or  $t_c$  is  $\alpha/2$  at n-2 degree of freedom. The probability of obtaining

any value of t which is equal to  $\frac{\hat{\beta} - \beta}{SE(\hat{\beta})}$  at n-2 degree of freedom is

$1 - (\alpha/2 + \alpha/2)$  i.e.  $1 - \alpha$ .

$$\text{i.e. } \Pr\{-t_c < t^* < t_c\} = 1 - \alpha \quad \dots\dots\dots(2.57)$$

$$\text{but } t^* = \frac{\hat{\beta} - \beta}{SE(\hat{\beta})} \quad \dots\dots\dots(2.58)$$

Substitute (2.58) in (2.57) we obtain the following expression.

$$\Pr\left\{-t_c < \frac{\hat{\beta} - \beta}{SE(\hat{\beta})} < t_c\right\} = 1 - \alpha \quad \dots\dots\dots(2.59)$$

$$\Pr\{-SE(\hat{\beta})t_c < \hat{\beta} - \beta < SE(\hat{\beta})t_c\} = 1 - \alpha \quad \text{--- by multiplying } SE(\hat{\beta})$$

$$\Pr\{-\hat{\beta} - SE(\hat{\beta})t_c < -\beta < -\hat{\beta} + SE(\hat{\beta})t_c\} = 1 - \alpha \quad \text{--- by subtracting } \hat{\beta}$$

$$\Pr\{+\hat{\beta} + SE(\hat{\beta}) > \beta > \hat{\beta} - SE(\hat{\beta})t_c\} = 1 - \alpha \quad \text{--- by multiplying by } -1$$

$$\Pr\{\hat{\beta} - SE(\hat{\beta})t_c < \beta < \hat{\beta} + SE(\hat{\beta})t_c\} = 1 - \alpha \quad \text{--- interchanging}$$

The limit within which the true  $\beta$  lies at  $(1 - \alpha)\%$  degree of confidence is:

$[\hat{\beta} - SE(\hat{\beta})t_c, \hat{\beta} + SE(\hat{\beta})t_c]$ ; where  $t_c$  is the critical value of t at  $\alpha/2$  confidence interval and n-2 degree of freedom.

The test procedure is outlined as follows.

$$H_0 : \beta = 0$$

$$H_1 : \beta \neq 0$$

**Decision rule:** If the hypothesized value of  $\beta$  in the null hypothesis is within the confidence interval, accept  $H_0$  and reject  $H_1$ . The implication is that  $\hat{\beta}$  is

statistically insignificant; while if the hypothesized value of  $\beta$  in the null hypothesis is outside the limit, reject  $H_0$  and accept  $H_1$ . This indicates  $\hat{\beta}$  is statistically significant.

Numerical Example:

Suppose we have estimated the following regression line from a sample of 20 observations.

$$Y = 128.5 + 2.88X + e$$

$$(38.2) \quad (0.85)$$

The values in the bracket are standard errors.

- a. Construct 95% confidence interval for the slope of parameter
- b. Test the significance of the slope parameter using constructed confidence interval.

Solution:

- a. The limit within which the true  $\beta$  lies at 95% confidence interval is:

$$\hat{\beta} \pm SE(\hat{\beta})t_c$$

$$\hat{\beta} = 2.88$$

$$SE(\hat{\beta}) = 0.85$$

$t_c$  at 0.025 level of significance and 18 degree of freedom is 2.10.

$$\Rightarrow \hat{\beta} \pm SE(\hat{\beta})t_c = 2.88 \pm 2.10(0.85) = 2.88 \pm 1.79.$$

The confidence interval is:

$$(1.09, 4.67)$$

- b. The value of  $\beta$  in the null hypothesis is zero which implies it is out side the confidence interval. Hence  $\beta$  is statistically significant.

### 2.2.3 Reporting the Results of Regression Analysis

The results of the regression analysis derived are reported in conventional formats. It is not sufficient merely to report the estimates of  $\beta$  's. In practice we report

regression coefficients together with their standard errors and the value of  $R^2$ . It has become customary to present the estimated equations with standard errors placed in parenthesis below the estimated parameter values. Sometimes, the estimated coefficients, the corresponding standard errors, the p-values, and some other indicators are presented in tabular form.

These results are supplemented by  $R^2$  on ( to the right side of the regression equation).

Example: 
$$\begin{array}{c} Y = 128.5 + 2.88 X \\ (38.2) \quad (0.85) \end{array}, \quad R^2 = 0.93.$$
 The numbers in the

parenthesis below the parameter estimates are the standard errors. Some econometricians report the t-values of the estimated coefficients in place of the standard errors.

Review Questions

### Review Questions

1. Econometrics deals with the measurement of economic relationships which are stochastic or random. The simplest form of economic relationships between two variables X and Y can be represented by:

$$Y_i = \beta_0 + \beta_1 X_i + U_i ; \text{ where } \beta_0 \text{ and } \beta_1 = \text{are regression parameters and } U_i = \text{the stochastic disturbance term}$$

What are the reasons for the insertion of U-term in the model?

2. The following data refers to the demand for money (M) and the rate of interest (R) in for eight different economics:

|                 |     |     |     |     |     |     |     |     |
|-----------------|-----|-----|-----|-----|-----|-----|-----|-----|
| M (In billions) | 56  | 50  | 46  | 30  | 20  | 35  | 37  | 61  |
| R%              | 6.3 | 4.6 | 5.1 | 7.3 | 8.9 | 5.3 | 6.7 | 3.5 |

- a. Assuming a relationship  $M = \alpha + \beta R + U_i$ , obtain the OLS estimators of  $\alpha$  and  $\beta$
- b. Calculate the coefficient of determination for the data and interpret its value
- c. If in a 9<sup>th</sup> economy the rate of interest is  $R=8.1$ , predict the demand for money(M) in this economy.

3. The following data refers to the price of a good 'P' and the quantity of the good supplied, 'S'.

|   |    |    |    |   |    |    |    |    |
|---|----|----|----|---|----|----|----|----|
| P | 2  | 7  | 5  | 1 | 4  | 8  | 2  | 8  |
| S | 15 | 41 | 32 | 9 | 28 | 43 | 17 | 40 |

- Estimate the linear regression line  $E(S) = \alpha + \beta P$
  - Estimate the standard errors of  $\hat{\alpha}$  and  $\hat{\beta}$
  - Test the hypothesis that price influences supply
  - Obtain a 95% confidence interval for
4. The following results have been obtained from a simple of 11 observations on the values of sales (Y) of a firm and the corresponding prices (X).

$$\bar{X} = 519.18$$

$$\bar{Y} = 217.82$$

$$\sum X_i^2 = 3,134,543$$

$$\sum X_i Y_i = 1,296,836$$

$$\sum Y_i^2 = 539,512$$

- Estimate the regression line of sale on price and interpret the results
  - What is the part of the variation in sales which is not explained by the regression line?
  - Estimate the price elasticity of sales.
5. The following table includes the GNP(X) and the demand for food (Y) for a country over ten years period.

| year | 1980 | 1981 | 1982 | 1983 | 1984 | 1985 | 1986 | 1987 | 1988 | 1989 |
|------|------|------|------|------|------|------|------|------|------|------|
| Y    | 6    | 7    | 8    | 10   | 8    | 9    | 10   | 9    | 11   | 10   |
| X    | 50   | 52   | 55   | 59   | 57   | 58   | 62   | 65   | 68   | 70   |

- Estimate the food function
  - Compute the coefficient of determination and find the explained and unexplained variation in the food expenditure.
  - Compute the standard error of the regression coefficients and conduct test of significance at the 5% level of significance.
6. A sample of 20 observation corresponding to the regression model  $Y_i = \alpha + \beta X_i + U_i$  gave the following data.

$$\begin{aligned}\sum Y_i &= 21.9 & \sum (Y_i - \bar{Y})^2 &= 86.9 \\ \sum X_i &= 186.2 & \sum (X_i - \bar{X})^2 &= 215.4 \\ \sum (X_i - \bar{X})(Y_i - \bar{Y}) &= 106.4\end{aligned}$$

- a. Estimate  $\alpha$  and  $\beta$
  - b. Calculate the variance of our estimates
  - c. Estimate the conditional mean of Y corresponding to a value of X fixed at X=10.
7. Suppose that a researcher estimates a consumption function and obtains the following results:

$$\begin{aligned}C &= 15 + 0.81Y_d & n &= 19 \\ (3.1) & \quad (18.7) & R^2 &= 0.99\end{aligned}$$

where C=Consumption,  $Y_d$ =disposable income, and numbers in the parenthesis are the 't-ratios'

- a. Test the significance of  $Y_d$  statistically using t-ratios
  - b. Determine the estimated standard deviations of the parameter estimates
8. State and prove Gauss-Markov theorem
9. Given the model:
- $$Y_i = \beta_0 + \beta_1 X_i + U_i \text{ with usual OLS assumptions. Derive the expression for the error variance.}$$

# Chapter Three

## THE CLASSICAL REGRESSION ANALYSIS

### [The Multiple Linear Regression Model]

#### 3.1 Introduction

In simple regression we study the relationship between a dependent variable and a single explanatory (independent variable). But it is rarely the case that economic relationships involve just two variables. Rather a dependent variable  $Y$  can depend on a whole series of explanatory variables or regressors. For instance, in demand studies we study the relationship between quantity demanded of a good and price of the good, price of substitute goods and the consumer's income. The model we assume is:

$$Y_i = \beta_0 + \beta_1 P_1 + \beta_2 P_2 + \beta_3 X_i + u_i \text{.....} (3.1)$$

Where  $Y_i$  = quantity demanded,  $P_1$  is price of the good,  $P_2$  is price of substitute goods,  $X_i$  is consumer's income, and  $\beta$ 's are unknown parameters and  $u_i$  is the disturbance.

Equation (3.1) is a multiple regression with three explanatory variables. In general for  $K$ -explanatory variable we can write the model as follows:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \text{.....} + \beta_k X_{ki} + u_i \text{.....} (3.2)$$

Where  $X_{ki} = (i = 1, 2, 3, \dots, K)$  are explanatory variables,  $Y_i$  is the dependent variable and  $\beta_j (j = 0, 1, 2, \dots, (k+1))$  are unknown parameters and  $u_i$  is the disturbance term. The disturbance term is of similar nature to that in simple regression, reflecting:

- the basic random nature of human responses
- errors of aggregation
- errors of measurement
- errors in specification of the mathematical form of the model

and any other (minor) factors, other than  $x_i$  that might influence  $Y$ .

In this chapter we will first start our discussion with the assumptions of the multiple regressions and we will proceed our analysis with the case of two explanatory variables and then we will generalize the multiple regression model in the case of  $k$ -explanatory variables using matrix algebra.

### 3.2 Assumptions of Multiple Regression Model

In order to specify our multiple linear regression model and proceed our analysis with regard to this model, some assumptions are compulsory. But these assumptions are the same as in the single explanatory variable model developed earlier except the assumption of no perfect multicollinearity. These assumptions are:

1. *Randomness of the error term:* The variable  $u$  is a real random variable.
2. *Zero mean of the error term:*  $E(u_i) = 0$
3. *Homoscedasticity:* The variance of each  $u_i$  is the same for all the  $x_i$  values.

$$\text{i.e. } E(u_i^2) = \sigma_u^2 \text{ (constant)}$$

4. *Normality of  $u$ :* The values of each  $u_i$  are normally distributed.  
i.e.  $U_i \sim N(0, \sigma_u^2)$
5. *No auto or serial correlation:* The values of  $u_i$  (corresponding to  $X_i$ ) are independent from the values of any other  $u_i$  (corresponding to  $X_j$ ) for  $i \neq j$ .

$$\text{i.e. } E(u_i u_j) = 0 \text{ for } x_i \neq x_j$$

6. *Independence of  $u_i$  and  $X_i$ :* Every disturbance term  $u_i$  is independent of the explanatory variables. i.e.  $E(u_i X_{1i}) = E(u_i X_{2i}) = 0$

This condition is automatically fulfilled if we assume that the values of the  $X$ 's are a set of fixed numbers in all (hypothetical) samples.

7. *No perfect multicollinearity:* The explanatory variables are not perfectly linearly correlated.

We can't exclusively list all the assumptions but the above assumptions are some of the basic assumptions that enable us to proceed our analysis.

### 3.3 A Model With Two Explanatory Variables

In order to understand the nature of multiple regression model easily, we start our analysis with the case of two explanatory variables, then extend this to the case of k-explanatory variables.

#### 3.3.1 Estimation of parameters of two-explanatory variables model

The model:  $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + U_i$  .....(3.3)

is multiple regression with two explanatory variables. The expected value of the above model is called population regression equation i.e.

$$E(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2, \text{ Since } E(U_i) = 0 \text{.....(3.4)}$$

where  $\beta_i$  is the population parameters.  $\beta_0$  is referred to as the intercept and  $\beta_1$  and  $\beta_2$  are also some times known as regression slopes of the regression. Note that,  $\beta_2$  for example measures the effect on  $E(Y)$  of a unit change in  $X_2$  when  $X_1$  is held constant.

Since the population regression equation is unknown to any investigator, it has to be estimated from sample data. Let us suppose that the sample data has been used to estimate the population regression equation. We leave the method of estimation unspecified for the present and merely assume that equation (3.4) has been estimated by sample regression equation, which we write as:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 \text{ .....(3.5)}$$

Where  $\hat{\beta}_j$  are estimates of the  $\beta_j$  and  $\hat{Y}$  is known as the predicted value of Y.

Now it is time to state how (3.3) is estimated. Given sample observation on  $Y, X_1$  &  $X_2$ , we estimate (3.3) using the method of least square (OLS).

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + e_i \dots\dots\dots (3.6)$$

is sample relation between  $Y, X_1$  &  $X_2$ .

$$e_i = Y_i - \hat{Y} = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i} \dots\dots\dots (3.7)$$

To obtain expressions for the least square estimators, we partially differentiate  $\sum e_i^2$  with respect to  $\hat{\beta}_0, \hat{\beta}_1$  and  $\hat{\beta}_2$  and set the partial derivatives equal to zero.

$$\frac{\partial [\sum e_i^2]}{\partial \hat{\beta}_0} = -2 \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i}) = 0 \dots\dots\dots (3.8)$$

$$\frac{\partial [\sum e_i^2]}{\partial \hat{\beta}_1} = -2 \sum X_{1i} (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i}) = 0 \dots\dots\dots (3.9)$$

$$\frac{\partial [\sum e_i^2]}{\partial \hat{\beta}_2} = -2 \sum X_{2i} (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i}) = 0 \dots\dots\dots (3.10)$$

Summing from 1 to n, the multiple regression equation produces three Normal Equations:

$$\sum Y = n\hat{\beta}_0 + \hat{\beta}_1 \sum X_{1i} + \hat{\beta}_2 \sum X_{2i} \dots\dots\dots (3.11)$$

$$\sum X_{1i} Y_i = \hat{\beta}_0 \sum X_{1i} + \hat{\beta}_1 \sum X_{1i}^2 + \hat{\beta}_2 \sum X_{1i} X_{2i} \dots\dots\dots (3.12)$$

$$\sum X_{2i} Y_i = \hat{\beta}_0 \sum X_{2i} + \hat{\beta}_1 \sum X_{1i} X_{2i} + \hat{\beta}_2 \sum X_{2i}^2 \dots\dots\dots (3.13)$$

From (3.11) we obtain  $\hat{\beta}_0$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}_1 - \hat{\beta}_2 \bar{X}_2 \dots\dots\dots (3.14)$$

Substituting (3.14) in (3.12), we get:

$$\begin{aligned} \sum X_{1i} Y_i &= (\bar{Y} - \hat{\beta}_1 \bar{X}_1 - \hat{\beta}_2 \bar{X}_2) \sum X_{1i} + \hat{\beta}_1 \sum X_{1i}^2 + \hat{\beta}_2 \sum X_{2i} \\ \Rightarrow \sum X_{1i} Y_i - \bar{Y} \sum X_{1i} &= \hat{\beta}_1 (\sum X_{1i}^2 - \bar{X}_1 \sum X_{1i}) + \hat{\beta}_2 (\sum X_{1i} X_{2i} - \bar{X}_2 \sum X_{1i}) \\ \Rightarrow \sum X_{1i} Y_i - n \bar{Y} \bar{X}_1 &= \hat{\beta}_1 (\sum X_{1i}^2 - n \bar{X}_1^2) + \hat{\beta}_2 (\sum X_{1i} X_{2i} - n \bar{X}_1 \bar{X}_2) \dots\dots\dots (3.15) \end{aligned}$$

We know that

$$\sum (X_i - Y_i)^2 = (\sum X_i Y_i - n \bar{X}_i \bar{Y}_i) = \sum x_i y_i$$

$$\sum (X_i - \bar{X}_i)^2 = (\sum X_i^2 - n \bar{X}_i^2) = \sum x_i^2$$

Substituting the above equations in equation (3.14), the normal equation (3.12) can be written in deviation form as follows:

$$\sum x_1 y = \hat{\beta}_1 \sum x_1^2 + \hat{\beta}_2 \sum x_1 x_2 \dots\dots\dots(3.16)$$

Using the above procedure if we substitute (3.14) in (3.13), we get

$$\sum x_2 y = \hat{\beta}_1 \sum x_1 x_2 + \hat{\beta}_2 \sum x_2^2 \dots\dots\dots(3.17)$$

Let's bring (2.17) and (2.18) together

$$\sum x_1 y = \hat{\beta}_1 \sum x_1^2 + \hat{\beta}_2 \sum x_1 x_2 \dots\dots\dots(3.18)$$

$$\sum x_2 y = \hat{\beta}_1 \sum x_1 x_2 + \hat{\beta}_2 \sum x_2^2 \dots\dots\dots(3.19)$$

$\hat{\beta}_1$  and  $\hat{\beta}_2$  can easily be solved using matrix

We can rewrite the above two equations in matrix form as follows.

$$\begin{array}{ccc} \sum x_1^2 & \sum x_1 x_2 & \hat{\beta}_1 \\ \sum x_1 x_2 & \sum x_2^2 & \hat{\beta}_2 \end{array} = \begin{array}{c} \sum x_2 y \\ \sum x_3 y \end{array} \dots\dots\dots(3.20)$$

If we use Cramer's rule to solve the above matrix we obtain

$$\hat{\beta}_1 = \frac{\sum x_1 y \cdot \sum x_2^2 - \sum x_1 x_2 \cdot \sum x_2 y}{\sum x_1^2 \cdot \sum x_2^2 - (\sum x_1 x_2)^2} \dots\dots\dots(3.21)$$

$$\hat{\beta}_2 = \frac{\sum x_2 y \cdot \sum x_1^2 - \sum x_1 x_2 \cdot \sum x_1 y}{\sum x_1^2 \cdot \sum x_2^2 - (\sum x_1 x_2)^2} \dots\dots\dots(3.22)$$

We can also express  $\hat{\beta}_1$  and  $\hat{\beta}_2$  in terms of covariance and variances of  $Y, X_1$  and  $X_2$

$$\hat{\beta}_1 = \frac{\text{Cov}(X_1, Y) \cdot \text{Var}(X_2) - \text{Cov}(X_1, X_2) \cdot \text{Cov}(X_2, Y)}{\text{Var}(X_1) \cdot \text{Var}(X_2) - [\text{Cov}(X_1, X_2)]^2} \dots\dots\dots(3.23)$$

$$\hat{\beta}_2 = \frac{\text{Cov}(X_2, Y) \cdot \text{Var}(X_1) - \text{Cov}(X_1, X_2) \cdot \text{Cov}(X_1, Y)}{\text{Var}(X_1) \cdot \text{Var}(X_2) - [\text{Cov}(X_1, X_2)]^2} \dots\dots\dots(3.24)$$

### 3.3.2 The coefficient of determination ( $R^2$ ): two explanatory variables case

In the simple regression model, we introduced  $R^2$  as a measure of the proportion of variation in the dependent variable that is explained by variation in the explanatory variable. In multiple regression model the same measure is relevant, and the same formulas are valid but now we talk of the proportion of variation in the dependent variable explained by all explanatory variables included in the model. The coefficient of determination is:

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum e_i^2}{\sum y_i^2} \quad (3.25)$$

In the present model of two explanatory variables:

$$\begin{aligned} \sum e_i^2 &= \sum (y_i - \hat{\beta}_1 x_{1i} - \hat{\beta}_2 x_{2i})^2 \\ &= \sum e_i (y_i - \hat{\beta}_1 x_{1i} - \hat{\beta}_2 x_{2i}) \\ &= \sum e_i y_i - \hat{\beta}_1 \sum e_i x_{1i} - \hat{\beta}_2 \sum e_i x_{2i} \\ &= \sum e_i y_i \text{ since } \sum e_i x_{1i} = \sum e_i x_{2i} = 0 \\ &= \sum y_i (y_i - \hat{\beta}_1 x_{1i} - \hat{\beta}_2 x_{2i}) \\ \text{i.e. } \sum e_i^2 &= \sum y_i^2 - \hat{\beta}_1 \sum x_{1i} y_i - \hat{\beta}_2 \sum x_{2i} y_i \\ \Rightarrow \sum y_i^2 &= \hat{\beta}_1 \sum x_{1i} y_i + \hat{\beta}_2 \sum x_{2i} y_i + \sum e_i^2 \quad (3.26) \end{aligned}$$

Total sum of square (Total variation)
Explained sum of square (Explained variation)
Residual sum of squares (unexplained variation)

$$\therefore R^2 = \frac{ESS}{TSS} = \frac{\hat{\beta}_1 \sum x_{1i} y_i + \hat{\beta}_2 \sum x_{2i} y_i}{\sum y_i^2} \quad (3.27)$$

As in simple regression,  $R^2$  is also viewed as a measure of the prediction ability of the model over the sample period, or as a measure of how well the estimated regression fits the data. The value of  $R^2$  is also equal to the squared sample correlation coefficient between  $\hat{Y}_i$  &  $Y_i$ . Since the sample correlation coefficient measures the linear association between two variables, if  $R^2$  is high, that means there is a close association between the values of  $Y_i$  and the values of predicted by

the model,  $\hat{Y}_t$ . In this case, the model is said to “fit” the data well. If  $R^2$  is low, there is no association between the values of  $Y_t$  and the values predicted by the model,  $\hat{Y}_t$  and the model does not fit the data well.

### 3.3.3 Adjusted Coefficient of Determination ( $\bar{R}^2$ )

One difficulty with  $R^2$  is that it can be made large by adding more and more variables, even if the variables added have no economic justification. Algebraically, it is the fact that as the variables are added the sum of squared errors (RSS) goes down (it can remain unchanged, but this is rare) and thus  $R^2$  goes up. If the model contains  $n-1$  variables then  $R^2=1$ . The manipulation of model just to obtain a high  $R^2$  is not wise. An alternative measure of goodness of fit, called the adjusted  $R^2$  and often symbolized as  $\bar{R}^2$ , is usually reported by regression programs. It is computed as:

$$\bar{R}^2 = 1 - \frac{\sum e_i^2 / n - k}{\sum y^2 / n - 1} = 1 - (1 - R^2) \left( \frac{n - 1}{n - k} \right) \text{-----} (3.28)$$

This measure does not always goes up when a variable is added because of the degree of freedom term  $n-k$  is the numerator. As the number of variables  $k$  increases, RSS goes down, but so does  $n-k$ . The effect on  $\bar{R}^2$  depends on the amount by which  $R^2$  falls. While solving one problem, this corrected measure of goodness of fit unfortunately introduces another one. It losses its interpretation;  $\bar{R}^2$  is no longer the percent of variation explained. This modified  $\bar{R}^2$  is sometimes used and misused as a device for selecting the appropriate set of explanatory variables.

### 3.4. General Linear Regression Model and Matrix Approach

So far we have discussed the regression models containing one or two explanatory variables. Let us now generalize the model assuming that it contains  $k$  variables. It will be of the form:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + U$$

There are k parameters to be estimated. The system of normal equations consist of k+1 equations, in which the unknowns are the parameters  $\beta_0, \beta_1, \beta_2, \dots, \beta_k$  and the known terms will be the sums of squares and the sums of products of all variables in the structural equations.

Least square estimators of the unknown parameters are obtained by minimizing the sum of the squared residuals.

$$\Sigma e_i^2 = \Sigma (y_i - \hat{\beta}_0 - \hat{\beta}_1 X_1 - \hat{\beta}_2 X_2 - \dots - \hat{\beta}_k X_k)^2$$

With respect to  $\beta_j (j = 0, 1, 2, \dots (k+1))$

The partial derivations are equated to zero to obtain normal equations.

$$\frac{\partial \Sigma e_i^2}{\partial \hat{\beta}_0} = -2\Sigma (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_1 - \hat{\beta}_2 X_2 - \dots - \hat{\beta}_k X_k) = 0$$

$$\frac{\partial \Sigma e_i^2}{\partial \hat{\beta}_1} = -2\Sigma (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_1 - \hat{\beta}_2 X_2 - \dots - \hat{\beta}_k X_k)(x_i) = 0$$

.....

$$\frac{\partial \Sigma e_i^2}{\partial \hat{\beta}_k} = -2\Sigma (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_1 - \hat{\beta}_2 X_2 - \dots - \hat{\beta}_k X_k)(x_{ki}) = 0$$

The general form of the above equations (except first ) may be written as:

$$\frac{\partial \Sigma e_i^2}{\partial \hat{\beta}_j} = -2\Sigma (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \dots - \hat{\beta}_k X_{ki}) = 0; \quad \text{where } (j = 1, 2, \dots k)$$

The normal equations of the general linear regression model are

$$\begin{aligned} \Sigma Y_i &= n\hat{\beta}_0 + \hat{\beta}_1 \Sigma X_{1i} + \hat{\beta}_2 \Sigma X_{2i} + \dots + \hat{\beta}_k \Sigma X_{ki} \\ \Sigma Y_i X_{1i} &= \hat{\beta}_0 \Sigma X_{1i} + \hat{\beta}_1 \Sigma X_{1i}^2 + \dots + \hat{\beta}_k \Sigma X_{1i} X_{ki} \\ \Sigma Y_i X_{2i} &= \hat{\beta}_0 \Sigma X_{2i} + \hat{\beta}_1 \Sigma X_{1i} X_{2i} + \hat{\beta}_2 \Sigma X_{2i}^2 + \dots + \hat{\beta}_k \Sigma X_{2i} X_{ki} \\ &\vdots \\ &\vdots \\ \Sigma Y_i X_{ki} &= \hat{\beta}_0 \Sigma X_{ki} + \hat{\beta}_1 \Sigma X_{1i} X_{ki} + \sum X_{2i} X_{ki} \dots + \hat{\beta}_k \Sigma X_{ki}^2 \end{aligned}$$

Solving the above normal equations will result in algebraic complexity. But we can solve this easily using matrix. Hence in the next section we will discuss the matrix approach to linear regression model.

### 3.4.1 Matrix Approach to Linear Regression Model

The general linear regression model with k explanatory variables is written in the form:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + U_i$$

where  $(i = 1, 2, 3, \dots, n)$  and  $\beta_0$  = the intercept,  $\beta_1$  to  $\beta_k$  = partial slope coefficients  
 $U$  = stochastic disturbance term and  $i = i^{\text{th}}$  observation, 'n' being the size of the observation. Since i represents the  $i^{\text{th}}$  observation, we shall have 'n' number of equations with 'n' number of observations on each variable.

$$\begin{aligned} Y_1 &= \beta_0 + \beta_1 X_{11} + \beta_2 X_{21} + \beta_3 X_{31} \dots + \beta_k X_{k1} + U_1 \\ Y_2 &= \beta_0 + \beta_1 X_{12} + \beta_2 X_{22} + \beta_3 X_{32} \dots + \beta_k X_{k2} + U_2 \\ Y_3 &= \beta_0 + \beta_1 X_{13} + \beta_2 X_{23} + \beta_3 X_{33} \dots + \beta_k X_{k3} + U_3 \\ &\dots \dots \dots \\ Y_n &= \beta_0 + \beta_1 X_{1n} + \beta_2 X_{2n} + \beta_3 X_{3n} \dots + \beta_k X_{kn} + U_n \end{aligned}$$

These equations are put in matrix form as:

$$\begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & X_{11} & X_{21} & \dots & X_{k1} \\ 1 & X_{12} & X_{22} & \dots & X_{k2} \\ 1 & X_{13} & X_{23} & \dots & X_{k3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{1n} & X_{2n} & \dots & X_{kn} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix} + \begin{bmatrix} U_1 \\ U_2 \\ U_3 \\ \vdots \\ U_n \end{bmatrix}$$

$$Y = X \cdot \beta + U$$

In short  $Y = X\beta + U$  .....(3.29)

The order of matrix and vectors involved are:

$$Y = (n \times 1), \quad X = \{(n \times (k+1))\}, \quad \beta = \{(k+1) \times 1\} \quad \text{and} \quad U = (n \times 1)$$

To derive the OLS estimators of  $\beta$ , under the usual (classical) assumptions mentioned earlier, we define two vectors  $\hat{\beta}$  and 'e' as:

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_k \end{bmatrix} \quad \text{and} \quad e = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}$$

Thus we can write:  $Y = X\hat{\beta} + e$  and  $e = Y - X\hat{\beta}$

We have to minimize:

$$\begin{aligned} \sum_{i=1}^n e_i^2 &= e_1^2 + e_2^2 + e_3^2 + \dots + e_n^2 \\ &= [e_1, e_2, \dots, e_n] \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix} = e'e \\ &= \sum e_i^2 = e'e \\ e'e &= (Y - X\hat{\beta})'(Y - X\hat{\beta}) \\ &= YY' - \hat{\beta}'X'Y - Y'X\hat{\beta} + \hat{\beta}'X'X\hat{\beta} \dots \dots \dots (3.30) \end{aligned}$$

Since  $\hat{\beta}'X'Y$  is scalar (1x1), it is equal to its transpose;

$$\begin{aligned} \hat{\beta}'X'Y &= Y'X\hat{\beta} \\ e'e &= Y'Y - 2\hat{\beta}'X'Y + \hat{\beta}'X'X\hat{\beta} \dots \dots \dots (3.31) \end{aligned}$$

Minimizing  $e'e$  with respect to the elements in  $\hat{\beta}$

$$\begin{aligned} \frac{\partial \sum e_i^2}{\partial \hat{\beta}} &= \frac{\partial (e'e)}{\partial \hat{\beta}} = -2X'Y + 2X'X\hat{\beta} \\ \text{Since } \frac{\partial (X'AX)}{\partial \hat{\beta}} &= 2AX \text{ and also too } 2X'A \end{aligned}$$

Equating the expression to null vector 0, we obtain:

$$-2X'Y + 2X'X\hat{\beta} = 0 \Rightarrow X'X\hat{\beta} = X'Y$$

$$\boxed{\hat{\beta} = (X'X)^{-1}X'Y} \dots\dots\dots (3.32)$$

Hence  $\hat{\beta}$  is the vector of required least square estimators,  $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_k$ .

### 3.4.2. Statistical Properties of the Parameters (Matrix) Approach

We have seen, in simple linear regression that the OLS estimators ( $\hat{\alpha}$  &  $\hat{\beta}$ ) satisfy the small sample property of an estimator i.e. BLUE property. In multiple regression, the OLS estimators also satisfy the BLUE property. Now we proceed to examine the desired properties of the estimators in matrix notations:

#### 1. Linearity

We know that:  $\hat{\beta} = (X'X)^{-1}X'Y$

Let  $C = (X'X)^{-1}X'$

$$\Rightarrow \hat{\beta} = CY \dots\dots\dots (3.33)$$

Since C is a matrix of fixed variables, equation (3.33) indicates us  $\hat{\beta}$  is linear in Y.

#### 2. Unbiased ness

$$\hat{\beta} = (X'X)^{-1}X'Y$$

$$\hat{\beta} = (X'X)^{-1}X'(X\beta + U)$$

$$\hat{\beta} = \beta + (X'X)^{-1}X'U \dots\dots\dots (3.34)$$

$$\text{since } [(X'X)^{-1}X'X = I]$$

$$E(\hat{\beta}) = E\{\beta + (X'X)^{-1}X'U\}$$

$$= E(\beta) + E[(X'X)^{-1}X'U]$$

$$= \beta + E(X'X)^{-1}X'E(U)$$

$$= \beta, \text{ since } E(U) = 0$$

Thus, least square estimators are unbiased.

### 3. Minimum variance

Before showing all the OLS estimators are best (possess the minimum variance property), it is important to derive their variance.

We know that,  $\text{var}(\hat{\beta}) = E[(\hat{\beta} - \beta)^2] = E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)']$

$$E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'] = \begin{bmatrix} E(\hat{\beta}_1 - \beta_1)^2 & E[(\hat{\beta}_1 - \beta_1)(\hat{\beta}_2 - \beta_2)] & \dots & E[(\hat{\beta}_1 - \beta_1)(\hat{\beta}_k - \beta_k)] \\ E[(\hat{\beta}_2 - \beta_2)(\hat{\beta}_1 - \beta_1)] & E(\hat{\beta}_2 - \beta_2)^2 & \dots & E[(\hat{\beta}_2 - \beta_2)(\hat{\beta}_k - \beta_k)] \\ \vdots & \vdots & \ddots & \vdots \\ E[(\hat{\beta}_k - \beta_k)(\hat{\beta}_1 - \beta_1)] & E[(\hat{\beta}_k - \beta_k)(\hat{\beta}_2 - \beta_2)] & \dots & E(\hat{\beta}_k - \beta_k)^2 \end{bmatrix}$$

$$= \begin{bmatrix} \text{var}(\hat{\beta}_1) & \text{cov}(\hat{\beta}_1, \hat{\beta}_2) & \dots & \text{cov}(\hat{\beta}_1, \hat{\beta}_k) \\ \text{cov}(\hat{\beta}_2, \hat{\beta}_1) & \text{var}(\hat{\beta}_2) & \dots & \text{cov}(\hat{\beta}_2, \hat{\beta}_k) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(\hat{\beta}_k, \hat{\beta}_1) & \text{cov}(\hat{\beta}_k, \hat{\beta}_2) & \dots & \text{var}(\hat{\beta}_k) \end{bmatrix}$$

The above matrix is a symmetric matrix containing variances along its main diagonal and covariance of the estimators every where else. This matrix is, therefore, called the *Variance-covariance* matrix of least squares estimators of the regression slopes. Thus,

$$\text{var}(\hat{\beta}) = E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'] \dots \dots \dots (3.35)$$

$$\text{From (3.15)} \quad \hat{\beta} = \beta + (X'X)^{-1}X'U$$

$$\Rightarrow \hat{\beta} - \beta = (X'X)^{-1}X'U \dots \dots \dots (3.36)$$

Substituting (3.17) in (3.16)

$$\text{var}(\hat{\beta}) = E\left[\left\{(X'X)^{-1}X'U\right\}\left\{(X'X)^{-1}X'U\right\}'\right]$$

$$\text{var}(\hat{\beta}) = E\left[(X'X)^{-1}X'UU'X(X'X)^{-1}\right]$$

$$= (X'X)^{-1}X'E(UU')X(X'X)^{-1}$$

$$= (X'X)^{-1}X'\sigma_u^2 I_n X(X'X)^{-1}$$

$$= \sigma_u^2 (X'X)^{-1}X'X(X'X)^{-1}$$

$$\boxed{\text{var}(\hat{\beta}) = \sigma_u^2 (X'X)^{-1}} \dots \dots \dots (3.37)$$

Note: ( $\sigma_u^2$  being a scalar can be moved in front or behind of a matrix while identity matrix  $I_n$  can be suppressed).

Thus we obtain,  $\text{var}(\hat{\beta}) = \sigma_u^2 (X'X)^{-1}$

$$\text{Where, } (X'X)^{-1} = \begin{bmatrix} n & \sum X_{1n} & \dots & \sum X_{kn} \\ \sum X_{1n} & \sum X_{1n}^2 & \dots & \sum X_{1n} X_{kn} \\ \vdots & \vdots & \ddots & \vdots \\ \sum X_{kn} & \sum X_{1n} X_{kn} & \dots & \sum X_{kn}^2 \end{bmatrix}$$

We can, therefore, obtain the variance of any estimator say  $\hat{\beta}_i$  by taking the  $i^{\text{th}}$  term from the principal diagonal of  $(X'X)^{-1}$  and then multiplying it by  $\sigma_u^2$

Where the X's are in their absolute form. When the x's are in deviation form we can write the multiple regression in matrix form as ;

$$\hat{\beta} = (x'x)^{-1} x'y$$

$$\text{where } \hat{\beta} = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \vdots \\ \hat{\beta}_k \end{bmatrix} \quad \text{and} \quad (x'x) = \begin{bmatrix} \sum x_1^2 & \sum x_1 x_2 & \dots & \sum x_1 x_k \\ \sum x_2 x_1 & \sum x_2^2 & \dots & \sum x_2 x_k \\ \vdots & \vdots & \ddots & \vdots \\ \sum x_k x_1 & \sum x_k x_2 & \dots & \sum x_k^2 \end{bmatrix}$$

The above column matrix  $\hat{\beta}$  doesn't include the constant term  $\hat{\beta}_0$ . Under such conditions the variances of slope parameters in deviation form can be written as:  $\text{var}(\hat{\beta}) = \sigma_u^2 (x'x)^{-1} \dots \dots \dots (2.38)$

(the proof is the same as (3.37) above). In general we can illustrate the variance of the parameters by taking two explanatory variables.

The multiple regression when written in deviation form that has two explanatory variables is

$$y_1 = \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

$$\text{var}(\hat{\beta}) = E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)']$$

In this model;  $(\hat{\beta} - \beta) = \begin{bmatrix} (\hat{\beta}_1 - \beta_1) \\ (\hat{\beta}_2 - \beta_2) \end{bmatrix}$

$$(\hat{\beta} - \beta)' = [(\hat{\beta}_1 - \beta_1)(\hat{\beta}_2 - \beta_2)]$$

$$\therefore (\hat{\beta} - \beta)(\hat{\beta} - \beta)' = \begin{bmatrix} (\hat{\beta}_1 - \beta_1) \\ (\hat{\beta}_2 - \beta_2) \end{bmatrix} [(\hat{\beta}_1 - \beta_1)(\hat{\beta}_2 - \beta_2)]$$

and  $E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'] = E \begin{bmatrix} (\hat{\beta}_1 - \beta_1)^2 & (\hat{\beta}_1 - \beta_1)(\hat{\beta}_2 - \beta_2) \\ (\hat{\beta}_1 - \beta_1)(\hat{\beta}_2 - \beta_2) & (\hat{\beta}_2 - \beta_2)^2 \end{bmatrix}$

$$= \begin{bmatrix} \text{var}(\hat{\beta}_1) & \text{cov}(\hat{\beta}_1, \hat{\beta}_2) \\ \text{cov}(\hat{\beta}_1, \hat{\beta}_2) & \text{var}(\hat{\beta}_2) \end{bmatrix}$$

In case of two explanatory variables,  $x$  in the deviation form shall be:

$$x = \begin{bmatrix} x_{11} & x_{21} \\ x_{12} & x_{22} \\ \vdots & \vdots \\ x_{1n} & x_{2n} \end{bmatrix} \text{ and } x' = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \end{bmatrix}$$

$$\therefore \sigma_u^2 (x'x)^{-1} = \sigma_u^2 \begin{bmatrix} \sum x_1^2 & \sum x_1 x_2 \\ \sum x_1 x_2 & \sum x_2^2 \end{bmatrix}^{-1}$$

Or  $\sigma_u^2 (x'x)^{-1} = \frac{\sigma_u^2 \begin{bmatrix} \sum x_2^2 & -\sum x_1 x_2 \\ -\sum x_1 x_2 & \sum x_1^2 \end{bmatrix}}{\begin{vmatrix} \sum x_1^2 & \sum x_1 x_2 \\ \sum x_1 x_2 & \sum x_2^2 \end{vmatrix}}}$

$$\text{i.e., } \text{var}(\hat{\beta}_1) = \frac{\sigma_u^2 \sum x_2^2}{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2} \dots \dots \dots (3.39)$$

$$\text{and, } \text{var}(\hat{\beta}_2) = \frac{\sigma_u^2 \sum x_1^2}{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2} \dots \dots \dots (3.40)$$

$$\text{cov}(\hat{\beta}_1, \hat{\beta}_2) = \frac{(-)\sigma_u^2 \sum x_1 x_2}{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2} \dots \dots \dots (3.41)$$

The only unknown part in variances and covariance of the estimators is  $\sigma_u^2$ .

As we have seen in simple regression model  $\hat{\sigma}^2 = \left\{ \frac{\sum e_i^2}{n-2} \right\}$ . For k-parameters

(including the constant parameter)  $\hat{\sigma}^2 = \left\{ \frac{\sum e_i^2}{n-k} \right\}$ .

In the above model we have three parameters including the constant term and

$$\hat{\sigma}^2 = \left\{ \frac{\sum e_i^2}{n-3} \right\}$$

$$\sum e_i^2 = \sum y_i^2 - \beta_1 \sum x_{1i} y_i - \beta_2 \sum x_{2i} y_i \dots \dots \dots + \beta_k \sum x_{ki} y_i \dots \dots \dots (3.42)$$

this is for k explanatory variables. For two explanatory variables

$$\sum e_i^2 = \sum y_i^2 - \beta_1 \sum x_{1i} y_i - \beta_2 \sum x_{2i} y_i \dots \dots \dots (3.43)$$

This is all about the variance covariance of the parameters. Now it is time to see the minimum variance property.

### Minimum variance of $\hat{\beta}$

To show that all the  $\beta_i$ 's in the  $\hat{\beta}$  vector are Best Estimators, we have also to prove that the variances obtained in (3.37) are the smallest amongst all other possible linear unbiased estimators. We follow the same procedure as followed in case of single explanatory variable model where, we first assumed an alternative linear unbiased estimator and then it was established that its variance is greater than the estimator of the regression model.

Assume that  $\hat{\beta}$  is an alternative unbiased and linear estimator of  $\beta$ . Suppose that

$$\hat{\beta} = [(X'X)^{-1}X' + B]Y$$

Where B is (k x n) matrix of known constants.

$$\therefore \hat{\beta} = [(X'X)^{-1}X' + B][X\beta + U]$$

$$\hat{\beta} = (X'X)^{-1}X'(X\beta + U) + B(X\beta + U)$$

$$\begin{aligned}
E(\hat{\beta}) &= E[(X'X)^{-1}X'(X\beta + U) + B(X\beta + U)] \\
&= E[(X'X)^{-1}X'X\beta + (X'X)^{-1}X'U + BX\beta + BU] \\
&= \beta + BX\beta, \quad [\text{since } E(U) = 0] \dots\dots\dots (3.44)
\end{aligned}$$

Since our assumption regarding an alternative  $\hat{\beta}$  is that it is to be an unbiased estimator of  $\beta$ , therefore,  $E(\hat{\beta})$  should be equal to  $\beta$ ; in other words  $(\beta XB)$  should be a null matrix.

Thus we say,  $BX$  should be  $=0$  if  $(\hat{\beta}) = [(X'X)^{-1}X' + B]Y$  is to be an unbiased estimator. Let us now find variance of this alternative estimator.

$$\begin{aligned}
\text{var}(\hat{\beta}) &= E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'] \\
&= E\left\{[(X'X)^{-1}X' + B]Y - \beta\right\}\left\{[(X'X)^{-1}X' + B]Y - \beta\right\}' \\
&= E\left\{[(X'X)^{-1}X' + B](X\beta + U) - \beta\right\}\left\{[(X'X)^{-1}X' + B](X\beta + U) - \beta\right\}' \\
&= E\left\{(X'X)^{-1}X'X\beta + (X'X)^{-1}X'U + BX\beta + BU - \beta\right\} \\
&\quad \left\{(X'X)^{-1}X'X\beta + (X'X)^{-1}X'U + BX\beta + BU - \beta\right\}' \\
&= E\left\{(X'X)^{-1}X'U + BU\right\}\left\{(X'X)^{-1}X'U + BU\right\}' \\
&\quad (QBX = 0) \\
&= E\left\{(X'X)^{-1}X'U + BU\right\}\left\{U'X(X'X)^{-1} + U'B'\right\} \\
&= E\left\{(X'X)^{-1}B\right\}UU'\left\{X(X'X)^{-1} + U'B'\right\} \\
&= [(X'X)^{-1}X' + B]E(UU')\left[X(X'X)^{-1} + B'\right] \\
&= \sigma_u^2 I_n [(X'X)^{-1}X' + B]\left[X(X'X)^{-1} + B'\right] \\
&= \sigma_u^2 [(X'X)^{-1}X'X(X'X)^{-1} + BX(X'X)^{-1} + (X'X)^{-1}] \\
&= \sigma_u^2 [(X'X)^{-1}X'X(X'X)^{-1} + BX(X'X)^{-1} + (X'X)^{-1}X'B' + BB'] \\
&= \sigma_u^2 [(X'X)^{-1} + BB'] [QBX = 0] \\
\boxed{\text{var}(\hat{\beta})} &= \sigma_u^2 (X'X)^{-1} + \sigma_u^2 BB' \dots\dots\dots (3.45)
\end{aligned}$$

Or, in other words,  $\text{var}(\hat{\beta})$  is greater than  $\text{var}(\beta)$  by an expression  $\sigma_u^2 BB'$  and it proves that  $\hat{\beta}$  is the best estimator.

### 3.4.3. Coefficient of Determination in Matrix Form

The coefficient of determination ( $R^2$ ) can be derived in matrix form as follows.

We know that  $\sum e_i^2 = e'e = Y'Y - 2\hat{\beta}'X'Y + \hat{\beta}'X'X\hat{\beta}$  since  $(X'X)\hat{\beta} = X'Y$  and

$$\sum Y_i^2 = Y'Y$$

$$\therefore e'e = Y'Y - 2\hat{\beta}'X'Y + \hat{\beta}'X'Y$$

$$e'e = Y'Y - \hat{\beta}'X'Y \dots \dots \dots (3.46)$$

$$\hat{\beta}'X'Y = e'e - Y'Y \dots \dots \dots (3.47)$$

We know,  $y_i = Y_i - \bar{Y}$

$$\therefore \sum y_i^2 = \sum Y_i^2 - \frac{1}{n}(\sum Y_i)^2$$

In matrix notation

$$\sum y_i^2 = Y'Y - \frac{1}{n}(\sum Y_i)^2 \dots \dots \dots (3.48)$$

Equation (3.48) gives the total sum of squares variations in the model.

$$\text{Explained sum of squares} = \sum y_i^2 - \sum e_i^2$$

$$= Y'Y - \frac{1}{n}(\sum y_i)^2 - e'e$$

$$= \hat{\beta}'X'Y - \frac{1}{n}(\sum Y_i)^2 \dots \dots \dots (3.49)$$

$$\text{Since } R^2 = \frac{\text{Explained sum of squares}}{\text{Total sum of squares}}$$

$$\therefore R^2 = \frac{\hat{\beta}'X'Y - \frac{1}{n}(\sum Y_i)^2}{Y'Y - \frac{1}{n}(\sum Y_i)^2} = \frac{\hat{\beta}'X'Y - n\bar{Y}^2}{Y'Y - n\bar{Y}^2} \dots \dots \dots (3.50)$$

Dear Students! We hope that from the discussion made so far on multiple regression model, in general, you may make the following summary of results.

|       |                          |                                                                                                                                             |
|-------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| (i)   | Model:                   | $Y = X\beta + U$                                                                                                                            |
| (ii)  | Estimators:              | $\hat{\beta} = (X'X)^{-1}X'Y$                                                                                                               |
| (iii) | Statistical properties:  | BLUE                                                                                                                                        |
| (iv)  | Variance-covariance:     | $\text{var}(\hat{\beta}) = \sigma_u^2 (X'X)^{-1}$                                                                                           |
| (v)   | Estimation of $(e'e)$ :  | $e'e = Y'Y - \hat{\beta}'X'Y$                                                                                                               |
| (vi)  | Coeff. of determination: | $R^2 = \frac{\hat{\beta}'X'Y - \frac{1}{n}(\sum Y_i)^2}{Y'Y - \frac{1}{n}(\sum Y_i)^2} = \frac{\hat{\beta}'X'Y - n\bar{Y}}{Y'Y - n\bar{Y}}$ |

### 3.5. Hypothesis Testing in Multiple Regression Model

In multiple regression models we will undertake two tests of significance. One is significance of individual parameters of the model. This test of significance is the same as the tests discussed in simple regression model. The second test is overall significance of the model.

#### 3.5.1. Tests of individual significance

If we invoke the assumption that  $U_i \sim N(0, \sigma^2)$ , then we can use either the t-test or standard error test to test a hypothesis about any individual partial regression coefficient. To illustrate consider the following example.

$$\text{Let } Y = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + e_i \dots \dots \dots (3.51)$$

$$\text{A. } H_0 : \beta_1 = 0$$

$$H_1 : \beta_1 \neq 0$$

$$\text{B. } H_0 : \beta_2 = 0$$

$$H_1 : \beta_2 \neq 0$$

The null hypothesis (A) states that, holding  $X_2$  constant  $X_1$  has no (linear) influence on  $Y$ . Similarly hypothesis (B) states that holding  $X_1$  constant,  $X_2$  has no influence on the dependent variable  $Y_i$ . To test these null hypothesis we will use the following tests:

- i- *Standard error test*: under this and the following testing methods we test only for  $\hat{\beta}_1$ . The test for  $\hat{\beta}_2$  will be done in the same way.

$$SE(\hat{\beta}_1) = \sqrt{\text{var}(\hat{\beta}_1)} = \sqrt{\frac{\hat{c}^2 \sum x_{2i}^2}{\sum x_{1i}^2 \sum x_{2i}^2 - (\sum x_1 x_2)^2}} \quad ; \text{ where } \hat{\sigma}^2 = \frac{\sum e^2}{n-3}$$

- If  $SE(\hat{\beta}_1) > \frac{1}{2}\hat{\beta}_1$ , we accept the null hypothesis that is, we can conclude that the estimate  $\beta_i$  is not statistically significant.
- If  $SE(\hat{\beta}_1) < \frac{1}{2}\hat{\beta}_1$ , we reject the null hypothesis that is, we can conclude that the estimate  $\beta_i$  is statistically significant.

Note: The smaller the standard errors, the stronger the evidence that the estimates are statistically reliable.

- ii. *The student's t-test*: We compute the t-ratio for each  $\hat{\beta}_i$

$$t^* = \frac{\hat{\beta}_i - \beta}{SE(\hat{\beta}_i)} \sim t_{n-k} \quad , \text{ where } n \text{ is number of observation and } k \text{ is number of}$$

parameters. If we have 3 parameters, the degree of freedom will be  $n-3$ . So;

$$t^* = \frac{\hat{\beta}_2 - \beta_2}{SE(\hat{\beta}_2)} \quad ; \text{ with } n-3 \text{ degree of freedom}$$

In our null hypothesis  $\beta_2 = 0$ , the  $t^*$  becomes:

$$t^* = \frac{\hat{\beta}_2}{SE(\hat{\beta}_2)}$$

- If  $t^* < t$  (tabulated), we accept the null hypothesis, i.e. we can conclude that  $\hat{\beta}_2$  is not significant and hence the regressor does not appear to contribute to the explanation of the variations in  $Y$ .

- If  $t^* > t$  (tabulated), we reject the null hypothesis and we accept the alternative one;  $\hat{\beta}_2$  is statistically significant. Thus, the greater the value of  $t^*$  the stronger the evidence that  $\beta_i$  is statistically significant.

### 3.5.2 Test of Overall Significance

Through out the previous section we were concerned with testing the significance of the estimated partial regression coefficients individually, i.e. under the separate hypothesis that each of the true population partial regression coefficient was zero.

In this section we extend this idea to joint test of the relevance of all the included explanatory variables. Now consider the following:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + U_i$$

$$H_0 : \beta_1 = \beta_2 = \beta_3 = \dots = \beta_k = 0$$

$$H_1 : \text{at least one of the } \beta_k \text{ is non-zero}$$

This null hypothesis is a joint hypothesis that  $\beta_1, \beta_2, \dots, \beta_k$  are jointly or simultaneously equal to zero. A test of such a hypothesis is called *a test of overall significance* of the observed or estimated regression line, that is, whether Y is linearly related to  $X_1, X_2, \dots, X_k$ .

Can the joint hypothesis be tested by testing the significance of individual significance of  $\hat{\beta}_i$ 's as the above? The answer is no, and the reasoning is as follows.

*In testing the individual significance of an observed partial regression coefficient, we assumed implicitly that each test of significance was based on different (i.e. independent) sample. Thus, in testing the significance of  $\hat{\beta}_2$  under the hypothesis that  $\beta_2 = 0$ , it was assumed tacitly that the testing was based on different sample*

*from the one used in testing the significance of  $\hat{\beta}_3$  under the null hypothesis that  $\beta_3 = 0$ . But to test the joint hypothesis of the above, we shall be violating the assumption underlying the test procedure.*

“.....testing a series of single (individual) hypothesis is not equivalent to testing those same hypothesis. The institutive reason for this is that in a joint test of several hypotheses any single hypothesis is affected by the information in the other hypothesis.”<sup>1</sup>

The test procedure for any set of hypothesis can be based on a *comparison of the sum of squared errors from the original, the unrestricted multiple regression model to the sum of squared errors from a regression model in which the null hypothesis is assumed to be true.* When a null hypothesis is assumed to be true, we in effect place conditions or constraints, on the values that the parameters can take, and the sum of squared errors increases. The idea of the test is that if these sum of squared errors are substantially different, then the assumption that the joint null hypothesis is true has significantly reduced the ability of the model to fit the data, and the data do not support the null hypothesis.

If the null hypothesis is true, we expect that the data are compliable with the conditions placed on the parameters. Thus, there would be little change in the sum of squared errors when the null hypothesis is assumed to be true.

Let the Restricted Residual Sum of Square (RRSS) be the sum of squared errors in the model obtained by assuming that the null hypothesis is true and URSS be the sum of the squared error of the original unrestricted model i.e. unrestricted residual sum of square (URSS). It is always true that  $RRSS - URSS \geq 0$ .

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<sup>1</sup> Gujarati, 3<sup>rd</sup> ed.pp

Consider  $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \dots + \hat{\beta}_k X_k + e_i$ .

This model is called unrestricted. The test of joint hypothesis is that:

$$H_0: \beta_1 = \beta_2 = \beta_3 = \dots = \beta_k = 0$$

$H_1$ : at least one of the  $\beta_k$  is different from zero.

We know that:  $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \dots + \hat{\beta}_k X_{ki}$

$$Y_i = \hat{Y} + e$$

$$e_i = Y_i - \hat{Y}_i$$

$$\sum_i e^2 = \sum_i (Y_i - \hat{Y}_i)^2$$

This sum of squared error is called unrestricted residual sum of square (URSS).

This is the case when the null hypothesis is not true. If the null hypothesis is assumed to be true, i.e. when all the slope coefficients are zero.

$$Y = \hat{\beta}_0 + e_i$$

$$\hat{\beta}_0 = \frac{\sum Y_i}{n} = \bar{Y} \rightarrow \quad (\text{applying OLS}) \dots \dots \dots (3.52)$$

$$e = Y - \hat{\beta}_0 \quad \text{but} \quad \hat{\beta}_0 = \bar{Y}$$

$$e = Y - \bar{Y}$$

$$\sum_i e^2 = \sum_i (Y_i - \bar{Y})^2 = \sum y^2 = TSS$$

The sum of squared error when the null hypothesis is assumed to be true is called Restricted Residual Sum of Square (RRSS) and this is equal to the total sum of square (TSS).

$$\text{The ratio: } \frac{RRSS - URSS / K - 1}{URSS / n - K} \sim F_{(k-1, n-k)} \dots \dots \dots (3.53);$$

(has an F-distribution with k-1 and n-k degrees of freedom for the numerator and denominator respectively)

$$RRSS = TSS$$

$$URSS = \sum_i e^2 = \sum y^2 - \hat{\beta}_1 \sum yx_1 - \hat{\beta}_2 \sum yx_2 + \dots + \hat{\beta}_k \sum yx_k = RSS$$

$$F = \frac{(TSS - RSS) / k - 1}{RSS / n - k}$$

$$F = \frac{ESS / k - 1}{RSS / n - k} \dots\dots\dots (3.54)$$

If we divide the above numerator and denominator by  $\Sigma y^2 = TSS$  then:

$$F = \frac{\frac{ESS}{TSS} / k - 1}{\frac{RSS}{TSS} / n - k}$$

$$F = \frac{R^2 / k - 1}{1 - R^2 / n - k} \dots\dots\dots (3.55)$$

This implies the computed value of F can be calculated either as a ratio of ESS & TSS or  $R^2$  &  $1-R^2$ . If the null hypothesis is not true, then the difference between RRSS and URSS (TSS & RSS) becomes large, implying that the constraints placed on the model by the null hypothesis have large effect on the ability of the model to fit the data, and the value of F tends to be large. Thus, we reject the null hypothesis if the F test static becomes too large. This value is compared with the critical value of F which leaves the probability of in the upper tail of the F-distribution with k-1 and n-k degree of freedom.

If the computed value of F is greater than the critical value of F (k-1, n-k), then the parameters of the model are jointly significant or the dependent variable Y is linearly related to the independent variables included in the model.

#### Application of Multiple Regression.

In order to help you understand the working of matrix algebra in the estimation of the regression coefficient, variance of the coefficients and testing of the parameters and the model, consider the following numerical example.

**Example 1.** Consider the data given in Table 2.1 below to fit a linear function:

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + U$$

Table: 2.1. Numerical example for the computation of the OLS estimators.

| $n$ | Y   | $X_1$ | $X_2$ | $X_3$ | $y_i$            | $x_1$            | $x_2$            | $x_3$            | $y_i^2$              | $x_1x_2$              | $x_2x_3$               | $x_1x_3$               | $x_1^2$              | $x_2^2$              | $x_3^2$              | $x_1y_i$              | $x_2y_i$              | $x_3y_i$               |
|-----|-----|-------|-------|-------|------------------|------------------|------------------|------------------|----------------------|-----------------------|------------------------|------------------------|----------------------|----------------------|----------------------|-----------------------|-----------------------|------------------------|
| 1   | 49  | 35    | 53    | 200   | -3               | -7               | -9               | 0                | 9                    | 63                    | 0                      | 0                      | 49                   | 81                   | 0                    | 21                    | 27                    | 0                      |
| 2   | 40  | 35    | 53    | 212   | -12              | -7               | -9               | 12               | 144                  | 63                    | -108                   | -84                    | 49                   | 81                   | 144                  | 84                    | 108                   | -144                   |
| 3   | 41  | 38    | 50    | 211   | -11              | -4               | -12              | 11               | 121                  | 48                    | -132                   | -44                    | 16                   | 144                  | 121                  | 44                    | 132                   | -121                   |
| 4   | 46  | 40    | 64    | 212   | -6               | -2               | 2                | 12               | 36                   | -4                    | 24                     | -24                    | 4                    | 4                    | 144                  | 12                    | -12                   | -72                    |
| 5   | 52  | 40    | 70    | 203   | 0                | -2               | 8                | 3                | 0                    | -16                   | 24                     | -6                     | 4                    | 64                   | 9                    | 0                     | 0                     | 0                      |
| 6   | 59  | 42    | 68    | 194   | 7                | 0                | 6                | -6               | 49                   | 0                     | -36                    | 0                      | 0                    | 36                   | 36                   | 0                     | 42                    | -42                    |
| 7   | 53  | 44    | 59    | 194   | 1                | 2                | -3               | -6               | 1                    | -6                    | 18                     | -12                    | 4                    | 9                    | 36                   | 2                     | -3                    | -06                    |
| 8   | 61  | 46    | 73    | 188   | 9                | 4                | 11               | -12              | 81                   | 44                    | -132                   | -48                    | 16                   | 121                  | 144                  | 36                    | 99                    | -108                   |
| 9   | 55  | 50    | 59    | 196   | 3                | 8                | -3               | -4               | 9                    | -24                   | 12                     | -32                    | 64                   | 9                    | 16                   | 24                    | -9                    | -12                    |
| 10  | 64  | 50    | 71    | 190   | 12               | 8                | 9                | -10              | 144                  | 72                    | -90                    | -80                    | 64                   | 81                   | 100                  | 96                    | 108                   | -120                   |
|     | 520 | 420   | 620   | 2000  | $\Sigma y_i = 0$ | $\Sigma x_1 = 0$ | $\Sigma x_2 = 0$ | $\Sigma x_3 = 0$ | $\Sigma y_i^2 = 534$ | $\Sigma x_1x_2 = 240$ | $\Sigma x_2x_3 = -420$ | $\Sigma x_1x_3 = -330$ | $\Sigma x_1^2 = 270$ | $\Sigma x_2^2 = 630$ | $\Sigma x_3^2 = 750$ | $\Sigma x_1y_i = 319$ | $\Sigma x_2y_i = 432$ | $\Sigma x_3y_i = -625$ |

From the table, the means of the variables are computed and given below:

$$\bar{Y} = 52; \quad \bar{X}_1 = 42; \quad \bar{X}_2 = 62; \quad \bar{X}_3 = 200$$

Based on the above table and model answer the following question.

- Estimate the parameter estimators using the matrix approach
- Compute the variance of the parameters.
- Compute the coefficient of determination ( $R^2$ )
- Report the regression result.

*Solution:*

In the matrix notation:  $\hat{\beta} = (x'x)^{-1}x'y$ ; (when we use the data in deviation form),

$$\text{Where, } \hat{\beta} = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \end{bmatrix}, x = \begin{bmatrix} x_{11} & x_{21} & x_{31} \\ x_{12} & x_{22} & x_{32} \\ \vdots & \vdots & \vdots \\ x_{1n} & x_{2n} & x_{3n} \end{bmatrix}; \text{ so that}$$

$$(x'x) = \begin{bmatrix} \sum x_1^2 & \sum x_1 x_2 & \sum x_1 x_3 \\ \sum x_1 x_2 & \sum x_2^2 & \sum x_2 x_3 \\ \sum x_1 x_3 & \sum x_2 x_3 & \sum x_3^2 \end{bmatrix} \text{ and } x'y = \begin{bmatrix} \sum x_1 y \\ \sum x_2 y \\ \sum x_3 y \end{bmatrix}$$

(i) Substituting the relevant quantities from table 2.1 we have;

$$(x'x) = \begin{bmatrix} 270 & 240 & -330 \\ 240 & 630 & -420 \\ -330 & -420 & 750 \end{bmatrix} \text{ and } x'y = \begin{bmatrix} 319 \\ 492 \\ -625 \end{bmatrix}$$

Note: the calculations may be made easier by taking 30 as common factor from all the elements of matrix  $(x'x)$ . This will not affect the final results.

$$|x'x| = \begin{vmatrix} 270 & 240 & -330 \\ 240 & 630 & -420 \\ -330 & -420 & 750 \end{vmatrix} = 4716000$$

$$(x'x)^{-1} = \begin{bmatrix} 0.0085 & -0.0012 & 0.0031 \\ -0.0012 & 0.0027 & 0.0009 \\ 0.0031 & 0.0009 & 0.0032 \end{bmatrix}$$

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \end{bmatrix} = (x'x)^{-1} x'y = \begin{bmatrix} 0.0085 & -0.0012 & 0.0031 \\ -0.0012 & 0.0027 & 0.0009 \\ 0.0031 & 0.0009 & 0.0032 \end{bmatrix} \begin{bmatrix} 319 \\ 492 \\ -625 \end{bmatrix} = \begin{bmatrix} 0.2063 \\ 0.3309 \\ -0.5572 \end{bmatrix}$$

And

$$\begin{aligned} \alpha &= \bar{Y} - \hat{\beta}_1 \bar{X}_1 - \hat{\beta}_2 \bar{X}_2 - \hat{\beta}_3 \bar{X}_3 \\ &= 52 - (0.2063)(42) - (0.3309)(62) - (-0.5572)(200) \\ &= 52 - 8.6633 - 20.5139 + 111.4562 = 134.2789 \end{aligned}$$

(ii) The elements in the principal diagonal of  $(x'x)^{-1}$  when multiplied

$\sigma_u^2$  give the variances of the regression parameters, i.e.,

$$\begin{aligned} \text{var}(\hat{\beta}_1) &= \sigma_u^2 (0.0085) \\ \text{var}(\hat{\beta}_2) &= \sigma_u^2 (0.0027) \\ \text{var}(\hat{\beta}_3) &= \sigma_u^2 (0.0032) \end{aligned} \quad \left\{ \sigma_u^2 = \frac{\sum e_i^2}{n-k} = \frac{17.11}{6} = 2.851 \right.$$

$$\text{var}(\hat{\beta}_1) = 0.0243, \quad SE(\hat{\beta}_1) = 0.1560$$

$$\text{var}(\hat{\beta}_2) = 0.0077, \quad SE(\hat{\beta}_2) = 0.0877$$

$$\text{var}(\hat{\beta}_3) = 0.0093, \quad SE(\hat{\beta}_3) = 0.0962$$

$$(iii) \quad R^2 = \frac{\hat{\beta}' X' Y - \frac{1}{n} (\sum Y_i)^2}{Y' Y - \frac{1}{n} (\sum Y_i)^2} = \frac{\hat{\beta}_1 \sum X_1 y + \hat{\beta}_2 \sum X_2 y + \hat{\beta}_3 \sum X_3 y}{\sum y_i^2} = \frac{575.98}{594} = 0.97$$

(iv) The estimated relation may be put in the following form:

$$\hat{Y} = 134.28 + 0.2063X_1 + 0.3309X_2 - 0.5572X_3$$

|                     |          |          |          |              |
|---------------------|----------|----------|----------|--------------|
| $SE(\hat{\beta}_i)$ | (0.1560) | (0.0877) | (0.0962) | $R^2 = 0.97$ |
| $t^*$               | (1.3221) | (3.7719) | (5.7949) |              |

The variables  $X_1, X_2$  and  $X_3$  explain 97 percent of total variations.

We can test the significance of individual parameters using the student's t-test.

The computed value of 't' is given above as  $t^*$ . these values indicates us only  $\hat{\beta}_1$  is insignificant.

**Example 2.** The following matrix gives the variances and covariance of the of the three variables:

$$\begin{array}{c} y \quad x_1 \quad x_2 \\ y \left[ \begin{array}{ccc} 7.59 & 3.12 & 26.99 \end{array} \right] \\ x_1 \left| \begin{array}{cc} - & 29.16 & 30.80 \end{array} \right| \\ x_2 \left| \begin{array}{cc} - & - & 133.00 \end{array} \right| \end{array}$$

The first row and the first column of the above matrix shows  $\sum y^2$  and the first row and the second column shows  $\sum yx_{1i}$  and so on.

Consider the following model

$$Y_1 = AY_2^{\beta_1} Y_3^{\beta_2} e^{v_i}$$

Where;  $Y_1$  is food consumption per capita

$Y_2$  is food price

$Y_3$  is disposable income per capita

And  $Y = \ln Y_1$ ,  $X_1 = \ln Y_2$  and  $X_2 = \ln Y_3$

$$y = Y - \bar{Y}, x_1 = X - \bar{X}, \text{ and } x_2 = X - \bar{X}$$

Using the values in above matrix answer the following questions.

- Estimate  $\beta_1$  and  $\beta_2$
- Compute variance of  $\hat{\beta}_1$  and  $\hat{\beta}_2$
- Compute coefficient of determination
- Report the regression result.

**Solution:** It is difficult to estimate the above model as it is, to estimate the above model easily let's take the natural log of the above model;

$$\ln Y_1 = \ln A + \beta_1 \ln Y_2 + \beta_2 \ln Y_3 + V_i$$

And let:  $\beta_0 = \ln A$ ,  $Y = \ln Y_1$ ,  $X_1 = \ln Y_2$  and  $X_2 = \ln Y_3$  the above model

becomes:  $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + V_i$

The above matrix is based on the transformed model. Using values in the matrix we can now estimate the parameters of the original model.

We know that  $\hat{\beta} = (x'x)^{-1}x'y$

In the present question:

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} \text{ and } x = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \\ \vdots & \vdots \\ x_{1n} & x_{nn} \end{bmatrix}$$

$$\therefore x'x = \begin{bmatrix} \sum x_1^2 & \sum x_1 x_2 \\ \sum x_1 x_2 & \sum x_2^2 \end{bmatrix} \text{ and } x'y = \begin{bmatrix} \sum x_1 y \\ \sum x_2 y \end{bmatrix}$$

Substituting the relevant quantities from the given variance-covariance matrix, we obtain:

$$x'x = \begin{bmatrix} 29.16 & 30.80 \\ 30.80 & 133.00 \end{bmatrix} \text{ and } x'y = \begin{bmatrix} 3.12 \\ 26.99 \end{bmatrix}$$

$$|x'x| = \begin{vmatrix} 29.16 & 30.80 \\ 30.80 & 133.00 \end{vmatrix} = 2929.64$$

$$\therefore (x'x)^{-1} = \frac{1}{2929.64} = \begin{bmatrix} 133.00 & -30.80 \\ -30.80 & 29.16 \end{bmatrix} \approx \begin{bmatrix} 0.0454 & -0.0105 \\ -0.0105 & 0.0099 \end{bmatrix}$$

$$(a) \hat{\beta} = (x'x)^{-1}x'y = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix}$$

$$\begin{bmatrix} 0.0454 & -0.0105 \\ -0.0105 & 0.0099 \end{bmatrix} \begin{bmatrix} 3.12 \\ 26.99 \end{bmatrix} \approx \begin{bmatrix} -0.1421 \\ 0.2358 \end{bmatrix}$$

(b). The element in the principal diagonal of  $(x'x)^{-1}$  when multiplied by  $\sigma_u^2$  give the variances of the  $\hat{\alpha}$  and  $\hat{\beta}$ .

$$(c) \quad R^2 = \frac{\hat{\alpha}\sum x_2 y + \hat{\beta}\sum x_3 y}{\sum y_i^2}$$

$$= \frac{-(0.1421)(3.12) + (0.2358)(26.99)}{7.59}$$

$$\therefore R^2 = 0.78; \Sigma e_i^2 = (1 - R^2)(\sum y_i^2) \approx 1.6680$$

$$\therefore \sigma_u^2 = \frac{1.6680}{17} = 0.0981$$

$$\text{var}(\hat{\alpha}) = (0.0981)(0.0454) \approx 0.0045, \therefore (\hat{\alpha})SE = 0.0667$$

$$\text{var}(\hat{\beta}) = (0.0981)(0.0099) \approx 0.0009, \therefore (\hat{\beta})SE = 0.0312$$

(d). The results may be put in the following form:

$$\begin{array}{lll} \hat{Y}_1 = AY_1^{(-0.1421)} Y_3^{(0.2358)} & & \\ SE & (0.0667)(0.0312) & R^2 = 0.78 \\ t^* & (-2.13) (7.55) & \end{array}$$

The (constant) food price elasticity is negative but income elasticity is positive. Also income elasticity is highly significant. About 78 percent of the variations in the consumption of food are explained by its price and income of the consumer.

### Example 3:

Consider the model:  $Y = \alpha + \beta_1 X_{1i} + \beta_2 X_{2i} + U_i$

On the basis of the information given below answer the following question

$$\begin{array}{lll} \sum X_1^2 = 3200 & \sum X_1 X_2 = 4300 & \sum X_2 = 400 \\ \sum X_2^2 = 7300 & \sum X_1 Y = 8400 & \sum X_2 Y = 13500 \\ \sum Y = 800 & \sum X_1 = 250 & n = 25 \\ \sum Y_i^2 = 28,000 & & \end{array}$$

- Find the OLS estimate of the slope coefficient
- Compute variance of  $\hat{\beta}_2$
- Test the significant of  $\beta_2$  slope parameter at 5% level of significant
- Compute  $R^2$  and  $\bar{R}^2$  and interpret the result
- Test the overall significance of the model

**Solution:**

- Since the above model is a two explanatory variable model, we can estimate

$\hat{\beta}_1$  and  $\hat{\beta}_2$  using the formula in equation (3.21) and (3.22) i.e.

$$\hat{\beta}_1 = \frac{\sum X_2 Y \sum X_2^2 - \sum X_2 Y \sum X_1 X_2}{\sum X_1^2 \sum X_2^2 - (\sum X_1 X_2)^2}$$

$$\hat{\beta}_2 = \frac{\sum x_2 y \sum x_1^2 - \sum x_1 y \sum x_2 x_1}{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2}$$

Since the x's and y's in the above formula are in deviation form we have to find the corresponding deviation forms of the above given values.

We know that:

$$\begin{aligned}\sum x_1 x_2 &= \sum X_1 X_2 - n \bar{X}_1 \bar{X}_2 \\ &= 4300 - (25)(10)(16) \\ &= \underline{300}\end{aligned}$$

$$\begin{aligned}\sum x_1 y &= \sum X_1 Y - n \bar{X}_1 \bar{Y} \\ &= 8400 - 25(10)(32) \\ &= 400\end{aligned}$$

$$\begin{aligned}\sum x_2 y &= \sum X_2 Y - n \bar{X}_2 \bar{Y} \\ &= 13500 - 25(16)(32) \\ &= 700\end{aligned}$$

$$\begin{aligned}\sum x_1^2 &= \sum X_1^2 - n \bar{X}_1^2 \\ &= 3200 - 25(10)^2 \\ &= 700\end{aligned}$$

$$\begin{aligned}\sum x_2^2 &= \sum X_2^2 - n \bar{X}_2^2 \\ &= 7300 - 25(16)^2 \\ &= 900\end{aligned}$$

Now we can compute the parameters.

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum x_2 y \sum x_1^2 - \sum x_1 y \sum x_2 x_1}{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2} \\ &= \frac{(400)(900) - (700)(300)}{(900)(700) - (300)^2} \\ &= \underline{0.278}\end{aligned}$$

$$\hat{\beta}_2 = \frac{\sum x_2 y \sum x_1^2 - \sum x_1 y \sum x_2 x_1}{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2}$$

$$\begin{aligned}
&= \frac{(700)(700) - (400)(300)}{(900)(700) - (300)^2} \\
&= \underline{0.685}
\end{aligned}$$

The intercept parameter can be computed using the following formula.

$$\begin{aligned}
\hat{\alpha} &= \bar{Y} - \hat{\beta}_1 \bar{X}_1 - \hat{\beta}_2 \bar{X}_2 \\
&= 32 - (0.278)(10) - (0.685)(16) \\
&= \underline{18.26}
\end{aligned}$$

$$\begin{aligned}
\text{b. } \text{var}(\hat{\beta}_1) &= \frac{\hat{\sigma}^2 \sum x^2}{\sum x^2 \sum x^2 - (\sum x_1 x_2)^2} \\
&\Rightarrow \hat{\sigma}^2 = \frac{\sum e^2}{n - k} \text{ Where } k \text{ is the number of parameter}
\end{aligned}$$

In our case  $k=3$

$$\begin{aligned}
&\Rightarrow \hat{\sigma}^2 = \frac{\sum e^2}{n - 3} \\
\sum e^2 &= \sum y^2 - \hat{\beta}_1 \sum x_1 y - \hat{\beta}_2 \sum x_2 y \\
&= 2400 - 0.278(400) - (0.685)(700) \\
&= 1809.3
\end{aligned}$$

$$\begin{aligned}
\hat{\sigma}^2 &= \frac{\sum e^2}{n - 3} \\
&= \frac{1809.3}{25 - 3} \\
&= 82.24
\end{aligned}$$

$$\Rightarrow \text{var}(\hat{\beta}_1) = \frac{(82.24)(900)}{540,000} = 0.137$$

$$SE(\hat{\beta}_1) = \sqrt{\text{var}(\hat{\beta}_1)} = \sqrt{0.137} = 0.370$$

$$\begin{aligned}
\text{var}(\hat{\beta}_2) &= \frac{\hat{\sigma}^2 \sum x^2}{\sum x^2 \sum x^2 - (\sum x_1 x_2)^2} \\
&= \frac{(82.24)(700)}{540,000} = 0.1067
\end{aligned}$$

$$SE(\hat{\beta}_2) = \sqrt{\text{var}(\hat{\beta}_2)} = \sqrt{0.1067} = 0.327$$

c.  $\hat{\beta}_1$  can be tested using students t-test

This is done by comparing the computed value of t and critical value of t which is obtained from the table at  $\alpha/2$  level of significance and n-k degree of freedom.

$$\text{Hence; } \frac{t^*}{SE(\hat{\beta}_1)} = \frac{0.278}{0.370} = 0.751$$

The critical value of t from the t-table at  $\alpha/2 = 0.05/2 = 0.025$  level of significance and 22 degree of freedom is 2.074.

$$t_c = 2.074$$

$$t^* = 0.755$$

$$\Rightarrow t^* < t_c$$

The decision rule if  $t^* < t_c$  is to reject the alternative hypothesis that says  $\beta$  is different from zero and to accept the null hypothesis that says  $\beta$  is equal to zero.

The conclusion is  $\hat{\beta}_1$  is statistically insignificant or the sample we use to estimate  $\hat{\beta}_1$  is drawn from the population of Y &  $X_1$  in which there is no relationship between Y and  $X_1$  (i.e.  $\beta_1 = 0$ ).

d.  $R^2$  can be easily using the following equation

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS}$$

We know that  $RSS = \sum e_i^2$

and  $TSS = \sum y^2$  and  $ESS = \sum \hat{y}^2 = \hat{\beta}_1 \sum x_1 y + \hat{\beta}_2 \sum x_2 y + \dots + \hat{\beta}_k \sum x_k y$

For two explanatory variable model:

$$\begin{aligned} R^2 &= 1 - \frac{RSS}{TSS} = 1 - \frac{10809.3}{2400} \\ &= 0.24 \end{aligned}$$

$\Rightarrow 24\%$  of the total variation in Y is explained by the regression line  $\hat{Y} = 18.26 + 0.278X_1 + 0.685X_2$  or by the explanatory variables ( $X_1$  and  $X_2$ ).

$$\begin{aligned}\text{Adjusted } R^2 &= 1 - \frac{\Sigma e_i^2 / n - k}{\Sigma y^2 / n - 1} = 1 - \frac{(1 - R^2)(n - 1)}{n - k} \\ &= 1 - \frac{(1 - 0.24)(24)}{22} \\ &= 0.178\end{aligned}$$

e. Let's set first the joint hypothesis as

$$H_0 : \beta_1 = \beta_2 = 0$$

against  $H_1$  : at least one of the slope parameter is different from zero.

The joint test hypothesis is testing using the F-test given below.

$$\begin{aligned}F^*_{[(k-1), (n-k)]} &= \frac{ESS / k - 1}{RSS / n - k} \\ &= \frac{R^2 / k - 1}{1 - R^2 / n - k}\end{aligned}$$

From (d)  $R^2 = 0.24$  and  $k = 3$

$F^*_{(2,22)} = 3.4736$  this is the computed value of F. Let's compare this with the critical value F at 5% level of significance and (3,.23) numerator and denominator respectively. F (2,22) at 5%level of significance = 3.44.

$$F^*_{(2,22)} = 3.47$$

$$F_c(2,22)=3.44$$

$\Rightarrow F^* > F_c$ , the decision rule is to reject  $H_0$  and accept  $H_1$ . We can say that the model is significant i.e. the dependent variable is, at least, linearly related to one of the explanatory variables.

Sample exam questions :  
Mid-exam (May, 2004/2005)

- **Instructions:**

Read the following instructions carefully.

- Make sure that your exam paper contains 4 pages
- The exam has four parts. Attempt
  - ✓ All questions of part one
  - ✓ Only two questions from part two
  - ✓ One question from part three
  - ✓ And the question in part four.
- Maximum weight of the exam is 40%

**Part One: Attempt all of the following questions (15pts).**

1. Discuss briefly the goals of econometrics.
2. Researcher is using data for a sample of 10 observations to estimate the relation between consumption expenditure and income. Preliminary analysis of the sample data produces the following data.

$$\sum xy = 700, \sum x^2 = 1000, \sum X = 100$$

$$\sum Y = 200$$

Where  $x = X_i - \bar{X}_i$  and  $y = Y_i - \bar{Y}$

- a. Use the above information to compute OLS estimates of the intercept and slope coefficients and interpret the result
- b. Calculate the variance of the slope parameter
- c. Compute the value  $R^2$  (coefficient of determination) and interpret the result
- d. Compute 95% confidence interval for the slope parameter
- e. Test the significance of the slope parameter at 5% level of confidence using t-test

3. If the model  $Y_i = \alpha + \beta_1 X_{1i} + \beta_2 X_{2i} + U_i$  is to be estimated from a sample of 20 observation using the semi-processed data given in matrix in deviation form.

$$(x'x)^{-1} = \begin{pmatrix} 0.5 & -0.08 \\ -0.08 & 0.6 \end{pmatrix}$$

$$x'y = \begin{pmatrix} 100 \\ 250 \end{pmatrix} \quad \bar{X}_1 = 10, \bar{X}_2 = 25 \text{ and } \bar{Y} = 30$$

Obtain the OLS estimate of the above parameters.

4. Linearity is one assumption of classicalist in simple regression analysis. Identify which of the following satisfies this assumption. Discuss why?

- a.  $\ln Y^2 = \alpha + \beta_1 X_{1i} + \beta_2 X_{2i} + U_i$
- b.  $Y = \alpha + (1/\beta_1) X_{1i} + \beta_2 X_{2i} + U_i$
- c.  $Y = \alpha + \beta_1 X_{1i}^2 + \beta_2 X_{2i} + U_i$
- d.  $\ln Y = \alpha + \beta_1 \ln X_{1i} + \beta_2 \ln X_{2i} + U_i$
- e.  $Y_i = \alpha + \beta^2 X_i + U_i$

**Part Two: Attempt any two of the following questions.(10pts).** 1. Consider the model

$$Y_i = \alpha + \beta X_i + U_i$$

Show that the OLS estimate of  $\alpha$  is unbiased.

2. Suppose  $\sigma^2$  is the population variance of the error term and  $\hat{\sigma}^2$  is estimator of  $\sigma^2$ .

Show that  $\hat{\sigma}^2$  of maximum likelihood is the biased estimator of the true  $\sigma^2$  for the model  $Y_i = \alpha + \beta X_i + U_i$ .

3. In the model

$$Y_i = \alpha + \beta X_i + U_i$$

Show that  $\hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X}$  possesses minimum variance.

4. using the assumptions of simple regression model show that

a.  $Y \sim N(\alpha + \beta X_i, \sigma^2)$

b.  $\text{Cov}(\hat{\alpha}, \hat{\beta}) = -\bar{X} \text{Var}(\hat{\beta})$

For the model  $Y_i = \alpha + \beta X_i + U_i$

**Part Three: Attempt any one the Following(10 pts.)**

1. The model  $Y_i = \alpha + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + U_i$  is to be estimated from a sample of 20 observations. Using the information below obtain the OLS estimate of the parameters of the above model.

$$(X'X)^{-1} = \begin{bmatrix} 0.1 & -0.12 & -0.03 \\ -0.12 & 0.04 & 0.02 \\ -0.03 & 0.02 & 0.08 \end{bmatrix} \quad X'Y = \begin{bmatrix} 10,000 \\ 20,300 \\ 10,100 \\ 30,200 \end{bmatrix}$$

$$\sum X_1 = 400, \sum X_2 = 200, \text{ and } \sum X_3 = 600$$

Where  $x = X_i - \bar{X}_i$  and  $y = Y_i - \bar{Y}$

2. In a study of 100 firms, the total cost(C) was assumed to be dependent on the rate of output ( $X_1$ ) and the rate of absenteeism ( $X_2$ ). The means were:  $\bar{C} = 6$ ,  $\bar{X}_1 = 3$  and  $\bar{X}_2 = 4$ . The matrix showing sums of squares and cross products adjusted for means is

$$\begin{array}{cc|cc} & C & x_1 & x_2 \\ \hline C & 100 & 50 & 40 \\ x_1 & 50 & 50 & -70 \\ x_2 & 40 & -70 & 900 \end{array} \quad \text{where, } x_i = X_i - \bar{X}_i \text{ and } c = C_i - \bar{C}$$

Estimate the linear relation ship between C and the other two variables. (10points)

3. Consider the linear regression model

$$Y_i = \alpha + \beta_1 X_i + U_i$$

Suppose that there are no data on  $X_i$  but we have data on  $Z_i = a_0 + a_1 X_i$ , whrer  $a_0$  and  $a_1$  are arbitrary known constants. Using data of variable  $Z_i$  we can estimate

$$Y_i = c_0 + c_1 Z_i + U_i$$

Show how from the estimates of  $\hat{c}_0$  and  $\hat{c}_1$  you can obtain the estimates of the original model.

## Chapter Four

### Violations of basic Classical Assumptions

#### 4.0 Introduction

In both the simple and multiple regression models, we made important assumptions about the distribution of  $Y_t$  and the random error term ' $u_t$ '. We assumed that ' $u_t$ ' was random variable with mean zero and  $\text{var}(u_t) = \sigma^2$ , and that the errors corresponding to different observation are uncorrelated,  $\text{cov}(u_t, u_s) = 0$  and in multiple regression we assumed there is no perfect correlation between the independent variables.

Now, we address the following 'what if' questions in this chapter. What if the error variance is not constant over all observations? What if the different errors are correlated? What if the explanatory variables are correlated? We need to ask whether and when such violations of the basic classical assumptions are likely to occur. What types of data are likely to lead to heteroscedasticity (different error variance)? What type of data is likely to lead to autocorrelation (correlated errors)? What types of data are likely to lead to multicollinearity? What are the consequences such violations on least square estimators? How do we detect the presence of autocorrelation, heteroscedasticity, or multicollinearity? What are the remedial measures? How do we build an alternative model and an alternative set of assumptions when these violations exist? Do we need to develop new estimation procedures to tackle the problems? In the subsequent sections (4.1, 4.2, and 4.3), we attempt to answer such questions.

#### 4.1 Heteroscedasticity

##### 4.1.1 The nature of Heteroscedasticity

In the classical linear regression model, one of the basic assumptions is that the probability distribution of the disturbance term remains same over all observations

of  $X$ ; i.e. the variance of each  $u_i$  is the same for all the values of the explanatory variable. Symbolically,

$$\text{var}(u_i) = E[u_i - E(u_i)]^2 = E(u_i^2) = \sigma_u^2; \text{ constant value.}$$

This feature of homogeneity of variance (or constant variance) is known as homoscedasticity. It may be the case, however, that all of the disturbance terms do not have the same variance. This condition of non-constant variance or non-homogeneity of variance is known as heteroscedasticity.

### Graphical representation of heteroscedasticity and homoscedasticity

The assumption of homoscedasticity states that the variation of each  $u_i$  around its zero mean does not depend on the value of explanatory variable. The variance of each  $u_i$  remains the same irrespective of small or large values of  $X$ ; the explanatory variable. Mathematically,  $\sigma^2$  is not a function of  $X$ ;

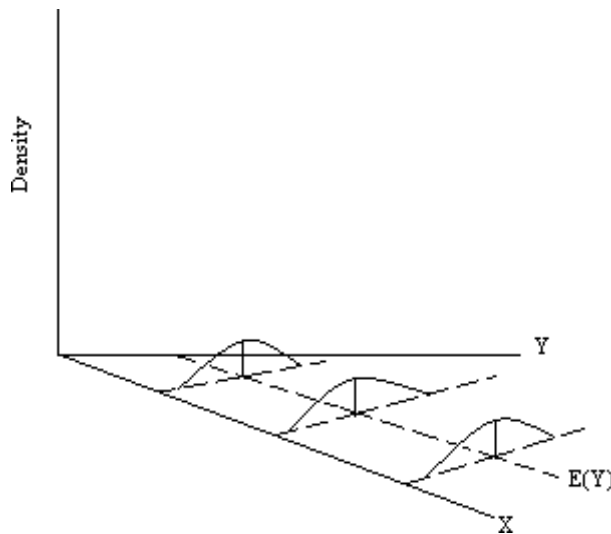


Fig. (a) Homoscedastic error variance

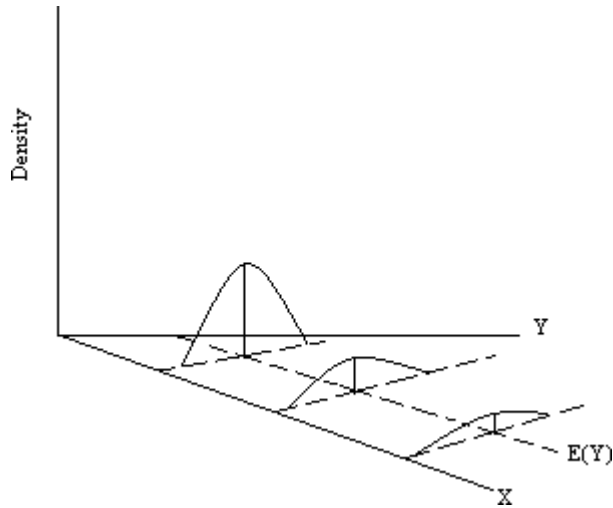
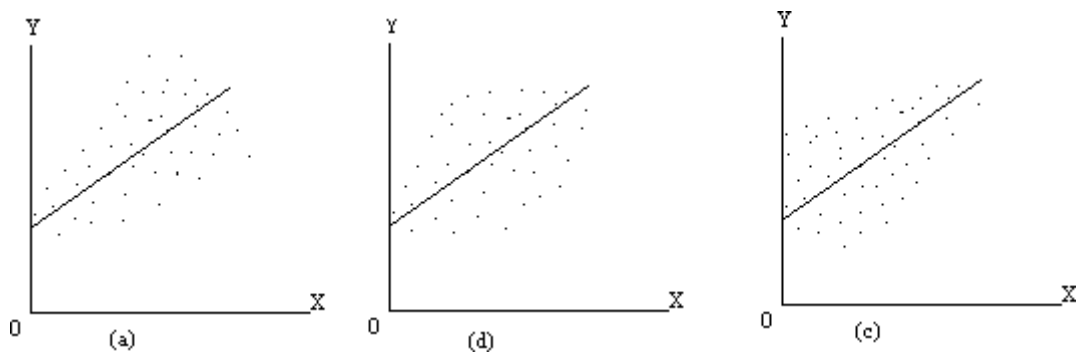


fig.(b) Heteroscedastic error variance

If the variance of  $U$  were the same at every point or for all values of  $X$ , definite restriction would be placed on the scatter of  $Y$  against  $X$  when plotted in the three dimensions, we should observe something approximating the pattern of fig (a). In contrast, consider fig (b) shows that the conditional variance of  $Y_i$  (which in fact is  $u_i$ ) increases as  $X$  increases.

$$\sigma_{ui}^2 = \sigma_u^2$$

Three cases of heteroscedasticity all shown by increasing or decreasing dispersion of the observation from the regression line.



In panel (a)  $\sigma_u^2$  seems to increase with  $X$ . in panel (b) the error variance appears greater in  $X$ 's middle range, tapering off toward the extremes. Finally, in panel

(c), the variance of the error term is greater for low values of X, declining and leveling off rapidly as X increases.

The pattern of heteroscedasticity would depend on the signs and values of the coefficients of the relationship  $\sigma_{ui}^2 = f(X_i)$ , but  $u_i$ 's are not observable. As such in applied research we make convenient assumptions that heteroscedasticity is of the forms:

- i.  $\sigma_{ui}^2 = K^2 (X_i^2)$
- ii.  $\sigma_{ui}^2 = K^2 (X_i)$
- iii.  $\sigma_{ui}^2 = \frac{K}{X_i}$  etc.

#### 4.1.2 Matrix Notation of Heteroscedasticity

The variance covariance matrix developed in chapter 2 is represented as:

$$E(UU') = \begin{bmatrix} E(U_1^2) & E(U_1U_2) & \dots & E(U_1U_n) \\ E(U_2U_1) & E(U_2^2) & \dots & E(U_2U_n) \\ \vdots & \vdots & \ddots & \vdots \\ E(U_nU_1) & E(U_nU_2) & \dots & E(U_n^2) \end{bmatrix}$$

If  $E(U_i^2) = \sigma_u^2$  (a constant value), then  $E(UU') = \sigma_u^2 I_n$  given that  $E(U_iU_j) = 0$ . If

$E(U_i^2) = \sigma_{ui}^2$  (a value that varies)

$$E(UU') = \begin{bmatrix} \sigma_{u1}^2 & 0 & \dots & 0 \\ 0 & \sigma_{u2}^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_{un}^2 \end{bmatrix}$$

.....3.10

Where  $\sigma_{ui}^2 = E(U_i^2)$ . In other words, variance covariance matrix in the present case is a diagonal matrix with unequal elements in the diagonal.

#### 4.1.4 Examples of Heteroscedastic functions

**I. Consumption Function:** Suppose we are to study the consumption expenditure from a given cross-section sample of family budgets:

$$C_i = \alpha + \beta Y_i + U_i ; \text{ where:}$$

$C_i$  = Consumption expenditure of  $i^{\text{th}}$  household;  $Y_i$  = Disposable income of  $i^{\text{th}}$  household

At low levels of income, the average consumption is low, and the variation below this level is less possible; consumption cannot fall too far below because this might mean starvation. On the other hand, it cannot rise too far above because money income does not permit it. Such constraints may not be found at higher income levels. Thus, consumption patterns are more regular at lower income levels than at higher levels. This implies that at high incomes the  $u$ 's will be high, while at low incomes the  $u$ 's will be small. The assumption of constant variance of  $u$ 's is therefore, does not hold when estimating the consumption function from across section of family budgets.

**ii. Production Function:** Suppose we are required to estimate the production function  $X = f(K, L)$  of the sugar industry from a cross-section random sample of firms of the industry. Disturbance terms in the production function would stand for many factors; like entrepreneurship, technological differences, selling and purchasing procedures, differences in organizations, etc. other than inputs, labor (L) and capital (K) considered in the production function. The factors mentioned above, which are not considered explicitly in the production function show considerable variance in large firms than in small ones. This leads to breakdown of our assumption on homogeneity of variance terms.

It should be noted that the problem of heteroscedasticity is likely to be more common in cross-sectional data than in time-series data. One deals with members of population at a given point of time, such as individual consumers or their families, firms, industries. These members may be of different size such as small, medium or large firms or low, medium or high income. In time series data on the other hand, the variables tend to be of similar orders of magnitude because one generally collects data for the same entity over a period of time.

#### 4.1.5 Reasons for Heteroscedasticity

There are several reasons why the variances of  $u_i$  may be variable. Some of these are:

1. Error learning model: it states that as people learn their error of behavior become smaller over time. In this case  $\sigma^2_i$  is expected to decrease.

*Example: as the number of hours of typing practice increases, the average number of typing errors and as well as their variance decreases.*

2. As data collection technique improves,  $\sigma^2_{ui}$  is likely to decrease. Thus banks that have sophisticated data processing equipment are likely to commit fewer errors in the monthly or quarterly statements of their customers than banks without such facilities.

3. Heteroscedasticity can also arise as a result of the presence of outliers. An outlier is an observation that is much different (either very small or very large) in relation to the other observation in the sample.

#### 4.1.6 Consequences of Heteroscedasticity for the Least Square estimates

What happens when we use ordinary least squares procedure to a model with heteroscedastic disturbance terms?

**1. The OLS estimators will have no bias**

$$\hat{\rho} = \frac{\sum xy}{\sum x^2} = \rho + \frac{\sum x_i u_i}{\sum x^2}$$

$$E(\hat{\rho}) = \rho + \frac{\sum x E(u_i)}{\sum x^2} = \rho$$

Similarly ;  $\hat{\alpha} = \bar{Y} - \hat{\rho} \bar{X} = (\alpha + \rho \bar{X} + \bar{U}) - \hat{\rho} \bar{X}$

$$E(\hat{\alpha}) = \alpha + \rho \bar{X} + E(\bar{U}) - E(\hat{\rho}) \bar{X} = \alpha$$

i.e., the least square estimators are unbiased even under the condition of heteroscedasticity. It is because we do not make use of assumption of homoscedasticity here.

**2. Variance of OLS coefficients will be incorrect**

Under homoscedasticity,  $\text{var}(\hat{p}) = \frac{\sigma^2}{\sum x_i^2} = \frac{\sigma^2}{K^2}$ , but under heteroscedastic assumption we shall have:  $\text{var}(\hat{p}) = \sum_i K_i^2 E(Y_i^2) = \sum_i K_i^2 \sigma_{ui}^2$

$\sigma_{ui}^2$  is no more a finite constant figure, but rather it tends to change with an increasing range of value of X and hence cannot be taken out of the summation (notation).

**3.OLS estimators shall be inefficient:** in other words, the OLS estimators do not have the smallest variance in the class of unbiased estimators and, therefore, they are not efficient both in small and large samples. Under the heteroscedastic assumption, therefore:

$$\text{var}(\hat{p}) = \sum_i \frac{K_i^2}{\sum x_i^2} E(Y_i^2) = \frac{\sum_i K_i^2 \sigma_{ui}^2}{(\sum x_i^2)^2} \quad \text{--- 3.11}$$

$$\text{Under homoscedasticity, } \text{var}(\hat{p}) = \frac{\sigma^2}{\sum x_i^2} \quad \text{--- 3.12}$$

These two variances are different. This implies that, under heteroscedastic assumption although the OLS estimator is unbiased, but it is inefficient. Its variance is larger than necessary.

To see the consequence of using (3.12) instead of (3.11), let us assume that:

$$\sigma_{ui}^2 = K_i \sigma^2$$

Where  $K_i$  are same non-stochastic constant weights. This assumption merely states that the heteroscedastic variances are proportional to  $K_i$ ;  $\sigma^2$  being factor of proportionality. Substituting this value of  $\sigma_{ui}^2$  in (3.11), we obtain:

$$\begin{aligned} \text{var}(\hat{p}) &= \frac{\sum_i K_i^2 \sigma^2 x_i^2}{(\sum_i x_i^2)^2} = \left| \frac{\sum_i K_i^2 x_i^2}{\sum_i x_i^2} \right| \left| \frac{\sigma^2}{\sum_i x_i^2} \right| \\ &= (\text{var}(\hat{p}))_{\text{Homo}} \left| \frac{\sum_i K_i x_i^2}{\sum_i x_i^2} \right| \quad \text{--- 3.13} \end{aligned}$$

That is to say if  $x^2$  and  $k_i$  are positively correlated and if and only if the second term of (3.13) is greater than 1, then  $\text{var}(\hat{\rho})$  under heteroscedasticity will be greater than its variance under homoscedasticity. As a result the true standard error of  $\hat{\rho}$  shall be underestimated. As such the t-value associated with it will be overestimated which might lead to the conclusion that in a specific case at hand  $\hat{\rho}$  is statistically significant (which in fact may not be true). Moreover, if we proceed with our model under false belief of homoscedasticity of the error variance, our inference and prediction about the population coefficients would be incorrect.

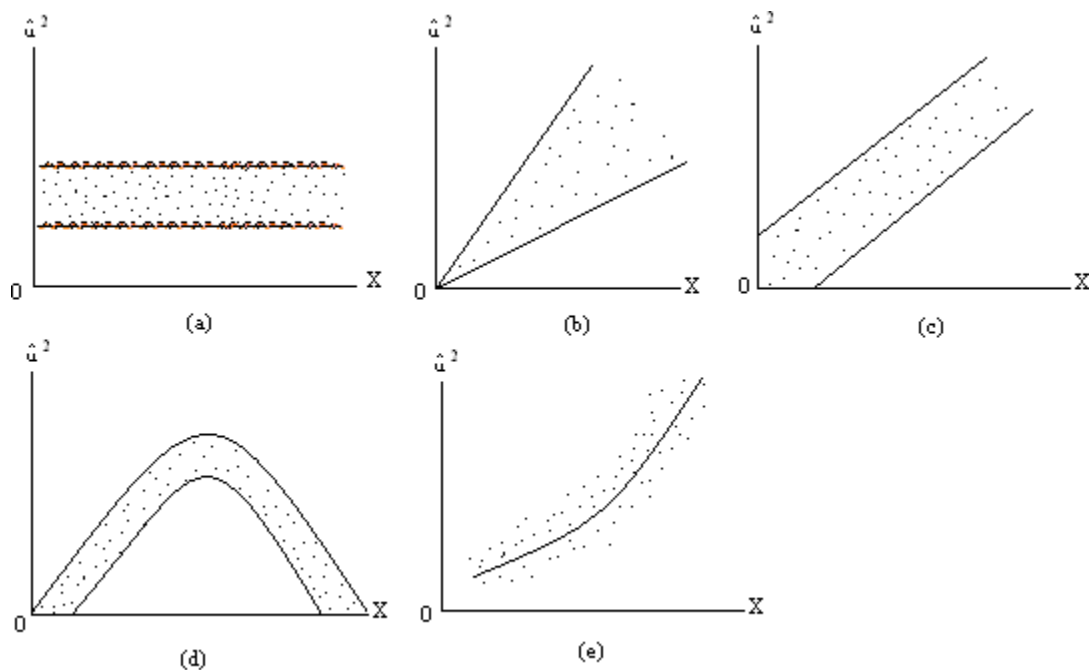
#### 4.1.7 Detecting Heteroscedasticity

We have observed that the consequences of heteroscedasticity are serious on OLS estimates. As such, it is desirable to examine whether or not the regression model is in fact homoscedastic. Hence there are two methods of testing or detecting heteroscedasticity. These are:

- i. ***Informal method***
- ii. ***Formal method***

##### i. ***Informal method***

This method is called informal because it does not undertake the formal testing procedures such as t-test, F-test and the like. It is a test based on the nature of the graph. In this method to check whether a given data exhibits heteroscedasticity or not we look on whether there is a systematic relation between residual squared  $e_i^2$  and the mean value of Y i.e.  $(\hat{Y})$  or with  $X_i$ . In the figure below  $e_i^2$  are plotted against  $\hat{Y}$  or  $(X_i)$ . In fig (a), we see there is no systematic pattern between the two variables, suggesting that perhaps no heteroscedasticity is present in the data. Figures b to e, however, exhibit definite patterns. . For instance, c suggests a linear relationship where as d and e indicate quadratic relationship between  $e_i^2$  and  $Y_i$ .



Scattergram of estimated squared residuals against X

## ii. Formal methods

There are several formal methods of testing heteroscedasticity which are based on the formal testing procedures mentioned earlier. In what follows, we will see some of the major ways of detecting heteroscedasticity.

### a. Park test

Park formalizes the graphical method by suggesting that the variance of random disturbance  $\sigma_i^2$  is some function of the explanatory variable  $X_i$ . The functional form he suggested was:  $\sigma_i^2 = \sigma^2 X_i^p e^{v_i}$

$$\text{Or } \ln \sigma_i^2 = \ln \sigma^2 + p \ln X_i + v_i \quad \text{--- 3.14}$$

where  $v$  is the stochastic disturbance term.

Since  $\sigma_i^2$  is generally not known, park suggests using  $e_i^2$  as proxy and running the following regression.

$$\ln e_i^2 = \ln \sigma^2 + p \ln X_i + v_i$$

$$\ln e_i^2 = \alpha + p \ln X_i + v_i \quad \text{--- 3.15 Since } \sigma^2 \text{ is constant.}$$

Equation (3.15) indicates on how to test hetroscedasticity by testing on whether there is a significant relation between  $X_i$  and  $e_i^2$ .

Let  $H_0 : \rho = 0$

Against  $H_1 : \rho \neq 0$

If  $\rho$  turns out to be statistically significant, it would suggest that hetroscedasticity is present in the data. If it turns out to be insignificant, we may accept the assumption of homoscedasticity. The park test is thus a two-stage test procedure; in the first stage, we run OLS regression disregarding the hetroscedasticity question. We obtain  $e_i$  from this regression and then in the second stage we run the regression in equation (3.15) above.

Example: Suppose that from a sample of size  $n=100$  we estimate the relation between compensation and productivity.

$$\begin{aligned} Y &= 1992.342 + 0.2329X_i + e_i \quad \text{--- (3.16)} \\ SE &= (936.479) \quad (0.0098) \\ t &= (2.1275) \quad (2.333) \quad R^2 = 0.4375 \end{aligned}$$

The results reveal that the estimated slope coefficient is significant at 5% level of significant on the bases of one tail t-test. The equation shows that as labour productivity increases by, say, a birr, labor compensation on the average increases by about 23 cents.

The residual obtained from regression (3.16) were regressed on  $X_i$  as suggested by equation (3.15) giving the following result.

$$\begin{aligned} \ln e_i^2 &= 35.817 - 2.8099 \ln X_i + v_i \quad \text{--- (3.17)} \\ SE &= (38.319) \quad (4.216) \\ t &= (0.934) \quad (-0.667) \quad R^2 = 0.0595 \end{aligned}$$

The above result revealed that the slope coefficient is statistically insignificant implying there is no statistically significant relationship between the two variables. Following the park test, one may conclude that there is no hetroscedasticity in the error variance. Although empirically appealing, the park test has some problems. Gold Feld and Quandt have argued that the error term  $v_i$  entering into

$\ln e_i^2 = \alpha + \beta \ln X_i + v_i$  may not satisfy the OLS assumptions and may itself be heteroscedastic. Nonetheless, as a strict explanatory method, one may use the park test.

### **b. Glejser test:**

The Glejser test is similar in spirit to the park test. After obtaining the residuals  $e_i$  from the OLS regression. Glejser suggest regressing the absolute value of  $U_i$  on the  $X_i$  variable that is thought to be closely associated with  $\sigma_i^2$ . In his experiment,

Glejser use the following functional forms:

$$\begin{array}{ll} |e_i| = \alpha + \beta X_i + v_i, & e_i = \alpha + \beta \frac{1}{\sqrt{X_i}} + v_i \\ |e_i| = \alpha + \beta \sqrt{X_i} + v_i, & |e_i| = \sqrt{\alpha + \beta X_i} + v_i \\ |e_i| = \alpha + \beta \frac{1}{X_i} + v_i, & |e_i| = \sqrt{\alpha + \beta X_i^2} + v_i \end{array} \quad ; \text{ where } v_i \text{ is error term.}$$

Goldfeld and Quandt point out that error term  $v_i$  has some problems in that its expected value is non-zero, i.e. it is serially correlated and irrorrically it is heteroscedstic. An additional difficulty with the Glejser method is that models such as:  $|e_i| = \sqrt{\alpha + \beta X_i} + v_i$  and  $|e_i| = \sqrt{\alpha + \beta X_i^2} + v_i$  are non-linear in parameters and therefore cannot be estimated with the usual OLS procedure. Glejester has found that for large samples the first four preceding models give generally satisfactory results in detecting heterosedasticity. As a practical matter, therefore, the Glejester technique may be used for large samples and may be used in small samples strictly as qualitative device to learn something about heterosedasticity.

### **c. Goldfield-Quandt test**

This popular method is applicable if one assumes that the heteroscedastic variance  $\sigma_i^2$  is positively related to one of the explanatory variables in the regression model. For simplicity, consider the usual two variable models:

$$Y_i = \alpha + \beta X_i + U_i$$

Suppose  $\sigma_i^2$  is positively related to  $X_i$  as:

$$\sigma_i^2 = \sigma^2 X_i^2 \text{ --- 3.18; where } \sigma^2 \text{ is constant.}$$

If the above equation is appropriate, it would mean  $\sigma_i^2$  would be larger, the larger values of  $X_i$ . If that turns out to be the case, heteroscedasticity is most likely to be present in the model. To test this explicitly, Goldfeld and Quandt suggest the following steps:

*Step 1: Order or rank the observations according to the values of  $X_i$  beginning with the lowest  $X$  value*

*Step 2: Omit  $C$  central observations where  $C$  is specified a priori, and divide the remaining  $(n-c)$  observations into two groups each of  $\frac{(n-c)}{2}$  observations*

*Step 3: Fit separate OLS regression to the first  $\frac{(n-c)}{2}$  observations and the last  $\frac{(n-c)}{2}$  observations, and obtain the respective residual sums of squares  $RSS_1$  and  $RSS_2$ , representing  $RSS$  from the regression corresponding to the smaller  $X_i$  values (the small variance group) and  $RSS_2$  that from the larger  $X_i$  values (the large variance group). These  $RSS$  each have*

$$\frac{(n-c)}{2} - K \text{ or } \frac{(n-c-2K)}{2} \text{ df, where: } K \text{ is the number of parameters}$$

*to be estimated, including the intercept term; and df is the degree of freedom.*

*Step 4: compute  $F = \frac{RSS_2 / df}{RSS_1 / df}$*

If  $U_i$  are assumed to be normally distributed (which we usually do), and if the assumption of homoscedasticity is valid, then it can be shown that  $F$  follows  $F$  distribution with numerator and denominator df each  $(n-c-2k)/2$ .

$$F = \frac{RSS_2 / (n-c-2K) / 2}{RSS_1 / (n-c-2K) / 2} \sim F_{\left[ \frac{(n-c)}{2} - K, \frac{(n-c)}{2} - K \right]}$$

If in application the computed  $F (=F)$  is greater than the critical  $F$  at the chosen level of significance, we can reject the hypothesis of homoscedasticity, i.e. we can say that heteroscedasticity is very likely.

*Example:* to illustrate the Goldfeld-Quandt test, we present in table 3.1 data on consumption expenditure in relation to income for a cross-section of 30 families.

Suppose we postulate that consumption expenditure is linearly related to income but that heteroscedasticity is present in the data. We further postulate that the nature of heteroscedastic is given in equation (3.15) above. The necessary reordering of the data for the application of the test is also presented in table 3.1.

Table 3.1 Hypothetical data on consumption expenditure Y(\$) and income X(\$). (Data ranked by X values)

| Y   | X   | Y   | X   |
|-----|-----|-----|-----|
| 55  | 80  | 55  | 80  |
| 65  | 100 | 70  | 85  |
| 70  | 85  | 75  | 90  |
| 80  | 110 | 65  | 100 |
| 79  | 120 | 74  | 105 |
| 84  | 115 | 80  | 110 |
| 98  | 130 | 84  | 115 |
| 95  | 140 | 79  | 120 |
| 90  | 125 | 90  | 125 |
| 75  | 90  | 98  | 130 |
| 74  | 105 | 95  | 140 |
| 110 | 160 | 108 | 145 |
| 113 | 150 | 113 | 150 |
| 125 | 165 | 110 | 160 |
| 108 | 145 | 125 | 165 |
| 115 | 180 | 115 | 180 |
| 140 | 225 | 130 | 185 |
| 120 | 200 | 135 | 190 |
| 145 | 240 | 120 | 200 |
| 130 | 185 | 140 | 205 |
| 152 | 220 | 144 | 210 |
| 144 | 210 | 152 | 220 |
| 175 | 245 | 140 | 225 |
| 180 | 260 | 137 | 230 |
| 135 | 190 | 145 | 240 |
| 140 | 205 | 175 | 245 |
| 178 | 265 | 189 | 250 |
| 191 | 270 | 180 | 260 |
| 137 | 230 | 178 | 265 |
| 189 | 250 | 191 | 270 |

Dropping the middle four observations, the OLS regression based on the first 13 and the last 13 observations and their associated sums of squares are shown next (standard errors in parentheses). Regression based on the last 13 observations

$$Y_i = 3.4094 + 0.6968X_i + e_i$$

$$(8.7049) \quad (0.0744)$$

$$R^2 = 0.8887$$

$$RSS_1 = 377.17$$

$$df = 11$$

Regression based on the last 13 observations

$$Y_i = -28.0272 + 0.7941X_i + e_i$$

$$(30.6421) \quad (0.1319)$$

$$R^2 = 0.7681$$

$$RSS_2 = 1536.8$$

$$df = 11$$

From these results we obtain:

$$F = \frac{RSS_2 / df}{RSS_1 / df} = \frac{1536.8 / 11}{377.17 / 11}$$

$$F = 4.07$$

The critical F-value for 11 numerator and 11 denominator for df at the 5% level is 2.82. Since the estimated  $F(=F)$  value exceeds the critical value, we may conclude that there is heteroscedasticity in the error variance. However, if the level of significance is fixed at 1%, we may not reject the assumption of homoscedasticity (why?) Note that the  $p$  value of the observed  $F$  is 0.014.

There are also other tests of heteroscedasticity like spearman's rank correlation test, Breusch-pagan-Goldfeldy test and white's general heteroscedastic test. But at these introductory level the above tests are enough.

#### 4.1.8 Remedial measures for the problems of heteroscedasticity

As we have seen, heteroscedasticity does not destroy the unbiasedness and consistency property of the OLS estimators, but they are no longer efficient. This lack of efficiency makes the usual hypothesis testing procedure of dubious value. Therefore, remedial measures concentrate on the variance of the error term.

*Consider the model*

$$Y = \alpha + \rho X_i + U_i, \quad \text{var}(u_i) = \sigma_i^2, \quad E(u_i) = 0 \quad E(u_i u_j) = 0$$

If we apply OLS to the above then it will result in inefficient parameters since  $\text{var}(u_i)$  is not constant.

The remedial measure is transforming the above model so that the transformed model satisfies all the assumptions of the classical regression model including homoscedasticity. Applying OLS to the transformed variables is known as the method of Generalized Least Squares (GLS). In short GLS is OLS on the

transformed variables that satisfy the standard least squares assumptions. The estimators thus obtained are known as GLS estimators, and it is these estimators that are BLUE.

#### 4.1.8.1 The Method of Generalized (Weight) Least Square

Assume that our original model is:  $Y = \alpha + \rho X_i + U_i$  where  $u_i$  satisfied all the assumptions except that  $u_i$  is heteroscedastic.

$$E(u_i)^2 = \sigma_i^2 = f(k_i)$$

If we apply OLS to the above model, the estimators are no more BLUE. To make them BLUE we have to transform the above model.

Let us assume the following types of heteroscedastic structures, under two conditions: heteroscedasticity when the population variance  $\sigma_i^2$  is known and when  $\sigma_i^2$  is not known.

**a. Assume  $\sigma_i^2$  known:**  $E(u_i)^2 = \sigma_i^2$

Given the model  $Y = \alpha + \rho X_i + U_i$

The transforming variable of the above model is  $\sqrt{\sigma_i^2} = \sigma_i$  so that the variance of the transformed error term is constant. Now divide the above model by  $\sigma_i$ .

$$\frac{Y}{\sigma_i} = \frac{\alpha}{\sigma_i} + \frac{\rho X_i}{\sigma_i} + \frac{U_i}{\sigma_i} \text{ - - - - - (3.19)}$$

The variance of the transformed error term is constant, i.e.

$$\text{var} \left| \frac{u_i}{\sigma_i} \right| = E \left| \frac{u_i}{\sigma_i} \right|^2 = \frac{1}{\sigma_i^2} E(u_i^2) = \frac{1}{\sigma_i^2} \sigma_i^2 = 1 \text{ Constant}$$

We can now apply OLS to the above model. The transformed parameters are BLUE. Because all the assumptions including homoscedasticity are satisfied to

$$(3.1). \frac{u_i}{\sigma_i} = \frac{1}{\sigma_i} (Y_i - \alpha - \rho X_i)$$

$$\left| \frac{u_i}{\sigma_i} \right|^2 = \frac{1}{\sigma_i^2} (Y_i - \alpha - \rho X_i)^2, \text{ Let } w_i = \frac{1}{\sigma_i^2}$$

$$w \hat{u}_i^2 = w (Y_i - \hat{\alpha} - \hat{\rho} X_i)^2$$

The method of GLS (WLS) minimizes the weighted residual sum of squares

$$\frac{\partial \sum w_i \hat{u}_i^2}{\partial \hat{\alpha}} = -2 \sum w_i (Y_i - \hat{\alpha} - \hat{\beta} X_i) = 0$$

$$\sum w_i (Y_i - \hat{\alpha} - \hat{\beta} X_i) = 0$$

$$\sum w_i Y_i - \hat{\alpha} \sum w_i - \hat{\beta} \sum w_i X_i = 0$$

$$\rightarrow \sum w_i Y_i - \hat{\beta} \sum w_i X_i = \hat{\alpha} \sum w_i$$

$$\hat{\alpha} = \frac{\sum w_i Y_i}{\sum w_i} - \frac{\hat{\beta} \sum w_i X_i}{\sum w_i} = \bar{Y}^* - \hat{\beta} \bar{X}^*$$

where  $\bar{Y}^*$  is the weighted mean and  $\bar{X}^*$  is the weighted mean which are different from the ordinary mean we discussed in 2.1 and 2.2.

$$2. \frac{\partial \sum w_i \hat{u}_i^2}{\partial \hat{\beta}} = -2 \sum w_i (Y_i - \hat{\alpha} - \hat{\beta} X_i) X_i = 0$$

$$\sum w_i (Y_i - \hat{\alpha} - \hat{\beta} X_i) X_i = 0$$

$$\sum w_i (Y_i X_i - \hat{\alpha} X_i - \hat{\beta} X_i^2) = 0$$

$$\sum w_i Y_i X_i - \hat{\alpha} \sum w_i X_i - \hat{\beta} \sum w_i X_i^2 = 0$$

$$\rightarrow \sum w_i Y_i X_i = \hat{\alpha} \sum w_i X_i + \hat{\beta} \sum w_i X_i^2$$

substituting  $\hat{\alpha} = \bar{Y}^* - \hat{\beta} \bar{X}^*$  in the above equation we get

$$\begin{aligned} \sum w_i Y_i X_i &= (\bar{Y}^* - \hat{\beta} \bar{X}^*) \sum w_i X_i + \hat{\beta} \sum w_i X_i^2 \\ &= \bar{Y}^* \sum w_i X_i - \hat{\beta} \bar{X}^* \sum w_i X_i + \hat{\beta} \sum w_i X_i^2 \end{aligned}$$

$$\sum w_i Y_i X_i - \bar{Y}^* \sum w_i X_i = \hat{\beta} (\sum w_i X_i^2 - \bar{X}^* \sum w_i X_i)$$

$$\sum w_i Y_i X_i - \bar{Y}^* \sum w_i X_i = \hat{\beta} (\sum w_i X_i^2 - \bar{X}^{*2} \sum w_i)$$

$$\hat{\beta} = \frac{\sum w_i Y_i X_i - \bar{Y}^* \sum w_i X_i}{\sum w_i X_i^2 - \bar{X}^{*2} \sum w_i} = \frac{\sum x_i^* y_i^*}{\sum x_i^{*2}}$$

where  $x_i^*$  and  $y_i^*$  are weighted deviations.

These parameters are now BLUE.

**b. Now assume that  $\sigma^2$  is not known**

Lets assume that  $E(u_i)^2 = \sigma_i^2 = k^2 f(X_i)$ , the transformed version of the model may be obtained by dividing through out the original model by  $\sqrt{f(X_i)}$ .

**Case a.** Suppose the heteroscedasticity is of the form

$E(u_i)^2 = \sigma_i^2 = k^2 X_i^2$ , the transforming variable is  $\sqrt{X_i^2} = X_i$  if  $Y = \alpha + \rho X_i + U_i$  where  $\text{var}(u_i) = \sigma_i^2 = K^2 X_i^2$ . The transformed model

$$\text{is: } \frac{Y}{X_i} = \frac{\alpha}{X_i} + \frac{\rho X_i}{X_i} + \frac{U_i}{X_i} = \frac{\alpha}{X_i} + \rho + \frac{U_i}{X_i}$$

$$E\left[\frac{u_i}{X_i}\right]^2 = \frac{1}{X_i^2} E(u_i^2) = \frac{K^2 X_i^2}{X_i^2} = K^2 \text{ constant}$$

which proves that the new random term in the model has a finite constant variance ( $=K^2$ ). We can, therefore, apply OLS to the transformed version of the model

$\frac{\alpha}{X_i} + \rho + \frac{U_i}{X_i}$ . Note that in this transformation the position of the coefficients has

changed: the parameter of the variable  $\frac{1}{X_i}$  in the transformed model is the constant intercept of the original model, while the constant of term of the transformed model is the parameter of the explanatory variable  $X$  in the original model. Therefore, to get back to the original model, we shall have to multiply the estimated regression by  $K_i$ .

**Case b.** Suppose the heteroscedasticity is of the form :  $E(u_i^2) = \sigma_i^2 = k^2 X_i$

The transforming variable is  $\sqrt{X_i}$

The transformed model is:  $\frac{Y}{\sqrt{X_i}} = \frac{\alpha}{\sqrt{X_i}} + \frac{\rho X_i}{\sqrt{X_i}} + \frac{U_i}{\sqrt{X_i}}$

$$= \frac{\alpha}{\sqrt{X_i}} + \rho \sqrt{X_i} + \frac{U_i}{\sqrt{X_i}}$$

$$E\left[\frac{u_i}{\sqrt{X_i}}\right]^2 = \frac{1}{X_i} E(u_i^2) = \frac{1}{X_i} k^2 X_i = k^2$$

→ Constant variance; thus we can apply OLS to the transformed model.

There is no intercept term in the transformed model. Therefore, one will have to use the ‘regression through the origin’ model to estimate  $\alpha$  and  $\rho$ . In this case, therefore, to get back to the original model, we shall have to multiply the estimated regression by  $\sqrt{X_i}$

**Case c.** suppose heteroscedasticity is of the form

$$E(u_i^2) = \sigma_i^2 = k^2 [E(Y_i)]^2$$

The transforming variable is  $\sqrt{E(u_i)^2} = E(Y_i) = \alpha + \rho X_i$

$$\frac{Y_i}{\alpha + \rho X_i} = \frac{\alpha}{\alpha + \rho X_i} + \frac{\rho X_i}{\alpha + \rho X_i} + \frac{U_i}{\alpha + \rho X_i} \text{----- (i)}$$

$$E\left[\frac{U_i}{\alpha + \rho X_i}\right]^2 = \frac{1}{(\alpha + \rho X_i)^2} E(u_i^2) = \frac{1}{[E(Y_i)]^2} K^2 [E(Y_i)]^2 = K^2$$

The transformed model described in (i) above is however not operational in this case. It is because values of  $\alpha$  and  $\rho$  are not known. But since we can obtain  $\hat{Y} = \hat{\alpha} + \hat{\rho}X_i$ , the transformation can be made through the following two steps.

**1<sup>st</sup>** : we run the usual OLS regression disregarding the heteroscedasticity problem in the data and obtain  $\hat{Y}$  using the estimated  $\hat{Y}$ , we transform the model as follows.

$$\frac{Y}{\hat{Y}} = \alpha \frac{1}{\hat{Y}} + \rho \frac{X_i}{\hat{Y}} + \frac{U_i}{\hat{Y}}$$

It should be, therefore, be clear that in order to adopt the necessary corrective measure (which is through transformation of the original data in such a way as to obtain a form in which the transformed disturbance terms possesses constant variance) we must have information on the form of heteroscedasticity. Also since our transformed data no more posses heteroscedasticity, it can be shown that the estimate of the transformed model are more efficient (i.e. they posses smaller variance) then the estimates obtained from the application of OLS to the original data.

Let's assume that a test reveals that original data possesses heteroscedasticity and that heteroscedasticity of the form  $\sigma_i^2 = K^2 X_i^2$  is being assumed.

Our original model is therefore:

$$Y_i = \alpha + \rho X_i + U_i, E(U_i) = 0, \sigma_i^2 = K^2 X_i^2$$

Apply OLS to the above heteroscedastic model

$$\hat{\rho} = \rho + \frac{\sum k_i u_i}{\sum X_i^2}$$

$$\begin{aligned} \text{var}(\hat{\rho}) &= E(\hat{\rho} - \rho)^2 = E\left[\frac{\sum k_i u_i}{\sum X_i^2}\right]^2 \\ &= \frac{E(\sum k_i^2 u_i^2 + 2 \sum X_i X_j u_i u_j)}{(\sum X_i^2)^2} \\ &= \frac{\sum X_i^2 E(u_i^2) + 2 \sum X_i X_j E(u_i u_j)}{(\sum X_i^2)^2} = \frac{K^2 \sum X_i^2 E(u_i^2)}{(\sum X_i^2)^2} \end{aligned}$$

On transforming the original model we obtain:  $\frac{Y_i}{X_i} = \rho + \frac{\alpha}{X_i} + \frac{U_i}{X_i}$

$$\hat{\rho} = \frac{\bar{Y}}{\bar{X}} - \alpha \frac{1}{\bar{X}}$$

$$\text{var}(\hat{\rho}) = \frac{\sum \left(\frac{1}{X_i}\right)^2 u_i^2}{n \left(\sum \left(\frac{1}{X_i}\right)^2 - \left(\frac{1}{\bar{X}}\right)^2\right)} = \frac{K^2 \sum \left(\frac{1}{X_i}\right)^2 u_i^2}{n \left(\sum \left(\frac{1}{X_i}\right)^2 - \left(\frac{1}{\bar{X}}\right)^2\right)}$$

Since  $\text{var}(\hat{\rho})$  in OLS =  $\frac{\sum \left(\frac{1}{X_i}\right)^2 u_i^2}{n \left(\sum \left(\frac{1}{X_i}\right)^2 - \left(\frac{1}{\bar{X}}\right)^2\right)}$

## 4.2 Autocorrelation

### 4.2.1 The nature of Autocorrelation

In our discussion of simple and multiple regression models, one of the assumptions of the classicalist is that the  $\text{cov}(u_i, u_j) = E(u_i u_j) = 0$  which implies that successive values of disturbance term  $U$  are temporarily independent, i.e. disturbance occurring at one point of observation is not related to any other disturbance. This means that when observations are made over time, the effect of disturbance occurring at one period does not carry over into another period.

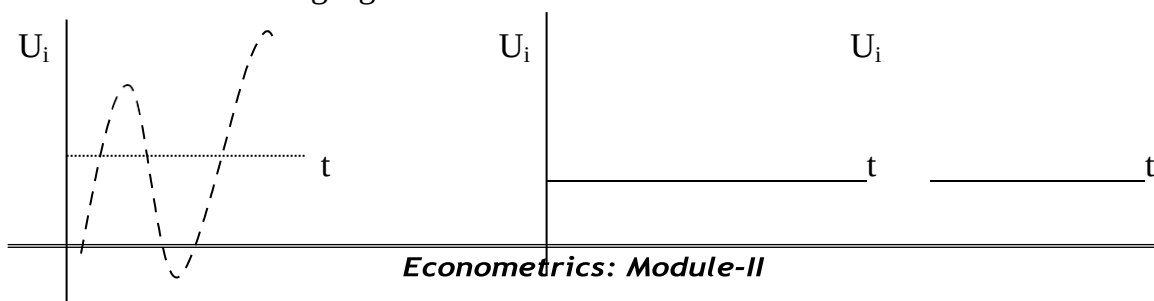
If the above assumption is not satisfied, that is, if the value of  $U$  in any particular period is correlated with its own preceding value(s), we say there is autocorrelation of the random variables. Hence, *autocorrelation is defined as a 'correlation' between members of series of observations ordered in time or space.*

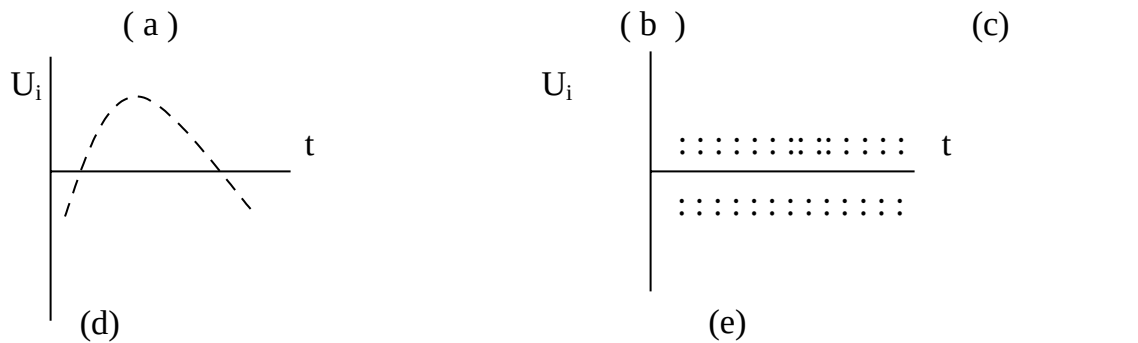
There is a difference between 'correlation' and autocorrelation. Autocorrelation is a special case of correlation which refers to the relationship between successive values of the same variable, while correlation may also refer to the relationship between two or more different variables. Autocorrelation is also sometimes called as serial correlation but some economists distinguish between these two terms. According to G.Tinner, autocorrelation is the lag correlation of a given series with itself but lagged by a number of time units. The term serial correlation is defined by him as "lag correlation between two different series." Thus, correlations between two time series such as  $u_1, u_2, \dots, u_{10}$  and  $u_2, u_3, \dots, u_{11}$ , where the former is the latter series lagged by one time period, is autocorrelation. Whereas correlation between time series such as  $u_1, u_2, \dots, u_{10}$  and  $v_2, v_2, \dots, v_{11}$  where  $U$  and  $V$  are two different time series, is called serial correlation. Although the distinction between the two terms may be useful, we shall treat these terms synonymously in our subsequent discussion.

#### **4.2.2 Graphical representation of Autocorrelation**

Since autocorrelation is correlation between members of series of observations ordered in time, we will see graphically the trend of the random variable by plotting time horizontally and the random variable ( $U_i$ ) vertically.

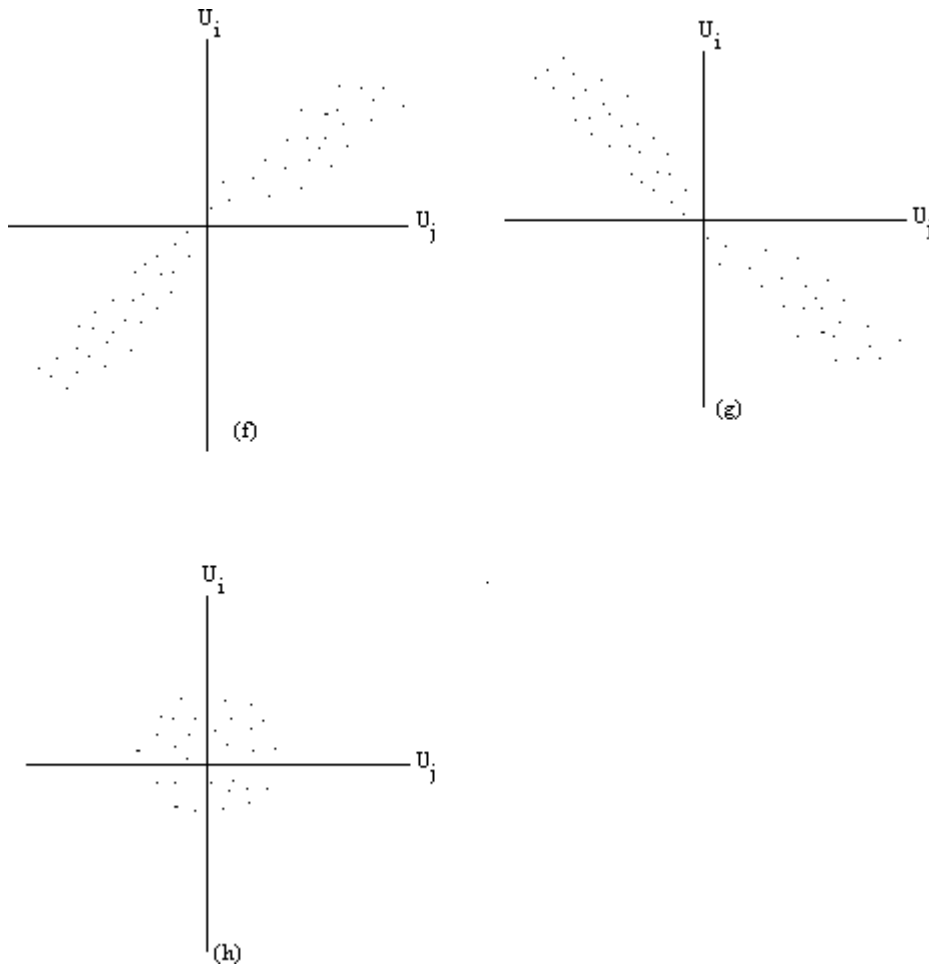
Consider the following figures





The figures (a) –(d) above, show a cyclical pattern among the  $U$ 's indicating autocorrelation i.e. figures (b) and (c) suggest an upward and downward linear trend and (d) indicates quadratic trend in the disturbance terms. Figure (e) indicates no systematic pattern supporting non-autocorrelation assumption of the classical linear regression model.

We can also show autocorrelation graphically by plotting successive values of the random disturbance term vertically ( $u_i$ ) and horizontally ( $u_j$ ).



The above figures **f** and **g** similarly indicates us positive and negative autocorrelation respectively while **h** indicates no autocorrelation.

In general, if the disturbance terms follow systematic pattern as in (f) and (g) there is autocorrelation or serial correlation and if there is no systematic pattern, this indicates no correlation.

### **42.3 Reasons for Autocorrelation**

There are several reasons why serial or autocorrelation a rises. Some of these are:

#### **a. Cyclical fluctuations**

Time series such as GNP, price index, production, employment and unemployment exhibit business cycle. Starting at the bottom of recession, when economic

recovery starts, most of these series move upward. In this upswing, the value of a series at one point in time is greater than its previous value. Thus, there is a momentum built in to them, and it continues until something happens (e.g. increase in interest rate or tax) to slowdown them. Therefore, regression involving time series data, successive observations are likely to be interdependent.

### ***b. Specification bias***

This arises because of the following.

- i. Exclusion of variables from the regression model*
- ii. Incorrect functional form of the model*
- iii. Neglecting lagged terms from the regression model*

Let's see one by one how the above specification biases causes autocorrelation.

- i. Exclusion of variables:* as we have discussed in chapter one (module I), there are several sources of the random disturbance term ( $u_i$ ). One of these is exclusion of variable(s) from the model. This error term will show a systematic change as this variable changes. For example, suppose the correct demand model is given by:

$$y_t = \alpha + p_1 x_{1t} + p_2 x_{2t} + p_3 x_{3t} + U_t \text{ - - - - - } 3.21 \quad \text{where}$$

$y$  = quantity of beef demanded,  $x_1$  = price of beef,  $x_2$  = consumer income,  $x_3$  = price of pork and  $t$  = time. Now, suppose we run the following regression in lieu of (3.21):

$$y_t = \alpha + p_1 x_{1t} + p_2 x_{2t} + V_t \text{-----} 3.22$$

Now, if equation 3.21 is the 'correct' model or true relation, running equation 3.22 is the tantamount to letting  $V_t = p_3 x_{3t} + U_t$ . And to the extent the price of pork affects the consumption of beef, the error or disturbance term  $V$  will reflect a systematic pattern, thus creating autocorrelation. A simple test of this would be to run both equation 3.21 and equation 3.22 and see whether autocorrelation, if

any, observed in equation 3.22 disappears when equation 3.21 is run. The actual mechanics of detecting autocorrelation will be discussed latter.

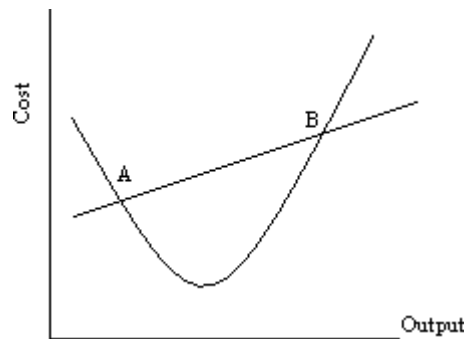
- ii. *Incorrect functional form*: This is also one source of the autocorrelation of error term. Suppose the ‘true’ or correct model in a cost-output study is as follows.

$$\text{Marginal cost} = \alpha_0 + \alpha_1 \text{output}_i + \alpha_2 \text{output}_i^2 + U_i \text{-----} 3.23$$

However, we incorrectly fit the following model.

$$\text{Marginal cost}_i = \alpha_1 + \alpha_2 \text{output}_i + V_i \text{-----} 3.24 \quad \text{The}$$

marginal cost curve corresponding to the ‘true’ model is shown in the figure below along with the ‘incorrect’ linear cost curve.



As the figure shows, between points A and B the linear marginal cost curve will consistently over estimate the true marginal cost; whereas, outside these points it will consistently underestimate the true marginal cost. This result is to be expected because the disturbance term  $V_i$  is, in fact, equal to  $(\text{output})^2 + u_i$ , and hence will catch the systematic effect of the  $(\text{output})^2$  term on the marginal cost. In this case,  $V_i$  will reflect autocorrelation because of the use of an incorrect functional form.

- iii. *Neglecting lagged term from the model*: - If the dependent variable of a certain regression model is to be affected by the lagged value of itself or the

explanatory variable and is not included in the model, the error term of the incorrect model will reflect a systematic pattern which indicates autocorrelation in the model. Suppose the correct model for consumption expenditure is:

$$C_t = \alpha + p_1 y_t + p_2 y_{t-1} + U_t \text{-----} 3.25 \quad \text{but}$$

again for some reason we incorrectly regress:

$$C_t = \alpha + p_1 y_t + V_t \text{-----} 3.26$$

As in the case in (3.21) and (3.22);  $V_t = p_2 y_{t-1} + U_t$

Hence,  $V_t$  shows systematic change reflecting autocorrelation.

#### 42.4 Matrix representation of autocorrelation

The

variance-covariance matrix of the error terms developed in chapter two (module I) is:

$$E(UU') = \begin{bmatrix} E(u_1^2) & E(u_1 u_2) & \dots & E(u_1 u_n) \\ E(u_2 u_1) & E(u_2^2) & \dots & E(u_2 u_n) \\ \vdots & \vdots & \ddots & \vdots \\ E(u_n u_1) & E(u_n u_2) & \dots & E(u_n^2) \end{bmatrix}$$

In the case of the assumption of non-autocorrelation and homoscedasticity.

$$E(UU') = \begin{bmatrix} \sigma^2 & 0 & \dots & 0 \\ 0 & \sigma^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma^2 \end{bmatrix} = \sigma^2 \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix} = \sigma^2 I_n \text{-----}$$

3.27 The assumption of no autocorrelation is responsible for the appearance of zero off- the diagonals, whereas the assumption of homoscedasticity establishes the equality of diagonal terms. The Following three examples of variance-covariance matrices help to understand the concept of autocorrelation and hetroscedasticity.

|                                                                     |                                                                                                             |                                                                                                                                 |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ | $\begin{bmatrix} 1 & \frac{1}{2} & 0 \\ \frac{1}{2} & 1 & \frac{1}{2} \\ 0 & \frac{1}{2} & 1 \end{bmatrix}$ | $\begin{bmatrix} 3 & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & 2 \end{bmatrix}$ |
| Heteroscedasticity with<br>no autocorrelation                       | Homoscedasticity<br>with autocorrelation                                                                    | Heteroscedasticity with<br>autocorrelation                                                                                      |

#### 4.2.5 The coefficient of autocorrelation

Autocorrelation, as stated earlier, is a kind of lag correlation between successive values of same variables. Thus, we treat autocorrelation in the same way as correlation in general. A simple case of linear correlation is termed here as autocorrelation of first order. In other words, if the value of  $U$  in any particular period depends on its own value in the preceding period alone, we say that  $U$ 's follow a first order autoregressive scheme AR(1) (or first order Markove scheme)

$$\text{i.e. } u_t = f(u_{t-1}) \quad \text{-----} \quad 3.28$$

If  $u_t$  depends on the values of the two previous periods, then:

$$u_t = f(u_{t-1}, u_{t-2}) \quad \text{-----} \quad 3.29 \quad \text{This}$$

form of autocorrelation is called a second order autoregressive scheme and so on.

Generally when autocorrelation is present, we assume simple first form of

autocorrelation:  $u_t = f(u_{t-1})$  and also in the linear form:

$$u_t = \rho u_{t-1} + v_t \quad \text{-----} \quad 3.30$$

where  $\rho$  the coefficient of autocorrelation and  $V$  is a random variable satisfying all the basic assumption of ordinary least square.

$$E(v) = 0, \quad E(v^2) = \sigma_v^2 \quad \text{and} \quad E(v_i v_j) = 0 \quad \text{for } i \neq j \quad \text{The}$$

above relationship states the simplest possible form of autocorrelation; if we apply OLS on the model given in ( 3.30) we obtain:

$$\hat{\rho} = \frac{\sum_{t=2}^n u_t u_{t-1}}{\sum_{t=2}^n u_{t-1}^2} \quad \text{3.31}$$

Given that for large samples:  $\sum_{t=2}^n u_t^2 \approx \sum_{t=2}^n u_{t-1}^2$ , we observe that coefficient of autocorrelation  $\hat{\rho}$  represents a simple correlation coefficient  $r$ .

$$\hat{\rho} = \frac{\sum_{t=2}^n u_t u_{t-1}}{\sum_{t=2}^n u_{t-1}^2} = \frac{\sum_{t=2}^n u_t u_{t-1}}{\sqrt{\sum_{t=2}^n u_{t-1}^2} \sqrt{\sum_{t=2}^n u_{t-1}^2}} = \frac{\sum_{t=2}^n u_t u_{t-1}}{\sqrt{\sum_{t=2}^n u_t^2 \sum_{t=2}^n u_{t-1}^2}} = r_{u_t, u_{t-1}} \text{ (Why?)} \quad \text{3.32}$$

$$\rightarrow -1 \leq \hat{\rho} \leq 1 \text{ since } -1 \leq r \leq 1 \quad \text{3.33}$$

This proves the statement “we can treat autocorrelation in the same way as correlation in general”. From our statistics background we know that:

- ✓ if the value of  $r$  is 1 we call it perfect positive correlation,
- ✓ if  $r$  is -1, perfect negative correlation and
- ✓ if the value of  $r$  is 0, there is no correlation.

By the same analogy if the value of  $\hat{\rho}$  is 1 it is called perfect positive autocorrelation, if  $\hat{\rho}$  is -1 it is called perfect negative autocorrelation and if  $\hat{\rho} = 0$ , no autocorrelation.

If  $\hat{\rho} = 0$  in  $u_t = \rho u_{t-1} + v_t$  i.e.  $u_t$  is not correlated.

#### 4.2.6 Mean, Variance and Covariance of Disturbance Terms in Autocorrelated

**Model:** To examine the consequences of autocorrelation on ordinary least square estimators, it is required to study the properties of  $U$ . If the values of  $U$  are found to be correlated with simple markove process, then it becomes:

$$U_t = \rho u_{t-1} + v_t \text{ with } |\rho| < 1$$

$v_t$  fulfilling all the usual assumptions of a disturbance term.

Our objective, here is to obtain value of  $u_t$  in terms of autocorrelation coefficient  $\rho$  and random variable  $v_t$ . The complete form of the first order autoregressive scheme may be discussed as under:

$$\begin{aligned}U_t &= f(U_{t-1}) = \rho U_{t-1} + v_t \\U_{t-1} &= f(U_{t-2}) = \rho U_{t-2} + v_{t-1} \\U_{t-2} &= f(U_{t-3}) = \rho U_{t-3} + v_{t-2} \\U_{t-r} &= f(U_{t-(r+1)}) = \rho U_{t-(r+1)} + v_{t-r}\end{aligned}$$

We make use of above relations to perform continuous substitutions in  $U_t = \rho u_{t-1} + v_t$  as follows.

$$\begin{aligned}U_t &= \rho U_{t-1} + v_t \\&= \rho(\rho U_{t-2} + v_{t-1}) + v_t, \quad u_{t-1} = \rho U_{t-2} + v_{t-1} \\&= \rho^2 U_{t-2} + \rho v_{t-1} + v_t \\&= \rho^2(\rho U_{t-3} + v_{t-3}) + (\rho v_{t-1} + v_t) \\U_t &= \rho^3 U_{t-3} + \rho^2 v_{t-3} + \rho v_{t-1} + v_t\end{aligned}$$

In this way, if we continue the substitution process for  $r$  periods (assuming that  $r$  is very large), we shall obtain:

$$U_t = v_t + \rho v_{t-1} + \rho^2 v_{t-2} + \rho^3 v_{t-3} + \dots \quad 3.35$$

$$\rho^r \rightarrow 0 \quad \text{since } |\rho| < 1$$

$$u = \sum_{r=0}^{\infty} \rho^r v_{t-r} \quad \dots \dots \dots 3.36$$

Now, using this value of  $u_t$ , let's compute its mean, variance and covariance

1. To obtain mean

$$E(U_t) = E\left[\sum_{r=0}^{\infty} \rho^r v_{t-r}\right] = \sum_{r=0}^{\infty} \rho^r E(v_{t-r}) = 0 \quad \text{since } E(v_{t-r}) = 0 \quad \dots \dots \dots 3.37$$

In other words, we found that the mean of autocorrelated  $U$ 's turns out to be zero.

2. To obtain variance

By the definition of variance

$$\begin{aligned}E(U_t^2) &= E\left[\sum_{r=0}^{\infty} \rho^r v_{t-r}\right]^2 = \sum_{r=0}^{\infty} (\rho^r)^2 E(v_{t-r})^2 = \sum_{r=0}^{\infty} (\rho^r)^2 \text{var}(v_{t-r}); \text{since} \\ \text{var}(v_{t-r}) &= E(V_{t-r})^2 \\ &= \sum_{r=0}^{\infty} \rho^{2r} = \sum_{r=0}^{\infty} (1 + \rho^2 + \rho^4 + \rho^6 + \dots \dots \dots + \rho^{2r}) = \frac{1}{1 - \rho^2}\end{aligned}$$

$$\text{var}(U_t) = \frac{\sigma^2}{(1 - \rho^2)} \text{-----} (3.38) ; \text{ Since } |\rho| \in 1$$

Thus, variance of autocorrelated  $u_i$  is  $\frac{\sigma^2}{1 - \rho^2}$  which is constant value.

From the above, the variance of  $U_i$  depends on the nature of variance of  $V_i$ . If the variance of  $V_i$  is homoscedastic,  $U_i$  is homoscedastic and if  $V_i$  is heteroscedastic,  $U_i$  is heteroscedastic.

### 3. To obtain covariance:

By the definition of covariance:

$$= E(U_t U_{t-1}) \text{-----} (3.39)$$

Since  $u_t = v_t + \rho v_{t-1} + \rho^2 v_{t-2} + \dots$

$$\rho U_{t-1} = \rho v_{t-1} + \rho^2 v_{t-2} + \rho^3 v_{t-3} + \dots$$

Substituting the above two equations in equation 3.39, we obtain

$$\text{cov}(U_t U_{t-1}) = E(v_t + \rho v_{t-1} + \rho^2 v_{t-2} + \dots)(v_{t-1} + \rho v_{t-2} + \rho^2 v_{t-3} + \dots)$$

$$= E\{v_t + \rho(v_{t-1} + \rho v_{t-2} + \dots)\}(v_{t-1} + \rho v_{t-2} + \rho^2 v_{t-3} + \dots)$$

$$= E[v_t(v_{t-1} + \rho v_{t-2} + \dots) + E(\rho(v_{t-1} + \rho v_{t-2} + \dots))^2] ; \text{ since}$$

$$E(v_t v_{t-r}) = 0$$

$$= 0 + E(\rho(v_{t-1} + \rho v_{t-2} + \dots)^2)$$

$$= E(\rho^2(v_{t-1} + \rho v_{t-2} + \dots)^2)$$

$$= \rho^2 E(v_{t-1}^2 + \rho^2 v_{t-2}^2 + \dots + 2 \text{ times cross products})$$

$$= \rho^2 (\sigma_v^2 + \rho^2 \sigma_v^2 + \dots + 0)$$

$$= \rho^2 \sigma_v^2 (1 + \rho^2 + \rho^4 + \dots)$$

$$= \frac{\rho^2 \sigma_v^2}{1 - \rho^2} \text{ since } |\rho| \in 1 \text{-----} 3.40$$

$$\rho \text{cov}(U_t, U_{t-1}) = \frac{\rho^2 \sigma_v^2}{1 - \rho^2} = \rho^2 \sigma_u^2 \text{-----} 3.41$$

$$\text{Similarly } \text{cov}(u_t, u_{t-2}) = \rho^2 \sigma_u^2 \text{-----} 3.42$$

$$\text{cov}(U_t, U_{t-3}) = \rho^3 \sigma_u^2 \text{-----} 3.43$$

and generalizing  $\text{cov}(U_t, U_{t-s}) = \sigma^2 \rho^s$  (for  $s \leq t$ ). Summarizing on the bases of the preceding discussions, we find that when  $u_t$ 's are autocorrelated, then:

$$U_t \sim N\left(0, \frac{\sigma^2}{1-\rho^2}\right) \text{ and; } E(U_t U_{t-r}) = \sigma^2 \rho^r \text{ -----3.44}$$

#### 4.2.7 Effect of Autocorrelation on OLS Estimators.

We have seen that ordinary least square technique is based on basic assumptions. Some of the basic assumptions are with respect to mean, variance and covariance of disturbance term. Naturally, therefore, if these assumptions do not hold good on what so ever account, the estimators derived by OLS procedure may not be efficient. Now, we are in a position to examine the effect of autocorrelation on OLS estimators. Following are effects on the estimators if OLS method is applied in presence of autocorrelation in the given data.

##### 1. OLS estimates are unbiased

We know that:  $\hat{p} = p + \sum k_i u_i$

$E(\hat{p}) = p + \sum k_i E(u_i) \rightarrow$  We proved  $E(u_i) = 0$  -- from (3.37). Therefore,  $E(\hat{p}) = p$

##### 2. The variance of OLS estimates is inefficient.

The variance of estimate  $\hat{p}$  in simple regression model will be biased down wards (i.e. underestimated) when  $u$ 's are auto correlated. It can be shown as follows.

We know that:  $\hat{p} = p + \sum k_i u_i$  ;  $\rightarrow \hat{p} - p = \sum k_i u_i$

$$\begin{aligned} \text{Var}(\hat{p}) &= E(\hat{p} - p)^2 = E\left(\sum k_i u_i\right)^2 \\ &= E\left(k_1 u_1 + k_2 u_2 + \dots + k_n u_n\right)^2 = E\left(k_1^2 u_1^2 + k_2^2 u_2^2 + \dots + k_n^2 u_n^2 + 2k_1 k_2 u_1 u_2 + \dots + 2k_{n-1} k_n u_{n-1} u_n\right) \\ &= E\left(\sum k_i^2 u_i^2 + 2 \sum k_i k_j u_i u_j\right) \\ &= \sum k_i^2 E(u_i^2) + 2 \sum k_i k_j E(u_i u_j) \end{aligned}$$

If  $E(u_i u_j) = 0$  which means if there is no autocorrelation, the last term disappears

so that:  $\text{var}(\hat{p}) = \sum k_i^2 = \frac{\sigma^2}{\sum x^2}$

However, we proved that  $E(u_i u_j) = 0$  but equal to  $\sigma_u^2$

$$\text{Var}(\hat{\beta}) = \frac{\sigma_u^2}{\sum x_i^2} + 2\sigma_u^2 \frac{\sum x_i x_j}{(\sum x_i^2)^2} \quad \text{In} \quad 3.45$$

the absence of autocorrelation  $\text{Var}(\hat{\beta}) = \frac{\sigma_u^2}{\sum x_i^2}$

But in the presence of autocorrelation

$$\text{var}(\hat{\beta})_{\text{auto}} = \text{var}(\hat{\beta})_{\text{nonauto}} + 2\sigma_u^2 \frac{\sum x_i x_j}{(\sum x_i^2)^2} \quad 3.46$$

-If  $\rho$  is positive and  $x$  is positively correlated

$$\text{Var}(\hat{\beta})_{\text{auto}} > \text{Var}(\hat{\beta})_{\text{nonauto}} .$$

The implication is if wrongly use  $\text{Var}(\hat{\beta}) = \frac{\sigma_u^2}{\sum x_i^2}$  while the data is autocorrelated.

$\text{var}(\hat{\beta})$  is underestimated because if the data is autocorrelated the true variance is  $\frac{\sigma_u^2}{\sum x_i^2} + 2\sigma_u^2 \frac{\sum x_i x_j}{(\sum x_i^2)^2}$  not  $\frac{\sigma_u^2}{\sum x_i^2}$ .

In the case the explanatory variable  $X$  of the model is random, the covariance of successive values is zero ( $\sum x_i x_j = 0$ ), under such circumstance the bias in  $\text{var}(\hat{\beta})$  will not be serious even though  $u$  is autocorrelated.

### 3. Wrong Testing Procedure

If  $\text{var}(\hat{\beta})$  is underestimated,  $SE(\hat{\beta})$  is also underestimated, this makes t-ratio large. This large t-ratio may make  $\hat{\beta}$  statistically significant while it may not.

4. Wrong testing procedure will make wrong prediction and inference about the characteristics of the population.

### 4.2.8 Detection (Testing) of Autocorrelation

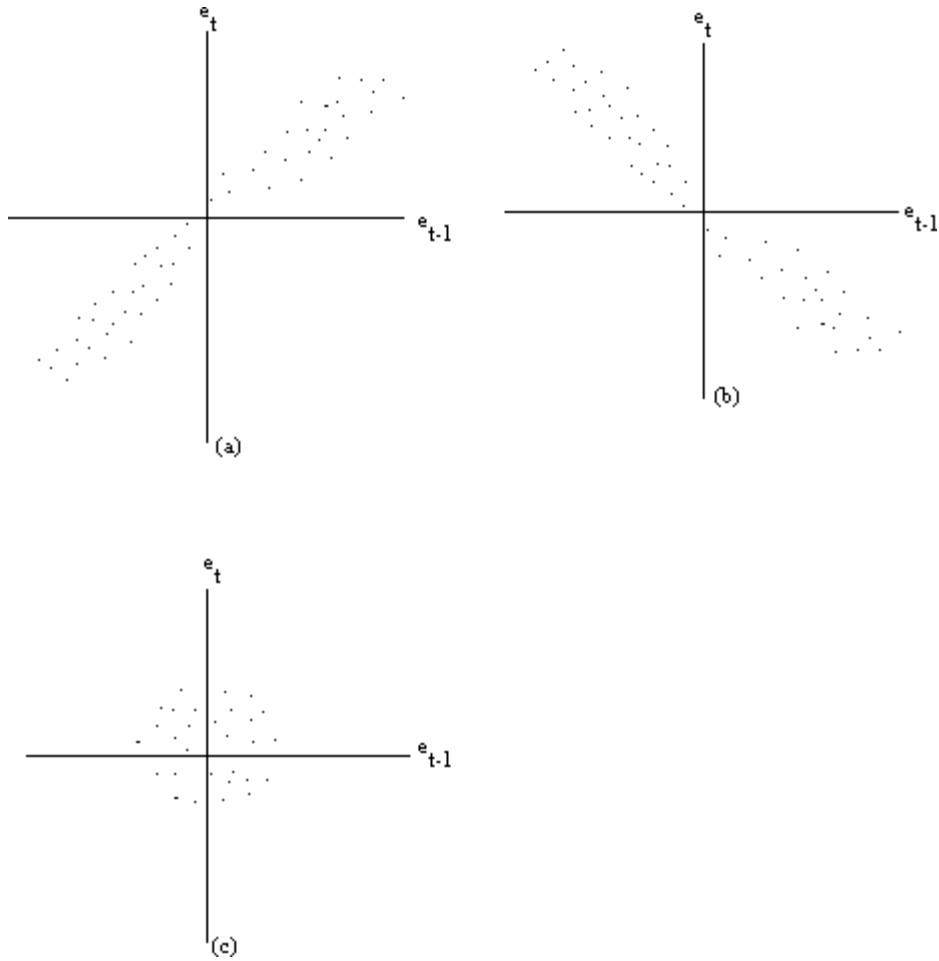
There are two methods that are commonly used to detect the existence or absence of autocorrelation in the disturbance terms. These are:

#### 1. Graphic method

Dear distance student, you recalled from section 3.2.2 that autocorrelation can be presented in graphs in two ways. Detection of autocorrelation using graphs will be based on these two ways.

Given a data of economic variables, autocorrelation can be detected in this data using graphs in the following two procedures.

- a. Apply OLS to the given data whether it is auto correlated or not and obtain the error terms. Plot  $e_t$  horizontally and  $e_{t-1}$  vertically. i.e. plot the following observations  $(e_1, e_2), (e_2, e_3), (e_3, e_4), \dots, (e_n, e_{n+1})$ . If on plotting, it is found that most of the points fall in quadrant I and III, as shown in fig (a) below, we say that the given data is autocorrelated and the type of autocorrelation is positive autocorrelation. If most of the points fall in quadrant II and IV, as shown in fig (b) below the autocorrelation is said to be negative. But if the points are scattered equally in all the quadrants as shown in fig (c) below, then we say there is no autocorrelation in the given data.



## 2. Formal testing method

This method is called formal because the testis based on the formal testing procedure you have seen in your statistics course. It is based on either the z-test, t-test, F-test or  $X^2$  test. If a test applies any of the above, it is called formal testing method. Different econometricians and statisticians suggest different types of testing methods. But, the most frequently and widely used testing methods by researchers are the following.

**A. Run test:** Before going to the detail analysis of this method, let us define what a run is in this context. *Run is the number of positive and negative signs of the*

error term arranged in sequence according to the values of the explanatory variables, like “++++++-----++++++-----+++++”

By examining how runs behave in a strictly random sequence of observations one can derive a test of randomness of runs. We ask this question: are the observed runs too many or too few compared with number of runs expected? If there are too many runs, it would mean the  $\hat{U}$ 's change sign frequently, thus indicating negative serial correlation. Similarly, if there are too few runs, they may suggest positive autocorrelation.

Now let:  $n$  = total number of observations  $= n_1 + n_2$

$n_1$  = number of + symbols;  $n_2$  = number of – symbols; and  $k$  = number of runs

Under the null hypothesis that successive outcomes (here, residuals) are independent, and assuming that  $n_1 > 10$  and  $n_2 > 10$ , the number of runs is distributed (asymptotically) normally with:

$$\text{Mean: } E(k) = \frac{2n_1n_2}{n_1 + n_2} + 1$$

$$\text{Variance: } \sigma_k^2 = \frac{2n_1n_2(2n_1n_2 - n_1 - n_2)}{(n_1 + n_2)^2 (n_1 + n_2 - 1)}$$

*Decision rule:* Do not reject the null hypothesis of randomness or independence with 95% confidence. If  $[E(k) - 1.96\sigma_k \leq k \leq 1.96\sigma_k]$ ; reject the null hypothesis if the estimated  $k$  lies outside these limits.

In a hypothetical example of  $n_1 = 14$ ,  $n_2 = 18$  and  $k = 5$  we obtain

$$E(k) = 16.75, \quad \sigma_k^2 = 7.49395 \rightarrow \sigma_k = 2.7375$$

Hence the 95% confidence interval is:

$$[16.75 \pm 1.96(2.7375)] = [11.3845, 22.1155]$$

since  $k=5$ , it clearly falls outside this interval. There fore we can reject the hypothesis that the observed sequence of residuals is random (are of independent) with 95% confidence.

**B. The Durbin-Watson  $d$  test:** The most celebrated test for detecting serial correlation is one that is developed by statisticians Durbin and Waston. It is popularly known as the Durbin-Waston  $d$  statistic, which is defined as:

$$d = \frac{\sum_{t=2}^{t=n} (e_t - e_{t-1})^2}{\sum_{t=1}^{t=n} e^2_t} \quad \text{3.47}$$

Note that, in the numerator of  $d$  statistic the number of observations is  $n - 1$  because one observation is lost in taking successive differences.

It is important to note the assumptions underlying the  $d$ -statistics

1. The regression model includes an intercept term. If such term is not present, as in the case of the regression through the origin, it is essential to rerun the regression including the intercept term to obtain the RSS.
2. The explanatory variables, the  $X$ 's, are non-stochastic, or fixed in repeated sampling.
3. The disturbances  $U_t$  are generated by the first order auto regressive scheme:  $V_t = \rho u_{t-1} + \varepsilon_t$
4. The regression model does not include lagged value of  $Y$  the dependent variable as one of the explanatory variables. Thus, the test is inapplicable to models of the following type

$$y_t = p_1 + p_2 X_{2t} + p_3 X_{3t} + \dots + p_k X_{kt} + \rho y_{t-1} + U_t$$

Where  $y_{t-1}$  the one period lagged value of  $y$  is such models are known as autoregressive models. If  $d$ -test is applied mistakenly, the value of  $d$  in such

cases will often be around 2, which is the value of  $d$  in the absence of first order autocorrelation. Durbin developed the so-called *h-statistic* to test serial correlation in such autoregressive.

5. There are no missing observations in the data.

In using the Durbin –Watson test, it is, there fore, to note that it can not be applied in violation of any of the above five assumptions.

Dear distance student, from equation 3.47 the value of  $d = \frac{\sum_{t=2}^{t=n} (e_t - e_{t-1})^2}{\sum_{t=1}^{t=n} e_t^2}$

Squaring the numerator of the above equation, we obtain

$$d = \frac{\sum_{t=2}^n e_t^2 + \sum_{t=1}^n e_{t-1}^2 - 2 \sum_{t=2}^n e_t e_{t-1}}{\sum_{t=1}^n e_t^2} \quad 3.48$$

However, for large samples  $\sum_{t=2}^n e_t^2 \approx \sum_{t=1}^n e_t^2$  because in both cases one observation is lost. Thus,

$$d = \frac{2 \sum_{t=1}^n e_t^2 - 2 \sum_{t=1}^n e_t e_{t-1}}{\sum_{t=1}^n e_t^2} = 2 \left( 1 - \frac{\sum_{t=1}^n e_t e_{t-1}}{\sum_{t=1}^n e_t^2} \right)$$

$$d = 2 \left( 1 - \frac{\sum_{t=1}^n e_t e_{t-1}}{\sum_{t=1}^n e_t^2} \right)$$

but  $\frac{\sum_{t=1}^n e_t e_{t-1}}{\sum_{t=1}^n e_t^2}$  from equation

$$d = 2(1 - \hat{\rho})$$

From the above relation, therefore

$$\begin{array}{l} \hat{\rho} = 0, \quad d \approx 2 \\ \text{if } \hat{\rho} = 1, \quad d \approx 0 \\ \hat{\rho} = -1, \quad d \approx 4 \end{array}$$

Thus we obtain two important conclusions

- i. Values of  $d$  lies between 0 and 4
- ii. If there is no autocorrelation  $\hat{\rho} = 0$ , then  $d = 2$

Whenever, therefore, the calculated value of  $d$  turns out to be sufficiently close to 2, we accept null hypothesis, and if it is close to zero or four, we reject the null hypothesis that there is no autocorrelation.

However, because the exact value of  $d$  is never known, there exist ranges of values with in which we can either accept or reject null hypothesis. We do not also have unique critical value of  $d$ -stastics. We have  $d_L$  -lower bound and  $d_u$  upper bound of he initial values of  $d$  to accept or reject the null hypothesis.

For the two-tailed Durbin Watson test, we have set five regions to the values of  $d$  as depicted in the figure below.

The mechanisms of the D.W test are as follows, assuming that the assumptions underlying the tests are fulfilled.

- Run the OLS regression and obtain the residuals
- Obtain the computed value of  $d$  using the formula given in equation 3.47
- For the given sample size and given number of explanatory variables, find out critical  $d_L$  and  $d_u$  values.
- Now follow the decision rules given below.

1. If  $d$  is less than  $d_L$  or greater than  $(4 - d_L)$  we reject the null hypothesis of no autocorrelation in favor of the alternative which implies existence of autocorrelation.
2. If  $d$  lies between  $d_U$  and  $(4 - d_U)$ , accept the null hypothesis of no autocorrelation
3. If however the value of  $d$  lies between  $d_L$  and  $d_U$  or between  $(4 - d_U)$  and  $(4 - d_L)$ , the D.W test is inconclusive.

*Example 1.* Suppose for a hypothetical model  $Y = \alpha + \rho X + U_i$ , if we found

$$d = 0.1380; \quad d_L = 1.37; \quad d_U = 1.50$$

Based on the above values test for autocorrelation

*Solution:* First compute  $(4 - d_L)$  and  $(4 - d_U)$  and compare the computed value

of  $d$  with  $d_L, d_U, (4 - d_L)$  and  $(4 - d_U)$

$$(4 - d_L) = 4 - 1.37 = 2.63$$

$$(4 - d_U) = 4 - 1.5 = 2.50$$

Since  $d$  is less than  $d_L$  we reject the null hypothesis of no autocorrelation

*Example 2.* Consider the model  $Y_t = \alpha + \rho X_t + U_t$  with the following observation on  $X$  and  $Y$

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|
| X | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9  | 10 | 11 | 12 | 13 | 14 | 15 |
| Y | 2 | 2 | 2 | 1 | 3 | 5 | 6 | 6 | 10 | 10 | 10 | 12 | 15 | 10 | 11 |

Test for autocorrelation using Durbin -Watson method

*Solution:*

1. regress  $Y$  on  $X$ : i.e.  $Y_t = \alpha + \rho X_t + U_t$ :

From the above table we can compute the following values.

$$\begin{aligned} \sum xy &= 255, & \bar{Y} &= 7, & \sum (e_i - e_{i-1})^2 &= 60.21 \\ \sum x^2 &= 280, & \bar{X} &= 8, & \sum e_i^2 &= 41.767 \\ \sum y^2 &= 274 \end{aligned}$$

$$\hat{\rho} = \frac{\sum xy}{\sum x^2} = \frac{255}{280} = 0.91$$

$$\hat{\alpha} = \bar{Y} - \hat{\rho}\bar{X} = 7 - 0.91(8) = -0.29$$

$$Y = -0.29 + 0.91X + U_i$$

$$\hat{Y} = 0.28 + 0.91X, \quad R^2 = 0.85$$

$$d = \frac{\sum (e - \bar{e})^2}{\sum e_i^2} = \frac{60.213}{41.767} = 1.442$$

Values of  $d_L$  and  $d_U$  on 5% level of significance, with  $n=15$  and one explanatory variable are:  $d_L=1.08$  and  $d_U=1.36$ .

$$(4 - d_U) = 2.64$$

$$d_U \in d \in 4 - d_U = (1.364 \ 2.64)$$

$$d^* = 1.442$$

Since  $d^*$  lies between  $d_U \in d \in 4 - d_U$ , accept  $H_0$ . This implies the data is autocorrelated.

Although D.W test is extremely popular, the d test has one great drawback in that

if it falls in the inconclusive zone or region, one cannot conclude whether autocorrelation does or does not exist. Several authors have proposed modifications of the D.W test.

In many situations, however, it has been found that the upper limit  $d_U$  is approximately the true significance limit. Thus, the modified DW test is based on  $d_U$  in case the estimated d value lies in the inconclusive zone, one can use the following modified d test procedure. Given the level of significance  $\alpha$  ; if

1.  $\rho = 0$  versus  $H_1 : \rho > 0$  if the estimated  $d \in d_U$ , reject  $H_0$  at level  $\alpha$  , that is there is statistically significant positive correlation.

2.  $H_0 : \rho = 0$  versus  $H_1 : \rho \neq 0$  if the estimated  $d \in d_U$  or  $(4 - d_u) \in d_U$  reject  $H_0$  at level  $2\alpha$  statistically there is significant evidence of autocorrelation, positive or negative.

#### 4.2.9 Remedial Measures for the problems of Autocorrelation

Since in the presence of serial correlation the OLS estimators are inefficient, it is essential to seek remedial measures. The remedy however depends on what knowledge one has about the nature of interdependence among the disturbances. : This means the remedy depends on whether the coefficient of autocorrelation is known or not known.

**A. when  $\rho$  is known-** When the structure of autocorrelation is known, i.e  $\rho$  is known, the appropriate corrective procedure is to transform the original model or data so that error term of the transformed model or data is non auto correlated. When we transform, we are wippiy of the effect of  $\rho$  .

Suppose that our model is

$$Y_t = \alpha + \rho X_t + U_t \dots\dots\dots 3.49$$

$$\text{and } U_t = \rho U_{t-1} + V_t, \quad |\rho| \in 1 - - - 3.50$$

Equation (3.50) indicates the existence of autocorrelation. If  $\rho$  is known, then transform Equation (3.49) into one that is not autocorrelated. The procedure of transformation will be given below.

Take the lagged form of equation (1) and multiply through by  $\rho$  ..

$$\rho_{y_{t-1}} = \rho\alpha + \rho\rho X_{t-1} + \rho U_{t-1} - - - - - 3.51$$

Subtracting (3) from (1), we have:

$$Y_t - \rho Y_{t-1} = (\alpha - \rho\alpha) + (\rho X_t - \rho\rho X_{t-1}) + (U_t - \rho U_{t-1}) - - - - - 3.52$$

By rearranging the terms in (3.50), we have

$$V_t = U_t - \rho U_{t-1}$$

which on substituting the last term of (3.52) gives

$$Y_t - \rho Y_{t-1} = (\alpha - \rho\alpha) + \rho(X_t - \rho X_{t-1}) + v_t \quad (3.53)$$

$$\text{Let: } Y_t^* = Y_t - \rho Y_{t-1}$$

$$a = \alpha - \rho\alpha = \alpha(1 - \rho)$$

$$X_t^* = X_t - \rho X_{t-1}$$

Equation (3.53) may be written as:

$$Y_t^* = a + \rho X_t^* + v_t \quad (3.54)$$

It may be noted that in transforming Equation (3.49) into (3.54) one observation shall be lost because of lagging and subtracting in (3.52). We can apply OLS to the transformed relation in (3.54) to obtain  $\hat{\alpha}$  and  $\hat{\rho}$  for our two parameters  $\alpha$  and  $\rho$ .

$$\hat{\alpha} = \frac{\hat{a}}{1 - \rho} \quad \text{and it can be shown that}$$

$$\text{var } \hat{\alpha} = \frac{1}{1 - \rho^2} \text{var}(\hat{a})$$

Because  $\hat{\alpha}$  is perfectly and linearly related to  $\hat{a}$ . Again since  $v_t$  satisfies all standards assumptions, the variance of  $\hat{\alpha}$  and  $\hat{\rho}$  would be given by our standard OLS formulae.

$$\text{var}(\hat{\alpha}) = \frac{\sum u^2 X_t^{*2}}{n \sum (X_t^* - \bar{X}^*)^2}, \quad \text{var}(\hat{\rho}) = \frac{\sum u^2}{n \sum (X_t^* - \bar{X}^*)^2}$$

Estimators obtained in equation 6 are efficient, only if our sample size is large so that loss of one observation becomes negligible.

### B. When $\rho$ is not known

When  $\rho$  is not known, we will describe the methods through which the coefficient of autocorrelation can be estimated.

#### Method I: A priori information on $\rho$

Many times an investigator makes some reasonable guess about the value of autoregressive coefficient by using his knowledge or institution about the relationship under study. Many researchers usually assume that  $\rho = 1$  or  $-1$ .

Under this method, the process of transformation is the same as when  $\rho$  is known.

When  $\rho = 1$ , the transformed model becomes;

$$(Y_t - Y_{t-1}) = (X_t - X_{t-1}) + V_t; \text{ where } V_t = U_t - U_{t-1}$$

Note that the constant term is suppressed in the above.  $\hat{B}$  is obtained by taking merely the first differences of the variable and obtaining line that passes through the origin. Suppose that one assumes  $\rho = -1$  instead of  $\rho = 1$ , i.e the case of perfect negative autocorrelation. In such a case, the transformed model becomes:

$$Y_t + Y_{t-1} = 2\alpha + \rho(X_t + X_{t-1}) + v_t \quad \text{Or} \quad \frac{Y_t + Y_{t-1}}{2} = \alpha + \rho \frac{(X_t + X_{t-1})}{2} + \frac{v_t}{2}$$

This model is then called two period moving average regression model because actually we are regressing the value of one moving average  $\left[ \frac{Y_t + Y_{t-1}}{2} \right]$  on another

$$\left[ \frac{(X_t + X_{t-1})}{2} \right]$$

This method of first difference is quite popular in applied research for its simplicity. But the method rests on the assumption that either there is perfect positive or perfect negative autocorrelation in the data.

**Method II: Estimation of  $\rho$  from d-statistic:** From equation ( 3.47 ), we obtained  $d \approx 2(1 - \hat{\rho})$ . Suppose we calculate certain value of d-statistic in the case

of certain data. Given the d-value we can estimate  $\rho$  from this.

$$\begin{aligned} d &\approx 2(1 - \hat{\rho}) \\ \rightarrow \hat{\rho} &\approx 1 - \frac{1}{2}d \end{aligned}$$

As already pointed out,  $\hat{\rho}$  will not be accurate if the sample size is small. The above relationship is true only for large samples. For small samples, Theil and Nagar have suggested the following relation:

$$\hat{\rho} = \frac{n^2(1 - \frac{d}{2}) + k^2}{n^2 - k^2} \dots\dots\dots 3.55$$

where n=total number of observation; d= Durbin Watson statistic ; k=number of coefficients (including intercept term). Using this value of  $\hat{\rho}$  we can perform the above transformation to avoid autocorrelation from the model.

**Method III: The Cochrane-Orcutt iterative procedure:** In this method, we remove autocorrelation gradually starting from the simplest form of a first order scheme. First we obtain the residuals and apply OLS to them;

$$e_t = \rho e_{t-1} + v_t \dots\dots\dots 3.56$$

We estimate  $\hat{\rho}$  from the above relation. With the estimated  $\hat{\rho}$ , we transform the original data and then apply OLS to the model.

$$(Y_t - \hat{\rho}Y_{t-1}) = \alpha(1 - \hat{\rho}) + \rho(X_t - \hat{\rho}X_{t-1}) + V_t - \hat{\rho}u_{t-1} \dots\dots\dots 3.57$$

we once again apply OLS to the newly obtained residuals

$$e_t^* = \rho e_{t-1}^* + w_t \dots\dots\dots 3.58$$

We use this second estimate  $\hat{\rho}$  to transform the original observations and so on we keep proceeding until the value of the estimate of  $\rho$  converges. It can be shown that the procedure is convergent. When the data is transformed only by using this second stage estimate of  $\rho$ , it is then called two stages Cochrane-Orcutt

method. However one can follow an alternative approach to use at each step of interaction, the Durbin Watson d-statistic to residuals for autocorrelation or till the estimates of  $\rho$  do not differ substantially from one another.

**Method IV: Durbin's two-stage method:** Assuming the first order autoregressive scheme, Durbin suggests a two-stage procedure for resolving the serial correlation problem. The steps under this method are:

$$\text{Given } Y_t = \alpha + \rho X_t + u_t \dots\dots\dots (3.59)$$

$$U_t = \rho U_{t-1} + v_t$$

1. Take the lagged term of the above and multiply by  $\rho$

$$\rho Y_{t-1} = \rho \alpha + \rho \rho X_{t-1} + \rho u_{t-1} \dots\dots\dots (3.60)$$

2. Subtract (3.60) from (3.59)

$$Y_t - \rho Y_{t-1} = \alpha(1 - \rho) + \rho(X_t - \rho X_{t-1}) + u_t - \rho u_{t-1} \dots\dots\dots (3.61)$$

3. Rewrite (3.61) in the following form

$$Y_t = \alpha(1 - \rho) + \rho Y_{t-1} + \rho X_t - \rho \rho X_{t-1} + v_t$$

$$Y_t = \alpha^* + \rho Y_{t-1} + \rho X_t - \rho \rho X_{t-1} + v_t$$

This equation is now treated as regression equation with three explanatory variables  $X_t, X_{t-1}$  and  $Y_{t-1}$ . This provides estimate of  $\rho$  which is used to construct new variables  $(Y_t - \hat{Y}_{t-1})$  and  $(X_t - \hat{X}_{t-1})$ . In the second step, estimators of  $\alpha$  and  $\rho$  are obtained from the regression equation:

$$(Y_t - \hat{Y}_{t-1}) = \alpha^* + \rho (X_t - \hat{X}_{t-1}) + u_t^* ; \text{ where } \alpha^* = \alpha(1 - \rho)$$

## 4.3 Multicollinearity

### 4.3.1 The nature of Multicollinearity

Originally, multicollinearity meant the existence of a “perfect” or exact, linear relationship among some or all explanatory variables of a regression model. For k-

variable regression involving explanatory variables  $x_1, x_2, \dots, x_k$ , an exact linear relationship is said to exist if the following condition is satisfied.

$$\beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + v_i = 0 \quad - - - - - (1)$$

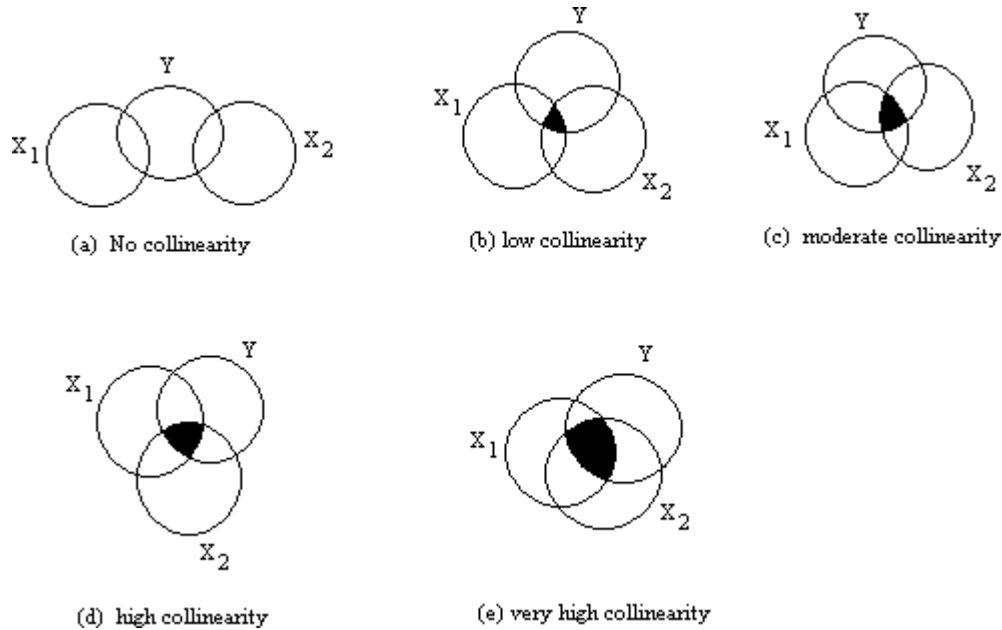
where  $\beta_1, \beta_2, \dots, \beta_k$  are constants such that not all of them are simultaneously zero.

Today, however, the term multicollinearity is used in a broader sense to include the case of perfect multicollinearity as shown by (1) as well as the case where the  $x$ -variables are inter-correlated but not perfectly so as follows

$$\beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + v_i = 0 \quad - - - - - (1)$$

where  $v_i$  is the stochastic error term.

The nature of multicollinearity can be illustrated using the figures below. Let, in the figures  $y, x_1$ , &  $x_2$  and represent respectively the variation in  $y$  (the dependent variable) and  $x_1$  and  $x_2$  (explanatory variables). The degree of collinearity can be measured by the extent of overlap (shaded area) of the  $x_1$  and  $x_2$ . In the fig.(a) below there is no overlap between  $x_1$  and  $x_2$  and hence no collinearity. In figs. 'b' through 'e', there is "low" to "high" degree of collinearity. In the extreme if  $x_1$  and  $x_2$  were to overlap completely (or if  $x_1$  is completely inside  $x_2$ , or vice versa), Collinearity would be perfect.



Note that: multicollinearity refers only to linear relationships among the x-variables. It does not rule out non-linear relationships among the x-variables.

For example:  $Y = \alpha + \rho_1 x_i + \rho_2 x_i^2 + \rho_3 x_i^3 + v_i$  - - - - - (3.31)

Where: Y-Total cost and X-output.

The variables  $x_i^2$  and  $x_i^3$  are obviously functionally related to  $x_i$  but the relationship is non-linear. Strictly, therefore, models such (3.31) do not violate the assumption of no multicollinearity. However, in concrete applications, the conventionally measured correlation coefficient will show  $x_i, x_i^2$  and  $x_i^3$  to be highly correlated, which as we shall show, will make it difficult to estimate the parameters with greater precision (i.e. with smaller standard errors).

#### 4.3.2 Reasons for Multicollinearity

1. *The data collection method employed:* Example: If we regress on small sample values of the population; there may be multicollinearity but if we take all the possible values, it may not show multicollinearity.
2. *Constraint over the model or in the population being sampled.*

For example: in the regression of electricity consumption on income ( $x_1$ ) and house size ( $x_2$ ), there is a physical constraint in the population in that families with higher income generally have larger homes than with lower incomes.

3. *Overdetermined model*: This happens when the model has more explanatory variables than the number of observations. This could happen in medical research where there may be a small number of patients about whom information is collected on a large number of variables.

### 4.3.3 Consequences of Multicollinearity

Why does the classical linear regression model put the assumption of no multicollinearity among the X's? It is because of the following consequences of multicollinearity on OLS estimators.

1. *If multicollinearity is perfect, the regression coefficients of the X variables are indeterminate and their standard errors are infinite.*

**Proof:** - Consider a multiple regression model with two explanatory variables, where the dependent and independent variables are given in deviation form as follows.  $y_i = \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + e_i$

Dear distance student, do you recall the formulas of  $\hat{\beta}_1$  and  $\hat{\beta}_2$  from our discussion of multiple regression?

$$\hat{\beta}_1 = \frac{\sum x_i y_i \sum x_i^2 - \sum x_i y_i \sum x_i x_i}{\sum x_i^2 \sum x_i^2 - (\sum x_i x_i)^2}$$

$$\hat{\beta}_2 = \frac{\sum x_i y_i \sum x_i^2 - \sum x_i y_i \sum x_i x_i}{\sum x_i^2 \sum x_i^2 - (\sum x_i x_i)^2}$$

Assume  $x_2 = \lambda x_1$  ..... 3.32

Where  $\lambda$  is non-zero constants. Substitute 3.32 in the above  $\hat{\beta}_1$  and  $\hat{\beta}_2$  formula:

$$\hat{\beta}_1 = \frac{\sum x_i y_i (\lambda x_i)^2 - \sum x_i y_i \sum x_i \lambda x_i}{\sum x_i^2 (\lambda x_i)^2 - (\sum x_i \lambda x_i)^2}$$

$$= \frac{\sum x y \sum x^2 - \sum x y \sum x^2}{\sum x^2 (\sum x^2)^2 - (\sum x x)^2} = \frac{0}{0} \rightarrow \text{indeterminate.}$$

Applying the same procedure, we obtain similar result (indeterminate value) for  $\hat{p}_2$ . Likewise, from our discussion of multiple regression model, variance of  $\hat{p}_1$

$$\text{is given by : } \text{var}(\hat{p}_1) = \frac{\sum x^2 \sum x^2}{\sum x^2 \sum x^2 - (\sum x x)^2}$$

Substituting  $x_2 = x_1$  in the above variance formula, we get:

$$= \frac{\sum x^2 \sum x^2}{\sum x^2 (\sum x^2)^2 - \sum x^2 (\sum x^2)^2} = \frac{\sum x^2 \sum x^2}{0} = \infty \rightarrow \text{infinite.}$$

These are the consequences of perfect multicollinearity. One may raise the question on consequences of less than perfect correlation. In cases of near or high multicollinearity, one is likely to encounter the following consequences.

**2. If multicollinearity is less than perfect (i.e near or high multicollinearity), the regression coefficients are determinate**

**Proof:** Consider the two explanatory variables model above in deviation form.

If we assume  $x_2 = x_1$  it indicates us perfect correlation between  $x_1$  and  $x_2$  because the change in  $x_2$  is completely because of the change in  $x_1$ . Instead of exact multicollinearity, we may have:  $x_{2i} = x_{1i} + v_i$  Where  $\sum v_i = 0$ ,  $v_i$  is stochastic error term such that  $\sum x_i v_i = 0$ . In this case  $x_2$  is not only determined by  $x_1$ , but also affected by some other variables given by  $v_i$  (stochastic error term).

Substitute  $x_2 = x_{1i} + v_i$  in the formula of  $\hat{p}_1$  above

$$\hat{p}_1 = \frac{\sum x y \sum x^2 - \sum x y \sum x x}{\sum x^2 \sum x^2 - (\sum x x)^2}$$

$$= \frac{\sum y^2 \sum x^2 + \sum v^2 - (\sum y x + \sum y v) \sum x^2}{\sum x^2 (\sum x^2 + \sum v^2) - (\sum x^2)^2} \rightarrow 0$$

determinate.

This proves that if we have less than perfect multicollinearity the OLS coefficients are determinate.

The implication of indetermination of regression coefficients in the case of perfect multicollinearity is that it is not possible to observe the separate influence of  $x_1$  and  $x_2$ . But such extreme case is not very frequent in practical applications. Most data exhibit less than perfect multicollinearity.

**3. If multicollinearity is less than perfect (i.e. near or high multicollinearity) , OLS estimators retain the property of BLUE**

**Explanation:**

Note: While we were proving the BLUE property of OLS estimators in simple and multiple regression models(module-I); we did not make use of the assumption of no multicollinearity. Hence, if the basic assumptions which are important to prove the BLUE property are not violated ,whether multicollinearity exist or not ,the OLS estimators are BLUE .

**3. Although BLUE, the OLS estimators have large variances and covariances.**

$$\text{var}(\hat{\beta}_2) = \frac{\sum x^2}{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2}$$

Multiply the numerator and the denominator by  $\frac{1}{\sum x_2^2}$

$$\begin{aligned} \text{var}(\hat{\beta}_2) &= \frac{\sum x^2 \frac{1}{\sum x_2^2}}{\sum x_1^2 \frac{1}{\sum x_2^2} - (\sum x_1 x_2 \frac{1}{\sum x_2^2})^2} = \frac{\sum x^2}{\sum x_1^2 - \frac{(\sum x_1 x_2)^2}{\sum x_2^2}} \\ &= \frac{\sum x^2}{\sum x_1^2 - \frac{(\sum x_1 x_2)^2}{\sum x_2^2}} = \frac{\sum x^2}{\sum x_1^2 (1 - r^2)} \end{aligned}$$

Where  $r_{12}^2$  is the square of correlation coefficient between  $x_1$  and  $x_2$ ,

If  $x_2 = x_{1i} + v_i$ , what happen to the variance of  $\hat{p}_1$  as  $r_{12}^2$  is line rises.

As  $r_{12}$  tends to 1 or as collinearity increases, the variance of the estimator increase and in the limit when  $r_{12} = 1$  variance of  $\hat{p}_1$  becomes infinite.

$$\text{Similarly } \text{cov}(\hat{p}_1, \hat{p}_2) = \frac{-r_{12}\sigma^2}{(1-r_{12}^2)\sqrt{\sigma_{x_1}^2\sigma_{x_2}^2}} \cdot \quad (\text{why?})$$

As  $r_{12}$  increases to ward one, the covariance of the two estimators increase in absolute value. The speed with which variances and covariance increase can be seen with the **variance-inflating factor (VIF)** which is defined as:

$$VIF = \frac{1}{1 - r_{12}^2}$$

VIF shows how the variance of an estimator is inflated by the presence of multicollinearity. As  $r_{12}^2$  approaches 1, the VIF approaches infinity. That is, as the extent of collinearity increase, the variance of an estimator increases and in the limit the variance becomes infinite. As can be seen, if there is no multicollinearity between  $x_1$  and  $x_2$ , VIF will be 1.

Using this definition we can express  $\text{var}(\hat{p}_1)$  and  $\text{var}(\hat{p}_2)$  interms of VIF

$$\text{var}(\hat{p}_1) = \frac{\sigma^2}{\sigma_{x_1}^2} VIF \text{ and } \text{var}(\hat{p}_2) = \frac{\sigma^2}{\sigma_{x_2}^2} VIF$$

Which shows that variances of  $\hat{p}_1$  and  $\hat{p}_2$  are directly proportional to the VIF.

*4. Because of the large variance of the estimators, which means large standard errors, the confidence interval tend to be much wider, leading to the acceptance of “zero null hypothesis” (i.e. the true population coefficient is zero) more readily.*

*5. Because of large standard error of the estimators, the computed t-ratio will be very small leading one or more of the coefficients tend to be statistically insignificant when tested individually.*

6. Although the *t*-ratio of one or more of the coefficients is very small (which makes the coefficients statistically insignificant individually),  $R^2$ , the overall measure of goodness of fit, can be very high.

Example: if  $y = \alpha + p_1x_1 + p_2x_2 + \dots + p_kx_k + v_i$

In the cases of high collinearity, it is possible to find that one or more of the partial slope coefficients are individually statistically insignificant on the basis of *t*-test. But the  $R^2$  in such situations may be so high say in excess of 0.9. In such a case on the basis of *F* test one can convincingly reject the hypothesis that

$p_1 = p_2 = \dots = p_k = 0$  Indeed, this is one of the signals of multicollinearity- insignificant *t*-values but a high overall  $R^2$  (i.e. a significant *F*-value).

7. The OLS estimators and their standard errors can be sensitive to small change in the data.

**Dear Readers!** These are the major consequences of near or high multicollinearity. If you have any comments or suggestion, you are welcome!

#### 4.3.4 Detection of Multicollinearity

A recognizable set of symptoms for the existence of multicollinearity on which one can rely are:

- a. High coefficient of determination ( $R^2$ )
- b. High correlation coefficients among the explanatory variables ( $r_{x_i x_j}$ 's)
- c. Large standard errors and smaller *t*-ratio of the regression parameters

Note: None of the symptoms by itself are a satisfactory indicator of multicollinearity. Because:

- i. Large standard errors may arise for various reasons and not only because of the presence so linear relationships among explanatory variables.

ii. A high  $r_{x_i x_j}$  is only sufficient but not a necessary condition (adequate condition) for the existence of multicollinearity because multicollinearity can also exist even if the correlation coefficient is low.

However, the combination of all these criteria should help the detection of multicollinearity.

#### 4.3.4.1 Test Based on Auxiliary Regressions:

Since multicollinearity arises because one or more of the regressors are exact or approximately linear combinations of the other regressors, one may of finding out which X variable is related to other X variables to regress each  $X_i$  on the remaining X variables and compute the corresponding  $R^2$ , which we designate as  $R_i^2$ , each one of these regressions is called auxiliary to the main regression of Y on the X's. Then, following the relationship between F and  $R^2$  established in chapter three under over all significance , the variable,

$$R_i = \frac{R^2_{X_1, X_2, \dots, X_k} / k - 2}{1 - R^2_{X_1, X_2, X_3, \dots, X_k} / (n - k + 1)} \sim F_{(k-2, n-k+1)}$$

where: - n is number of observation

- k is number of parameters including the intercept

If the computed F exceeds the critical F at the chosen level of significance, it is taken to mean that the particular  $X_i$  collinear with other X's; if it does not exceed the critical F, we say that it is not collinear with other X's in which case we may retain the variable in the model.

If  $F_i$  is statistically significant, we will have to decide whether the particular  $X_i$  should be dropped from the model.

Note that according tot Klieri's rule of thumb, which suggest that multicollinearity may be a trouble some problem only if  $R^2$  obtained from an auxiliary regression is greater than the overall  $R^2$ , that is obtained from the regression of Y on all regressors.

**4.3.4.2 The Farrar-Glauber test** - They use three statistics for testing multicollinearity. There are chi-square, F-ratio and t-ratio. This test may be outlined in three steps.

**A. Computation of  $R^2$  to test orthogonality:** two variables are called orthogonal if  $r_{x_i x_j} = 0$  i.e. if there is no any colinearity between them. In our discussion of multiple regression models, we have seen the matrix representation of a three explanatory variable model which is given by

$$X'X = \begin{bmatrix} \sum x_1^2 & \sum x_1 x_2 & \sum x_1 x_3 \\ \sum x_2 x_1 & \sum x_2^2 & \sum x_2 x_3 \\ \sum x_3 x_1 & \sum x_3 x_2 & \sum x_3^2 \end{bmatrix}$$

Divide each elements of  $X'X$  by  $\sqrt{\sum x_i^2} \sqrt{\sum x_j^2}$  and compute the determinant

$$\begin{aligned} & \begin{bmatrix} \sum x_1^2 & \sum x_1 x_2 & \sum x_1 x_3 \\ \sum x_2 x_1 & \sum x_2^2 & \sum x_2 x_3 \\ \sum x_3 x_1 & \sum x_3 x_2 & \sum x_3^2 \end{bmatrix} \\ & \rightarrow \begin{bmatrix} \frac{\sum x_1^2}{\sqrt{(\sum x_1^2)^2}} & \frac{\sum x_1 x_2}{\sqrt{\sum x_1^2} \sqrt{\sum x_2^2}} & \frac{\sum x_1 x_3}{\sqrt{\sum x_1^2} \sqrt{\sum x_3^2}} \\ \frac{\sum x_2 x_1}{\sqrt{\sum x_1^2} \sqrt{\sum x_2^2}} & \frac{\sum x_2^2}{\sqrt{(\sum x_2^2)^2}} & \frac{\sum x_2 x_3}{\sqrt{\sum x_2^2} \sqrt{\sum x_3^2}} \\ \frac{\sum x_3 x_1}{\sqrt{\sum x_1^2} \sqrt{\sum x_3^2}} & \frac{\sum x_3 x_2}{\sqrt{\sum x_2^2} \sqrt{\sum x_3^2}} & \frac{\sum x_3^2}{\sqrt{(\sum x_3^2)^2}} \end{bmatrix} \\ & \rightarrow \begin{bmatrix} 1 & r_{12} & r_{13} \\ r_{12} & 1 & r_{23} \\ r_{13} & r_{23} & 1 \end{bmatrix} \end{aligned}$$

The value of the determinant is equal to zero in the case of perfect multicollinearity. (since  $r_{12}, r_{23} = 1$ )

On the other hand, in the case of orthogonality of the  $x$ 's, therefore  $r_{ij} = 0$  and the value of the determinant is unity. It follows, therefore, that if the value of this determinant lies between zero and unity, there exists some degree of multicollinearity. For detecting the degree of multicollinearity over the whole set of explanatory variables, Glauber and Farrar suggests  $R^2$  to test in the following way.

$H_0 : x's \text{ are orthogonal (i.e. } r_{x_i x_j} = 0)$

$H_1 : x's \text{ are not orthogonal (i.e. } r_{x_i x_j} \neq 0)$

Glauber and Farrar have found that the quantity

$\Delta^2 = -[n - 1 - 1/6(2k+5)] \cdot \log_e \{\text{value of the standardized determinant}\}$ ; has a  $\chi^2$  distribution with  $1/2k(k-1)$  df. If the computed  $\Delta^2$  is greater than the critical value of  $\chi^2$ , reject  $H_0$  in favour of multicollinearity. But if it is less, then accept  $H_0$ .

### B. Computation of t-ratio to test the pattern of multicollinearity

The t-test helps to detect those variables which are the cause of multicollinearity. This test is performed based on the partial correlation coefficients through the following procedure of hypothesis.

$H_0 : r_{x_i x_j \cdot x_1, x_2, x_3, \dots, x_k} = 0$

$H_1 : r_{x_i x_j \cdot x_1, x_2, x_3, \dots, x_k} \neq 0$

In the three variable model

$$r_{x_1 x_2 \cdot x_3}^2 = \frac{(r_{12} - r_{13} r_{23})^2}{(1 - r_{23}^2)(1 - r_{12}^2)} \quad (\text{How?})$$

$$r_{x_1 x_3 \cdot x_2}^2 = \frac{(r_{13} - r_{12} r_{23})^2}{(1 - r_{23}^2)(1 - r_{12}^2)}$$

$$r_{x_2 x_3 \cdot x_1}^2 = \frac{(r_{23} - r_{12} r_{13})^2}{(1 - r_{12}^2)(1 - r_{13}^2)}$$

$$t^* = \frac{(r_{x_i x_j \cdot x_1, x_2, x_3, \dots, x_k}^2 - \chi_{1-\alpha, k-2}^2) \sqrt{n-k}}{(1 - r_{x_i x_j \cdot x_1, x_2, x_3, \dots, x_k}^2)} \quad (\text{How?})$$

if  $t^* > t$  (tabulate),  $H_0$  is rejected

$t^* < t$  (tabulated),  $H_0$  is accepted, we accept  $X_i$  and  $X_j$  are not the cause of multicollinearity since ( $r_{x_i x_j}$  is not significant)

### 4.3.4.3 Test of multicollinearity using Eigen values and condition index:

Using Eigen values we can drive a number called condition number K as

$$\text{follows: } k = \frac{\text{Maximum eigenvalue}}{\text{Minimum eigen value}}$$

In addition using these value we can drive the condition index (CI) defined as

$$CI = \sqrt{\frac{\text{Max.eigen value}}{\text{min.eigen value}}} = \sqrt{k}$$

Decision rule: if K is between 100 and 1000 there is moderate to strong multicollinearity and if it exceeds 1000 there is severe multicollinearity. Alternatively if  $CI(\approx \sqrt{k})$  is between 10 and 30, there is moderate to strong multicollinearity and if it exceeds 30 there is severe multicollinearity.

*Example .* If  $k=123,864$  and  $CI=352$  – This suggests existence of multicollinearity

#### 4.3.4.4 Test of multicollinearity using Tolerance and variance inflation factor

$$\text{var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum x_i^2} \left| \frac{1}{1 - R_i^2} \right| = \frac{\sigma^2}{\sum x_i^2} \text{VIF}$$

where  $R_i^2$  is the  $R^2$  in the auxiliary regression of  $X_j$  on the remaining  $(k-2)$  regressors and VIF is variance inflation factor.

Some authors therefore use the VIF as an indicator of multicollinearity: The larger is the value of  $\text{VIF}_j$ , the more “troublesome” or collinear is the variable  $X_j$ . However, how high should VIF be before a regressor becomes troublesome? As a rule of thumb, if VIF of a variable exceeds 10 (this will happen if  $R_i^2$  exceeds 0.9) the variable is said to be highly collinear.

Other authors use the measure of tolerance to detect multicollinearity. It is defined

$$\text{as } \text{TOL}_i = (1 - R_i^2) = \frac{1}{\text{VIF}}$$

Clearly,  $\text{TOL}_j = 1$  if  $X_j$  is not correlated with the other regressors, whereas it is zero if it is perfectly related to other regressors.

VIF (or tolerance) as a measure of collinearity, is not free of criticism. As we have

seen earlier  $\text{var}(\hat{\beta}) = \frac{\sigma^2}{\sum x_i^2} (VIF)$ ; depends on three factors  $\sigma^2$ ,  $\sum x_i^2$  and VIF. A

high VIF can be counter balanced by low  $\sigma^2$  or high  $\sum x_i^2$ . To put differently, a high VIF is neither necessary nor sufficient to get high variances and high standard errors. Therefore, high multicollinearity, as measured by a high VIF may not necessary cause high standard errors.

#### 4.3.5. Remedial measures

It is more difficult to deal with models indicating the existence of multicollinearity than detecting the problem of multicollinearity. Different remedial measures have been suggested by econometricians; depending on the severity of the problem, availability of other sources of data and the importance of the variables, which are found to be multicollinear in the model.

Some suggest that minor degree of multicollinearity can be tolerated although one should be a bit careful while interpreting the model under such conditions. Others suggest removing the variables that show multicollinearity if it is not important in the model. But, by doing so, the desired characteristics of the model may then get affected. However, following corrective procedures have been suggested if the problem of multicollinearity is found to be serious.

1. Increase the size of the sample: it is suggested that multicollinearity may be avoided or reduced if the size of the sample is increased. With increase in the size of the sample, the covariances are inversely related to the sample size. But we should remember that this will be true when intercorrelation happens to exist only in the sample but not in the population of the variables. If the variables are collinear in the population, the procedure of increasing the size of the sample will not help to reduce multicollinearity.

2. Introduce additional equation in the model: The problem of multicollinearity may be overcome by expressing explicitly the relationship between multicollinear variables. Such relation in a form of an equation may then be added to the original model. The addition of new equation transforms our single equation (original) model to simultaneous equation model. The reduced form method (which is usually applied for estimating simultaneous equation models) can then be applied to avoid multicollinearity.

3. Use extraneous information – Extraneous information is the information obtained from any other source outside the sample which is being used for the estimation. Extraneous information may be available from economic theory or from some empirical studies already conducted in the field in which we are interested. Three methods, through which extraneous information is utilized in order to deal with the problem of multicollinearity.

**a. Method of using prior information:** Suppose that the correct specification of the model is  $Y = \alpha + \rho_1 X_1 + \rho_2 X_2 + U_i$ , and also  $X_1$  and  $X_2$  are found to be collinear. If it is possible to gather information on the exact value of  $\rho_1$  or  $\rho_2$  from extraneous source, we then make use of such information in estimating the influence of the remaining variable of the model in the following way.

Suppose  $\rho_2^*$  known a priori, then:

$$Y - \rho_2^* X_2 = \alpha + \rho_1 X_1 + U$$

Applying OLS method: 
$$\hat{\rho}_1 = \frac{\sum (y - \hat{p}_2^* x_2) x_1}{\sum x_1^2} = \frac{\sum y x_1 - \hat{p}_2^* \sum x_2 x_1}{\sum x_1^2}$$

i.e.  $\hat{\rho}_1$  is the OLS estimator of the slope of the regression of  $(Y - \rho_2^* X_2)$  on  $X_1$ .

Thus, the estimating procedure described is equivalent to correcting the dependent variable for the influence of those explanatory variables with known coefficients

(from extraneous source of information) and regressing this residual on the remaining explanatory variables.

- b.** Methods of transforming variables: This method is used when the relationship between certain parameters is known as a priori. For instance, suppose that we want to estimate the production function expressed in the form  $Q = AL^\alpha K^\rho e^u$  where  $Q$  is quantity produced  $L$ -labor input and  $K$ - the input of capital. It is required to estimate  $\alpha$  and  $\rho$ . On logarithmic transformation, the function becomes:

$$\ln Q = \ln A + \alpha \ln L + \rho \ln K + u_i$$

$$Q^* = A^* + \alpha L^* + \rho K^* + U$$

The asterisk indicates logs of the variables. Suppose, it is observed that  $K$  and  $L$  move together so closely that it is difficult to separate the effect of changing quantities of labor inputs on output from the effect of variation in the use of capital. Again, let us assume that on the basis of information from some other source, we have a solid evidence that the present industry is characterized by constant returns to scale. This implies that  $\alpha + \rho = 1$ , we can therefore, on the basis of this information, substitute  $\rho = (1 - \alpha)$  in the transformed function. On combining the results, the relationship becomes:

$$\hat{D}_t = \alpha + \hat{p}_1 P_t + \hat{p}_2^* Y_t + \hat{u}_t$$

Where  $\hat{p}_1$  is derived from the time series data,  $\hat{p}_2^*$  is obtained by using the cross-section data. By following the pooling technique, we have skirted the multicollinearity between income and price.

The methods described above are no sure methods to get rid of the problem of multicollinearity. Which of these rules work in practice will depend on the nature of the data under investigation and severity of the multicollinearity problem.